Dictionary Learning as Unsupervised Steering-Vector Generation

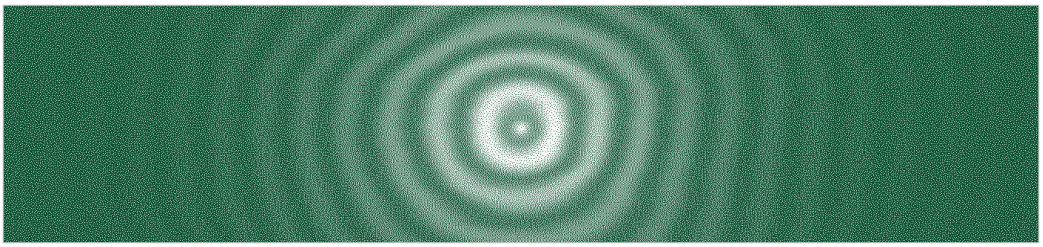
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ORBIT · SEED 1024 · FLOYD--STEINBERG ERROR DIFFUSION

ABSTRACT

The Ontologizer is a multihead, multilayer dictionary-learning architecture whose design goal is unsupervised steering-vector generation*: a stack of competitive classifier "heads" reconstructing an embedding space should discover the space's natural contrastive clusters, MoEfying the representation into small, nameable classification tasks whose within-head entry swaps are steering moves. We compare a softmax (soft-code) Ontologizer and a straight-through (hard-code) arm against* `TopK` *sparse autoencoders (SAEs) on SONAR sentence embeddings under the same whitened objective, and the hard code on GPT-2 small residuals. Soft heads are contrast sets — within-head decode directions repel — but their argmax is not the model's code, and the plain-SAE frontier dominates them in reconstruction. The straight-through arm makes the discrete code the model: at 1900 index bits it reaches* FVU_w 0.154*, and* 0.158 *with its four per-layer gains, its only continuous numbers, pinned to their means. A* `TopK` *SAE held to the same 1900 bits cannot get below* 0.236 *however its coefficients are coded, but an L1 SAE with its coefficients quantized and entropy-coded reaches* 0.128*: its latents fire unevenly, which makes them cheap to name, and the hard code's entries do not. At fixed capacity the residual cascade is worth* 1.79× *over a flat layer, but doubles the collateral of a single-head intervention (66% of other heads against 33%), and two seeds reach identical error while no head past the first layer reproduces. The strongest head encodes topic and genre rather than language, as a language-agnostic embedding should be organized. On GPT-2, an 800-bit hard code of signed activation-space directions recovers 97.6% of the model's loss, more than a* k_{SAE}=32 *SAE at nearly the same* FVU_w*.*

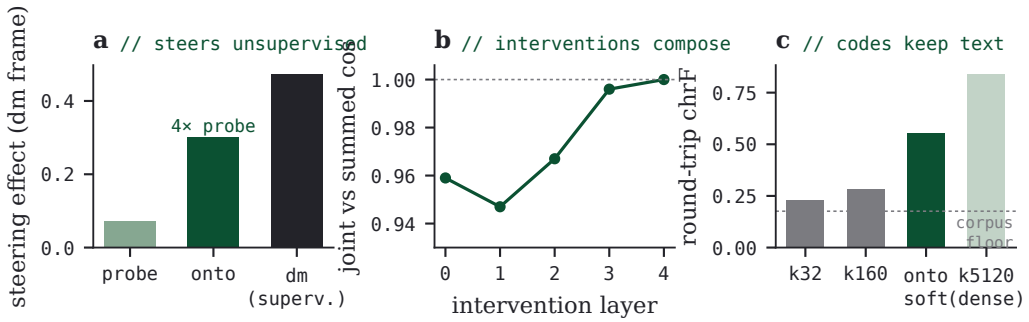


FIGURE 1 · The softmax Ontologizer's steering primitive, measured. (a) Forcing an entry moves decodes along the supervised difference-of-means direction 0.64× as far as supervised steering itself and 4× as far as probe directions (`resid_nc_hm`; Table 17). (b) Joint two-head interventions point where the summed single interventions point, exactly in the final layer (`resid_nc`; Section 8). (c) The sparse SAEs' codes lose most text before any intervention (round-trip chrF 0.23–0.28 against a 0.18 corpus floor) while the soft code keeps it (0.55); the dense `k5120` keeps more but is not a sparse code (Appendix C.5).

1 Evaluation frame

The comparisons below were largely built on SAE-native terms — per-feature monosemanticity, sparse probing, reconstruction at matched sparsity — and we report them unchanged. But the Ontologizer's goal is not a monosemantic dictionary. It is an *unsupervised generator of steering vectors*: each head is a k-way classifier over its input, trained only by reconstruction pressure, and the hope is that a series of such classifiers finds the most natural contrastive clusters in the embedding space. Interpretability value is then expected from *MoEfication*, in an extended sense of the term [1]: decomposing the space into small classification tasks — one per head — each nameable in isolation, with within-head entry swaps as the steering primitive (forcing head i from its argmax entry to entry j' moves the decode along exactly the dimension head i classifies).

Under this frame the evaluation questions are: (a) what does contrast-set structure cost in reconstruction (the Pareto and ladder); (b) do trained heads actually behave as contrastive classifiers (categorical form, semantic anti-coherence); (c) are the learned tasks *natural* — nameable, stable across runs, aligned with generative factors of the data (seed reproducibility, language and topic alignment); and (d) is the steering primitive good — consistent effect, low collateral, at least matching supervised diff-of-means vectors and SAE decoder-row addition (steering fidelity). Metrics that ask whether *individual latents* are concept-aligned remain informative about the SAE side but are the wrong lens for classifier codes, and we flag them as such where they appear. Figure 1 shows the softmax Ontologizer on three of question (d)'s criteria; the hard code steers less reliably than the soft one (Section 4.4).

Question (c) has a prerequisite that the original battery took for granted: a head can only be named if it is the same head on every run. Section 5.5 tests that directly and finds it holds only partly, and only in the first layer, which conditions every per-head result in this document.

2 Related work

No single line of prior work covers this combination; each commitment has a nearest neighbor.[1]

Quantizers and discrete latents. A `DictBlock` layer — h heads, each a softmax over k entries, their atoms summed — is a soft additive quantizer [3], the generalization of product quantization [4] in which every codebook spans the whole space rather than its own block of coordinates (`ConcatDictBlock` is a product quantizer in the dictionary space). In dictionary-learning terms a hard head is vector quantization, one atom at unit coefficient, and a sum of heads lies between vector quantization and free sparse coding, with exactly one unit-coefficient atom per codebook; under the gain–shape split (Eq. (14)) each layer is gain–shape VQ, its one coefficient the residual's measured norm rather than a quantized index [5]. The residual layer stack is a softmax relaxation of residual (multi-stage) VQ, the quantizer of neural codecs such as SoundStream [6], which picks each stage's code by nearest neighbor and judges codes only by rate and perceived quality, never by what a codeword means; residual VQ has also been applied to SONAR space itself, with 64 codebooks of 8192 entries [7]. The straight-through arm (Section 5) makes the analogy literal: with an argmax forward and the softmax's gradient in the backward pass — the straight-through estimator [8] in its categorical form [9] — it is a residual additive quantizer with non-negative atoms, whose codewords are picked by learned classifiers rather than by nearest-codeword search. Vectors of categorical variables trained with straight-through gradients are also the latent state of discrete world models [10, 11], which sample each categorical from the softmax the backward differentiates rather than take its argmax (DreamerV3 also mixes 1% uniform into it [11]); there the codes are compressed state, never read as features. Codebook Features [12] is the closest interpretability ancestor: it finetunes VQ bottlenecks into a network, one codebook per attention head ahead of the shared output projection (in effect `ConcatDictBlock` with its shared decoder), and steers by substituting codes, including swapping one state's code for another's in a synthetic finite-state task. But its codes are chosen by cosine similarity to the codes themselves rather than by a classifier, read one at a time as present-or-absent features, and scored on their target alone, with no defined set of operations and no measure of what else an intervention changes.

[1] Gene Ontology terms were a source of inspiration for the name, which this work shares with Ontologizer, a tool for Gene Ontology term enrichment analysis [2]; the two are separate projects.

Routed and mixture-of-experts dictionaries. Switch SAEs [13] route each activation to one of several expert `TopK` SAEs to cut training compute; routing gathers similar inputs, and experts duplicate some features (5–10% of decoder directions have a near-copy in another expert), which Xu et al. describe as experts collapsing into redundant ensembles and counter by activating several experts per input [14]. A head layer is the dual arrangement — every head fires, choosing one entry — and the hard stack's heads partition near-independently (Figure 31), while in the reference softmax model a head's decode directions repel (Section 4.3): a head there is a contrast set rather than the coherent cluster that MoE work counts as a specialized expert. MoEfication [1] supplies the frame's name but decomposes already-trained ReLU feed-forward layers post hoc into experts of co-activating neurons, several active per input; a head's entries instead exclude one another, and the partition is the training objective's own discovery.

Deep clustering. Each head is a k-way classifier trained without labels, the setting of deep clustering, which supplies two of the objective's terms. The batch-mean entropy term KL_m (Eq. (4)) is the anti-collapse term of contrastive clustering [15] and SCAN [16], which keeps a clusterer from assigning most inputs to a few clusters, and the partition-consensus statistic $\mathrm{KL}_{\mathrm{pwak}}$ (Eq. (22)) pulls each assignment toward a stop-gradient average of its neighbors', in the line of DEC's self-training target [17] and SCAN's neighbor agreement. Those methods learn one clustering of a learned embedding and are judged by agreement with class labels; here many heads partition a fixed embedding at once, trained only through reconstruction, and the partitions are the code.

Structured SAEs. Sparse autoencoders recover features in superposition [18] as sparse combinations of directions from an overcomplete dictionary [19, 20]; the `TopK` baselines here are k-sparse autoencoders [21] trained as in Gao et al. [22]. The field's main structure axis is nesting by scale: Matryoshka SAEs [23] carry the nested loss of Matryoshka representation learning [24] into a dictionary to curb feature absorption [25], and Tree SAEs [26] learn explicit parent–child pairs by gating each child latent on an assigned parent. Grouping is a second axis: subspace-aware SAEs [27] tie latents into rank-r blocks and sparsify across blocks, so that a group is a subspace with free signed coordinates, and OrtSAEs [28] penalize each decoder direction's cosine with its nearest neighbor. Rung 3 (`k32_p5`) imports the Matryoshka nested-prefix loss into the ladder, with five equal blocks rather than the doubling block sizes of [23], a log spacing that Matryoshka representation learning found better than uniform [24]. The Ontologizer's axis is instead *partition into exhaustive, mutually exclusive heads*, and the geometry such heads ask for has a theory: in Elhage et al.'s toy models, a set of features of which only one is active at a time shares a subspace with negative interference inside it, while separate sets take mutually orthogonal subspaces [18]. A head imposes the first arrangement by

construction and asks for the second (Section 3.1.4). Outside dictionary learning, Huang and Hahn [29] partition a residual stream without supervision into mutually independent orthogonal subspaces, on the hypothesis that each holds a group of mutually exclusive features; their subspaces carry continuous activations where a head names one of k values. Post hoc, co-occurring latents of LLM SAEs cluster in decoder space by document genre [30], the co-firing structure `headstruct.py` finds among SONAR SAE latents in place of categorical groups (Appendix C.1). Each of these asks whether an architectural commitment buys interpretability that code density alone does not explain. The autointerp results below answer "not per-latent describability"; the description-coherence view is where the partition begins to pay, suggestively and on the one Ontologizer it scores.

Embedding-space substrate. SAEs over dense text embeddings have become an active line of work: interpreting and discretizing dense-retrieval embeddings [31] and aligning sentence-embedding features with human concepts to steer retrieval [32], while a data-analysis toolkit builds interpretable document embeddings from a token-level LLM SAE instead [33]. All three rely on single-layer SAEs, `TopK` or its BatchTopK variant, the embedding SAEs on English retrieval encoders. None addresses SONAR, where the stakes are higher — Large Concept Models generate directly in SONAR space [7] — and where the paired M2M100 decoder reads reconstructions and interventions back as text without a trained inverter. Inverting a pooled sentence embedding otherwise needs one, which recovers exact text only by iteratively re-embedding and correcting its own decodes [34]; exact training-free inversion is known only from a causal language model's per-token states [35], which a bidirectional, mean-pooled encoder such as SONAR's does not provide. `textfid.py` and `steerfid.py` decode in a single greedy pass through a decoder that paraphrases and is fragile to small perturbations [36, 7], so their text-space numbers are lower bounds (Appendix C.8).

Evaluation stance. SAEBench [37] motivates judging a dictionary with many instruments rather than by reconstruction at fixed sparsity; of its eight metrics the battery adapts three to SONAR (a loss-recovered analogue through the paired decoder, `textfid.py`; LLM-judged detection, `autointerp.py`; k_{probe}-sparse probing on 86 language labels, `langprobe.py`) and omits absorption, RAVEL, unlearning, SCR and TPP. Description scoring uses the detection task of the scaled autointerp pipeline [38]; unlike that pipeline we score positives ranked just below the describing examples and report F1 rather than balanced accuracy, so our numbers are not on its scale. Capacity is compared in bits rather than L0, as MDL-SAEs [39] argue it should be, since an L0 axis rewards ever wider dictionaries with ever fewer active latents; their description length, summed per-latent usage entropy, is the realized-bits estimator applied here to heads (Eq. (36)). AxBench [40] found that, averaged over four Gemma-2 sites, prompting and then finetuning steer better than

every representation-based method. Among single directions only the difference of means beats SAE latents, while a linear probe that ties the best detectors steers worse than they do, and SAE latents chosen by their autointerp labels also trail at concept detection. Detection and steering dissociate, which supports judging a dictionary by what its directions do — the niche of `steerfid.py`'s effect-against-collateral design at matched magnitude, which follows Cunningham et al.'s scoring of feature edits against edit magnitude [19], where AxBench instead tunes each method's strength on held-out prompts. Song et al. [41] argue that run-to-run feature consistency should be an essential SAE evaluation criterion, measured as the mean absolute cosine between decoder directions matched one-to-one across seeds; `partition.py`'s Hungarian-matched head NMI is its partition-level analogue for classifier heads, and `splitting.py`'s match fractions a thresholded, activation-based one. Wang et al. [42] measure the same property as each latent's maximum activation correlation with any latent of another run, against model-seed and SAE-seed skylines. Korznikov et al. [43] retrain SAEs with the decoder frozen at a random initialization, or held near it, or with the encoder frozen at random, and find them within a few points of trained SAEs on autointerp, single-latent sparse probing and RAVEL; the nulls here permute a trained model's own entries, latents or rows, and do not measure what training adds over random components. Decoding parameter directions through the paired decoder, to see whether features can name themselves from weights alone, is a closed-form instance of dataset-free feature synthesis, whose evaluation against implanted ground truth finds that even the best such tools help humans identify a planted trigger less than half the time [44]; here the weights fail too. Two pieces have, to our knowledge, no close analog: reading per-head routing as contrastive classification with a causal intervention interface, and per-group coherence tests against size-matched draws from the same dictionary, applied to trained heads and to judge-written description texts as well as to geometry. Li et al. [30] test whole co-occurrence clusterings of SAE latents against permuted labels.

3 Methods

3.1 Architecture

We propose a novel method which encodes samples as classifications mapped to contrastive vectors. This directly constrains a model's representations. Moreover, classifications can natively be used for steering. Appendix A collects the notation.

The model (Figure 2) consists of l `DictEnc` layers writing into a shared residual accumulator, decoded by a single linear map. Each layer (Figure 3) is composed of a classifier layer followed by a dictionary layer. The classifier splits the input into h independent soft assignment vectors, each over one head's k entries. The dictionary maps each entry to an atom. Each layer writes a weighted average of these atoms to an accumulator. After each layer the accumulator is decoded, and

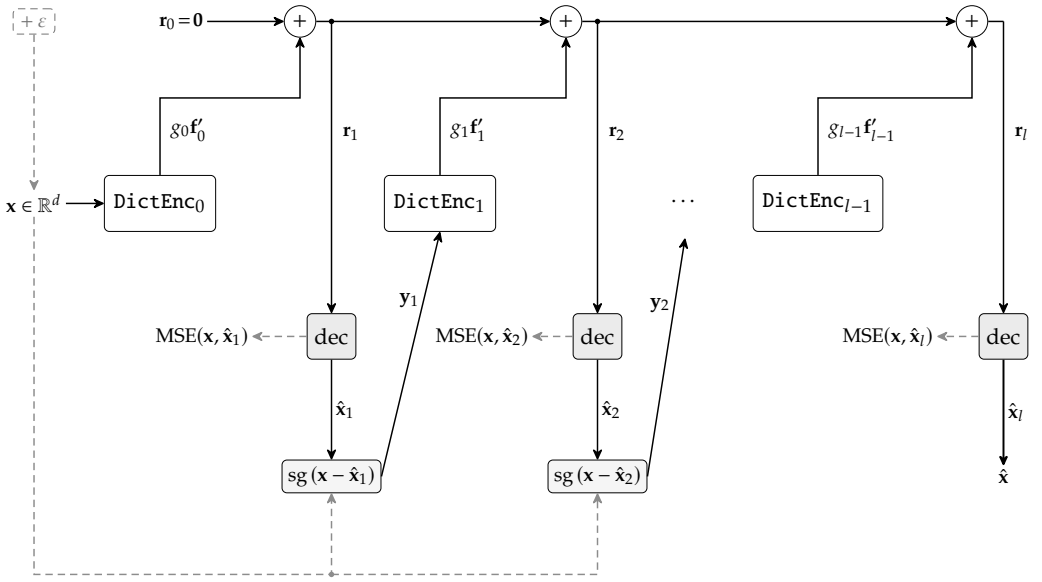


FIGURE 2 · **Ontologizer architecture.** Each `DictEnc` writes its gain-scaled output $g_\ell \mathbf{f}'_\ell$ into a shared accumulator. States count the layers applied: layer ℓ reads $\mathbf{r}_\ell$ and the residual $\mathbf{y}_\ell$, and writes $\mathbf{r}_{\ell+1}$. One decoder serves every prefix, and each prefix is scored against $\mathbf{x}$ (deep supervision). The dashed input-noise hook is off in every reported model. Section 3.1.5 gives the equations.

the residual is passed on: layer ℓ receives $\mathbf{x} - \mathrm{dec}(\mathbf{r}_\ell)$, where $\mathbf{r}_\ell$ holds the ℓ layers before it. Deep supervision [45] constrains every prefix of the stack to reconstruct the input, and because each layer reads the residual, what earlier layers encode is gone from what later layers see: the code is a successive refinement, coarse-to-fine in distortion. Unlike a Tree SAE [26], no later entry is tied to an earlier one, so whether the code is also coarse-to-fine in concept is an empirical question.

Because entries within a head are mutually exclusive rather than independent, each dictionary head serves as a contrastive set. An auxiliary loss term minimizing the heads' support overlap ω (Eq. (6)) pushes each head onto its own coordinates of the dictionary space (Figure 5). Because each head returns a convex combination of its atoms, a head's outputs fill the simplex the atoms span (Section 3.1.4).

DictEnc. Each `DictEnc` consists of a multihead bilinear classifier (`BilinearBlock`) with weights $[\mathsf{W}, \mathsf{V}] \in \mathbb{R}^{2\times h\times k\times d}$ ($d+1$ with the constant coordinate of Eq. (14)) and a multihead linear dictionary (`DictBlock`) with weights $\mathsf{D} \in \mathbb{R}^{h\times k\times e}$. Its input is the residual $\mathbf{y}_\ell$ left by the layers before it ($\mathbf{x}$ itself at layer 0), of which the classifier reads the shape $\mathbf{u}_\ell$ (Eq. (14)). Algorithm 1 gives one layer's training pass, and Algorithm 3 the stack and its loss.

Classifier. A `BilinearBlock` accepts an input $\mathbf{x} \in \mathbb{R}^d$ (in a layer, the shape $\mathbf{u}_\ell$). It returns logits $K \in \mathbb{R}^{h\times k}$ given by

Algorithm 1: `DictEnc`, one layer, in training (`DictEnc.withStats`). Without the router, Γ is all ones and $\mathrm{L1}_S = 0$; without fibers there are no coordinates C; without `resid_gain` the input is not split and $g = 1$.

Input: accumulator $R \in \mathbb{R}^{b \times e}$; layer input $E \in \mathbb{R}^{b \times d'}$, whose row n is $[\mathbf{y}_n; \mathbf{1}_{n_{\text{const}}}]$: the varying part $\mathbf{y}_n$ (the encoded input at layer 0, the residual above) and n_{const} constant coordinates (`resid_const`)

Data: bilinear classifier (Eq. (1)); router (`scaled`); fiber readouts $A_i \in \mathbb{R}^{r \times d'}$; `DictBlock` (Algorithm 2)

Hyperparameters: logit noise σ_K; fiber bound β; the `DictBlock` hyperparameters

Output: accumulator R; assignments $\mathsf{P} \in \mathbb{R}^{b \times h \times k}$; statistics row $\boldsymbol{\psi}$ (Eq. (22))

begin

$g_n \leftarrow \mathrm{sg}\big[\|\mathbf{y}_n\|\big]$; $\quad \mathbf{u}_n \leftarrow \big[\mathbf{y}_n / g_n; \mathbf{1}_{n_{\text{const}}}\big]$ `// gain and shape`

$\mathsf{K}, \mathrm{L1}_K \leftarrow \mathrm{classifier}(\mathbf{u}_{1:b})$ `// L1 of the gate factor`

$\mathsf{K} \leftarrow \mathrm{noise}_K(\mathsf{K})$ `// Eq. (16) or (17)`

if *router* **then**

$\Gamma, \mathrm{L1}_S \leftarrow \mathrm{router}(\mathbf{u}_{1:b})$; $\quad \Gamma \leftarrow |\Gamma|$ `// unless router_signed`

else

$\Gamma \leftarrow \mathbf{1}$; $\quad \mathrm{L1}_S \leftarrow 0$

end

$\mathbf{c}_{ni} \leftarrow \beta \tanh(A_i \mathbf{u}_n / \beta)$ `// fiber coordinates, Eq. (19)`

$F', \mathsf{P}, (\mathrm{L1}, H, \mathrm{cossim}_b, \mathbb{E}_i \kappa_i, \max_i \kappa_i, \mathrm{KL}_m, \omega) \leftarrow$ `DictBlock`$(\mathsf{K}, \Gamma, \mathsf{C})$

$\mathrm{KL}_{\text{pwak}}, \mathrm{L2}_{\text{pwak}} \leftarrow \mathrm{pwak}(\mathsf{P}, \mathbf{u}_{1:b})$ `// zero unless enabled`

$\mathbf{r}_n \leftarrow \mathbf{r}_n + g_n \, \mathbf{f}'_n$ `// row` n `of` R`: the contribution, at the input's gain`

return R, P,

$\boldsymbol{\psi} = (\mathrm{L1}_K, \mathrm{L1}, H, \mathrm{cossim}_b, \mathbb{E}_i \kappa_i, \max_i \kappa_i, \mathrm{KL}_m, \mathrm{KL}_{\text{pwak}}, \mathrm{L2}_{\text{pwak}}, \mathrm{L1}_S, \omega)$

end

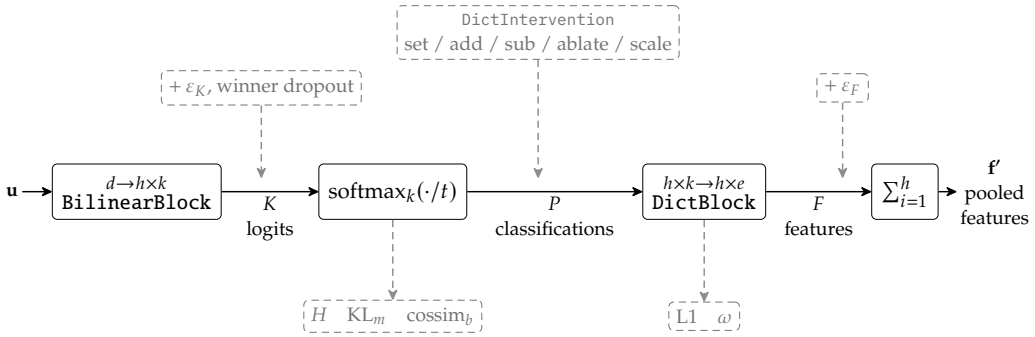


FIGURE 3 · **DictEnc architecture.** A bilinear classifier reads the layer's shape $\mathbf{u}$ and returns logits K; the assignments $P = \mathrm{softmax}_k(K/t)$ select atoms from the dictionary, and the per-head outputs F are pooled into $\mathbf{f}'$. Dashed boxes above an arrow perturb it in training or carry an intervention; dashed boxes below are statistics, usable as auxiliary losses (Sections 3.1.2–3.1.4).

$$\texttt{BilinearBlock}([\mathsf{W}, \mathsf{V}], \mathbf{x}) := \left(\sum_{q=1}^{d} W_{\cdot,\cdot,q}\, x_q\right) \odot \left(\sum_{q=1}^{d} V_{\cdot,\cdot,q}\, x_q\right) \tag{1}$$

An optional noise term ε_K may be added. Winner dropout sets each head's largest logit to $-\infty$ with probability p_{drop}.

Assignments. Soft assignments P are given by $\mathrm{softmax}_k(K/t)$ where t is a temperature hyperparameter. Annealing it sharpens the assignments; each model's schedule is in Appendix B.3. The architecture provides methods for causal interventions on assignments. From P we collect per-sample entropy H, in bits, so a head saturates at $\log_2 k = 5$ bits,

$$\mathrm{entropy}_k(\mathbf{p}) := -\sum_{j=1}^{k} p_j \log_2 p_j, \qquad H := \mathbb{E}_n\, \mathbb{E}_i \big[\mathrm{entropy}_k(\mathbf{p}_{ni})\big] \tag{2}$$

between-sample similarity cossim_b, the mean absolute cosine similarity of the per-head features F over all ordered pairs of samples,

$$\mathrm{cossim}_b := \mathbb{E}_{n,n'} \big|\cos \angle(\mathbf{f}_n, \mathbf{f}_{n'})\big|, \qquad \langle \mathbf{f}_n, \mathbf{f}_{n'} \rangle = \sum_{i=1}^{h} \gamma_{ni}\gamma_{n'i}\, \mathbf{p}_{ni}^{\top} G_i\, \mathbf{p}_{n'i} \tag{3}$$

where $\mathbf{f}_{ni} := \mathsf{F}_{ni\cdot} \in \mathbb{R}^e$ is head i's feature vector for sample n, $\mathbf{f}_n = \bigoplus_{i=1}^{h} \mathbf{f}_{ni}$ is their concatenation, γ_{ni} is head i's router scale on sample n (1 without a router; Section 3.1.7), and $G_i = |D_i||D_i|^{\top} \in \mathbb{R}^{k \times k}$ is head i's dictionary Gram, with $D_i := \mathsf{D}_{i\cdot\cdot}$ its $k \times e$ dictionary. The implementation evaluates Eq. (3) in entry space through the G_i rather than materializing F; the two are identical in value and gradient, but $O(hk^2e + b^2hk)$ instead of $O(b^2he)$.

Finally, KL_m is the divergence of the batch-mean assignment vector from uniform, in bits, averaged over heads.

$$\mathrm{KL}_m := \mathbb{E}_i\left[\log_2 k - \mathrm{entropy}_k(\bar{\mathbf{p}}_i)\right], \qquad \bar{\mathbf{p}}_i := \mathbb{E}_n\left[\mathbf{p}_{ni}\right] \tag{4}$$

In practice H is not needed, as including a cossim_b loss also minimizes H. KL_m serves to anchor the batch mean of each assignment vector $\mathbf{p}_{ni}$ to a uniform distribution. It keeps every entry in use, since a head that collapses onto one entry pays the full $\log_2 k$ bits, and it is meant to make a uniform $\mathbf{p}_{ni}$ code correspond to a maximally generic feature vector.

Dictionary. A `DictBlock` accepts assignments $P \in \mathbb{R}^{h\times k}$ and returns features $F \in \mathbb{R}^{h\times e}$, whose row $\mathbf{f}_i$ is head i's output, given by

$$\texttt{DictBlock}(\mathsf{D}, P)_{i\cdot} = \mathbf{f}_i := \sum_{j=1}^{k} p_{ij}\,\mathbf{a}_{ij}, \qquad \mathbf{a}_{ij} := |\mathsf{D}_{ij\cdot}| \tag{5}$$

where the atom $\mathbf{a}_{ij}$ is the absolute value of head i's dictionary row for entry j ($\mathsf{D}_{ij\cdot}$ itself when the dictionary is signed). Because atoms are absolute values and $p_{ij} \in [0, 1]$, features are strictly additive. Since each head's assignment sums to one, $\mathbf{f}_i$ lies in the simplex whose vertices are the head's atoms (Figure 5), the shape Park et al. [46] derive for the representation of a categorical concept: a head's entries are mutually exclusive alternatives, not independently firing features. The lookup has the form of attention over the head's own dictionary, a softmax over per-entry scores weighting the atoms as values [47, 48, 49], with scores computed from the input alone; with a linear classifier it is one update of a modern Hopfield layer at inverse temperature $1/t$ [50]. Algorithm 2 gives the full training pass, with the regularizing perturbations and statistics around this lookup.

From the assignments and the atoms we collect the heads' support overlap ω, the mean cosine over ordered pairs of distinct heads between their usage-weighted coordinate profiles $\boldsymbol{\pi}_i$,

$$\boldsymbol{\pi}_i := \sum_{j=1}^{k} \bar{p}_{ij}\,|\mathbf{a}_{ij}|, \qquad \omega := \mathbb{E}_{i\neq i'}\,\frac{\langle \boldsymbol{\pi}_i, \boldsymbol{\pi}_{i'}\rangle}{\|\boldsymbol{\pi}_i\|\|\boldsymbol{\pi}_{i'}\|} \tag{6}$$

with $\bar{\mathbf{p}}_i$ the batch-mean assignment of Eq. (4). The overlap is zero exactly when no two heads share a coordinate of the dictionary space and one when every head uses the same coordinates in the same proportions (Section 3.1.4). It costs $O(hke + h^2e)$, independent of the batch. Weighting by usage keeps dead entries out of it and gives the classifier a gradient path. Figure 4 shows the assignments, the atoms and both cosine statistics for one trained head. We also collect the sparsity penalty L1 after the noise ε_F is added, $\tilde{\mathsf{F}} := \mathsf{F} + \varepsilon_F$. L1 is the batch mean of each sample's norm, so it does not change with b but scales with h and e; its loss weight is correspondingly small.

Algorithm 2: `DictBlock` in training (`DictBlock.withStats`). At inference the four stochastic steps (revival, winner dropout, head dropout, fiber dropout) and the feature noise are skipped. Every statistic is stop-gradiented unless its loss term is enabled.

Input: logits $\mathsf{K} \in \mathbb{R}^{b\times h\times k}$; router scales $\Gamma \in \mathbb{R}^{b\times h}_{\geq 0}$ (all ones without a router); fiber coordinates $\mathsf{C} \in \mathbb{R}^{b\times h\times r}$ (none without fibers)
Data: dictionary $\mathsf{D} \in \mathbb{R}^{h\times k\times e}$; fiber bases $U_{ij} \in \mathbb{R}^{e\times r}$
Hyperparameters : temperature t; feature noise σ_F; winner dropout p_{drop}; revival rate p_{rev} and threshold τ_{rev}; head dropout p_{head}; fiber dropout p_{fib}; entries kept per head k_{sel} (1 under `ste`/`argmax`, k under softmax, and the rule's own value under `top`k_{sel}, e.g. 4 for `top4`)
Output: pooled features $F' \in \mathbb{R}^{b\times e}$; assignments $\mathsf{P} \in \mathbb{R}^{b\times h\times k}$; statistics $(\text{L1}, H, \text{cossim}_b, \mathbb{E}_i\kappa_i, \max_i \kappa_i, \text{KL}_m, \omega)$

begin

$\mathbf{a}_{ij} \leftarrow |\mathsf{D}_{ij\cdot}|$ `// atoms; D_ij. itself if signed`

if $k_{\text{sel}} < k$ **then** `// revive starved entries`

use$_{ij} \leftarrow$ share of rows on which entry j is among head i's top k_{sel} logits

foreach (n, i) *with probability* p_{rev} **do**

draw j uniformly from $\{j : \text{use}_{ij} < \tau_{\text{rev}}\, k_{\text{sel}}/k\}$, if any

raise entry j's logit on row n to head i's maximum `// tie the leader`

end

end

each (n, i): $\max_j[\mathsf{K}]_{nij} \leftarrow -\infty$ with probability p_{drop} `// winner dropout`

$\mathsf{P} \leftarrow \text{select}_k(\mathsf{K}/t)$ `// softmax, top`k_{sel} `or straight-through`

$H \leftarrow \mathbb{E}_{n,i}\, \text{entropy}_k(\mathbf{p}_{ni})$ `// Eq. (2)`

$\bar{\mathbf{p}}_i \leftarrow \text{sg}[\mathbb{E}_n\, \mathbf{p}_{ni}]$ `// head origin`

foreach (n, i) **do** `// head dropout`

$\mathbf{p}'_{ni} \leftarrow \bar{\mathbf{p}}_i$ with probability p_{head}, else $\bar{\mathbf{p}}_i + (\mathbf{p}_{ni} - \bar{\mathbf{p}}_i)/(1 - p_{\text{head}})$

end

each (n, i): $\mathbf{c}_{ni} \leftarrow \mathbf{0}$ with probability p_{fib} `// fiber dropout`

$\mathbf{f}_{ni} \leftarrow \gamma_{ni} \sum_j p'_{nij}\, \mathbf{a}_{ij}$ `// Eq. (5); interventions act on` $\mathbf{p}'$

$\tilde{\mathsf{F}} \leftarrow \text{noise}_{\text{featvar}}(\mathsf{F})$ `// Eq. (18)`

$\text{cossim}_b \leftarrow \mathbb{E}_{n,n'}|\cos\angle(\mathbf{f}_n, \mathbf{f}_{n'})|$ `// Eq. (3), from P through the Grams`

$\kappa_i \leftarrow \mathbb{E}_{j\neq j'} \cos\angle(\mathbf{a}_{ij}, \mathbf{a}_{ij'})$ `// within-head atom cosine`

$\text{KL}_m \leftarrow \mathbb{E}_i \left[\log_2 k - \text{entropy}_k(\bar{\mathbf{p}}_i)\right]$ `// Eq. (4)`

$\boldsymbol{\pi}_i \leftarrow \sum_j \bar{p}_{ij}\, |\mathbf{a}_{ij}|$; $\omega \leftarrow \mathbb{E}_{i\neq i'} \cos\angle(\boldsymbol{\pi}_i, \boldsymbol{\pi}_{i'})$ `// support overlap`

$\text{L1} \leftarrow \mathbb{E}_n \sum_i \|\tilde{\mathbf{f}}_{ni}\|_1$

$\mathbf{f}'_n \leftarrow \sum_i \tilde{\mathbf{f}}_{ni}$ `// concatenated, not summed, under` `ConcatDictBlock`

if *fibers* **then** `// Eq. (19), then any norm cap`

$\mathbf{f}'_n \leftarrow \mathbf{f}'_n + \sum_i \gamma_{ni} \sum_j p'_{nij}\, U_{ij}\, \mathbf{c}_{ni}$

end

return F', P, $(\text{L1}, H, \text{cossim}_b, \mathbb{E}_i\kappa_i, \max_i \kappa_i, \text{KL}_m, \omega)$

end

$$\mathrm{L1}(\tilde{\mathsf{F}}) := \frac{1}{b}\sum_{n=1}^{b}\sum_{i=1}^{h}\sum_{q=1}^{e}|\tilde{f}_{niq}| \tag{7}$$

L1 tends to decrease with only MSE loss, but incorporating it into the loss speeds up convergence.

Disjoint support Between heads, orthogonality of nonnegative features is not a rotation constraint but a support-partition constraint. Every $\mathbf{f}_{ni}$ lies in the nonnegative orthant of $\mathbb{R}^e$: the dictionary contributes $|\mathsf{D}| \geq 0$ and the assignments $P \geq 0$. For nonnegative $\mathbf{v}, \mathbf{v}'$ the inner product $\langle \mathbf{v}, \mathbf{v}' \rangle = \sum_q v_q v'_q$ is a sum of nonnegative terms, so it vanishes if and only if every term does,

$$\langle \mathbf{v}, \mathbf{v}' \rangle = 0 \iff \mathrm{supp}(\mathbf{v}) \cap \mathrm{supp}(\mathbf{v}') = \emptyset \tag{8}$$

Signed vectors admit no such conclusion: there orthogonality is cancellation, and two vectors with identical support can be orthogonal. The support overlap of Eq. (6) sidesteps this. It compares the absolute values of the atoms, which have disjoint supports exactly when they are orthogonal whatever the atoms' signs, so it measures disjointness under a signed dictionary too; and it is computed from the atoms and the batch-mean assignment rather than from noised features, which leave the orthant and the argument with it.

The constraint binds the dictionary, not merely one sample's features. Because the assignments are a softmax, $p_{nij} > 0$ for every entry, and expanding Eq. (5) gives

$$\langle \mathbf{f}_{ni}, \mathbf{f}_{ni'} \rangle = \sum_{j=1}^{k}\sum_{j'=1}^{k} p_{nij}\, p_{ni'j'} \left\langle \mathbf{a}_{ij}, \mathbf{a}_{i'j'} \right\rangle \tag{9}$$

a strictly positive combination of all k^2 cross-head atom overlaps. A single sample on which heads i and i' are orthogonal therefore forces $\langle \mathbf{a}_{ij}, \mathbf{a}_{i'j'} \rangle = 0$ for every pair of atoms (j, j'): the two heads own disjoint blocks of coordinates as a property of D, holding for all inputs rather than on the data that happened to be measured. That is the premise a static reading of the dictionary needs, and it is stronger than what any signed-feature architecture can ask for. The budget is not tight either. In the softmax configuration (Table 9), h=32 heads partitioning e=2048 coordinates leaves 64 dimensions per head to hold k=32 atoms, which need not be disjoint from each other within a head; head-disjointness is most of what the width $e > d$ buys.

Two different orthogonalities are in play and only the between-head one is architectural. Disjoint support assigns each head a block of coordinates; *within* a head the k atoms are contrastive alternatives spread apart inside that block (Figure 5). Section 4.3 measures the second. Neither is guaranteed at convergence: ω is a penalty, and the geometry is built to make a support partition the cheap

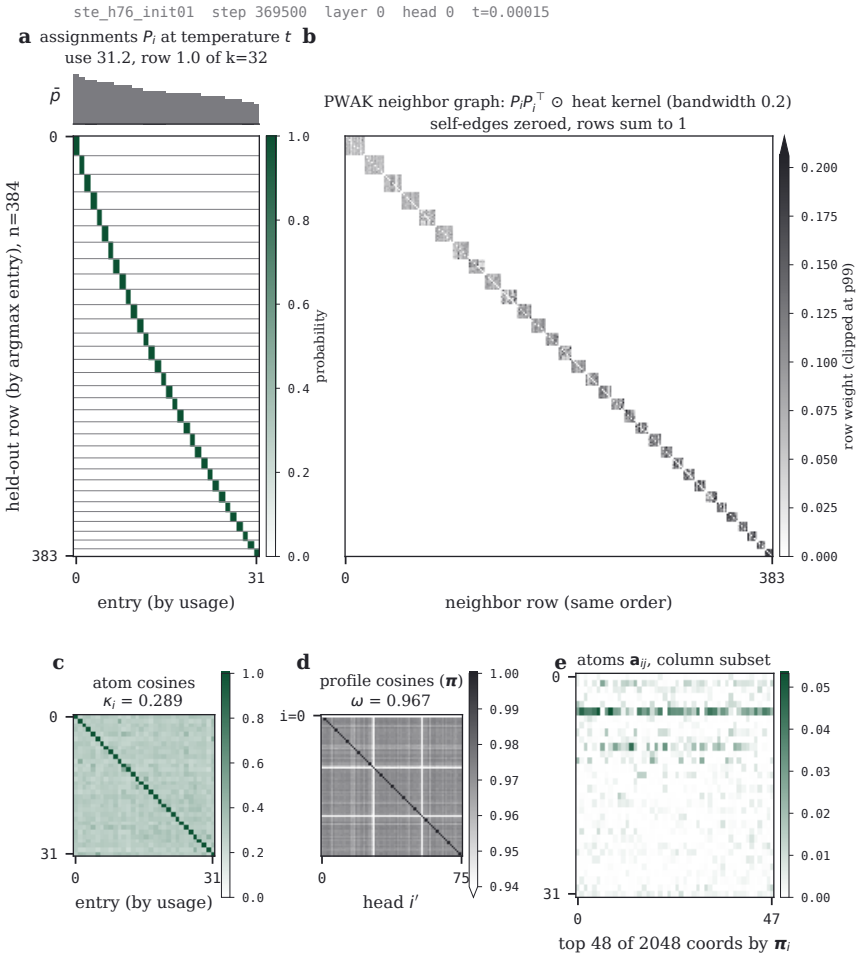
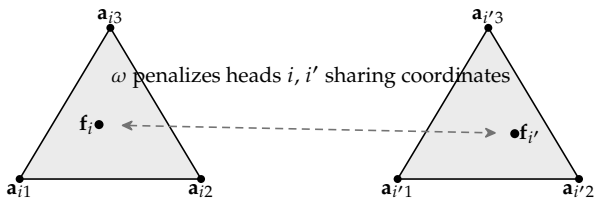


FIGURE 4 · **One trained head, block by block.** Head i=0 of layer 0 of the straight-through model `ste_h76_init01` (step 369,500, t=0.00015) on n=384 rows drawn from the eval tail; one seed. (a) The assignments P_i, rows sorted by winning entry and entries by mean usage $\bar{p}_{ij}$ (bars). "Use" is the effective number of entries across rows, $\exp H(\bar{\mathbf{p}}_i)$, and "row" the effective number within a row, the exponential of the mean per-row entropy: 31.2 and 1.0 of k=32, a hard code that uses nearly every entry. (b) The neighbor graph that $\mathrm{KL}_{\mathrm{pwak}}$ and $\mathrm{L2}_{\mathrm{pwak}}$ (Eq. (22)) diffuse over, in the same row order: the co-assignment matrix $P_iP_i^\top$ times the heat kernel $\exp((\langle \mathbf{u}_n, \mathbf{u}_{n'} \rangle - 1)/0.2)$ of the layer's shapes without their constant coordinate, self-edges removed and each row normalized to sum to one. A hard code makes it block-diagonal by cell. This model was not trained with either statistic, so the graph is the one they would build, not one that shaped it. (c) Cosines between head i's atoms, whose off-diagonal mean is κ_i (0.289). (d) Cosines between the heads' coordinate profiles $\boldsymbol{\pi}$ (Eq. (6)) with $\bar{\mathbf{p}}$ the mean over the n rows, whose off-diagonal mean is ω (0.967), head i's row outlined. (e) Head i's atoms on the 48 of e=2048 coordinates where $\boldsymbol{\pi}_i$ is largest, a column subset rather than a projection. Each block has its own scale, white at zero, except (d): ω is near 1, so its scale starts at the 5th percentile of the off-diagonal cosines, and the few below it show as white. Figure 15 is the same head of a softmax model.



$$\mathbf{f}' = \sum_{i=1}^{h} \mathbf{f}_i, \qquad \mathbf{f}_i = \sum_{j=1}^{k} p_{ij}\, \mathbf{a}_{ij}, \qquad \sum_{j=1}^{k} p_{ij} = 1$$

FIGURE 5 · `DictBlock` **head geometry** for h=2, k=3. Each head's output $\mathbf{f}_i$ lies in the simplex spanned by its atoms $\mathbf{a}_{ij}$, at the point its assignment picks. The support overlap ω compares heads: driving it to zero would give them disjoint coordinates (Section 3.1.4).

solution rather than to force one. Measured on the trained models, the cheap solution was not taken: the within-head orthogonality holds, and no model partitions its coordinates between heads (Appendix G.2). Under `ConcatDictBlock` heads write disjoint slices by construction and ω is identically zero.

We add noise ε_F to F, then sum over the head dimension to obtain pooled features $\mathbf{f}' \in \mathbb{R}^e$, which are added to the accumulator $\mathbf{r}$.

$$\mathbf{f}' = \sum_{i=1}^{h} \tilde{\mathbf{f}}_i \tag{10}$$

Decoder. Each `DictEnc` output $\mathbf{f}'_\ell$, scaled by its input's gain g_ℓ, is added to an accumulator $\mathbf{r}$, whose state after ℓ layers is

$$\mathbf{r}_\ell = \sum_{\ell'=0}^{\ell-1} g_{\ell'}\, \mathbf{f}'_{\ell'}, \qquad g_\ell := \mathrm{sg}[\mathrm{L2}(\mathbf{y}_\ell)] \tag{11}$$

Under `resid_gain` a layer classifies only the direction of its input, so the gain carries the magnitude it cannot see; without it $g_\ell = 1$. The empty sum makes $\mathbf{r}_0 = \mathbf{0}$, so layer 0's input is the encoded input itself, $\mathbf{y}_0 = \mathbf{x}$, whose gain is identically 1 on unit-norm SONAR embeddings.

A single-layer unbiased linear decoder dec projects $\mathbf{r}_\ell$ back to the input space $\mathbb{R}^d$. Weights $W_{\mathrm{dec}} \in \mathbb{R}^{d \times e}$ are shared for all $\texttt{DictEnc}_\ell$.

$$\hat{\mathbf{x}}_\ell = W_{\mathrm{dec}}\, \mathbf{r}_\ell \tag{12}$$

The residual $\mathbf{y}_\ell$, layer ℓ's input, is given by $\mathbf{x} - \hat{\mathbf{x}}_\ell$, stop-gradiented so that lower layers cannot write a communication code into the residual instead of reducing it. Layer ℓ splits it into the gain g_ℓ (Eq. (11)) and a shape $\mathbf{u}_\ell$, its unit direction with a constant coordinate appended, which is what the classifier, the router and the fibers read; ϵ is the floating-point epsilon of the working dtype.

$$\mathbf{y}_\ell = \mathrm{sg}[\mathbf{x} - \hat{\mathbf{x}}_\ell]\,, \qquad \mathrm{L2}(\mathbf{y}) := \sqrt{\sum_{q=1}^{d} y_q^2} \tag{13}$$

$$\mathbf{u}_\ell = \frac{\mathbf{y}_\ell}{\mathrm{L2}(\mathbf{y}_\ell) + \epsilon} \oplus [1] \tag{14}$$

Normalizing fixes the classifier's input scale: a bilinear classifier's logits scale with the square of its input's norm, so they would explode on large inputs (raw GPT-2 activations) and vanish on the small residuals deep layers see; the gain carries the magnitude instead. This is the re-scaling invariance of RMSNorm [51] without its learned gain, on the unit sphere rather than RMSNorm's sphere of radius $\sqrt{d}$, which is why the classifier's initial logits are small (Appendix E.1). A constant coordinate [1] is appended, giving a dimension of $d + 1$. This prevents the sign information of $\mathbf{y}_\ell$ from being lost by the bilinear transformation. Effectively, it adds a bias to an otherwise unbiased linear form while preserving eigendecomposition. This is the input augmentation by which Pearce et al. [52] give bilinear layers biases; it splits each output into an interaction, a linear and a constant term.

Noise. Noise is added to X, K, and F, controlled by hyperparameters σ_X, σ_K, and σ_F. Each draws a standard normal of the same shape as its target,

$$\varepsilon \sim \mathcal{N}(0, 1) \tag{15}$$

and scales it three different ways. Classifier logits take unscaled Gaussian noise,

$$\mathrm{noise}_{\mathrm{normal}}(\mathsf{K}) := \mathsf{K} + \sigma_K \varepsilon \tag{16}$$

inputs take noise proportional to their own norm, so the scale is a fraction of signal magnitude rather than an absolute quantity,

$$\mathrm{noise}_{\mathrm{batchnorm}}(X) := X + \sigma_X\, \varepsilon \odot \frac{\mathrm{sg}[\mathrm{L2}(X)]}{\sqrt{d}} \tag{17}$$

and features take noise proportional to each feature's standard deviation across the batch.

$$\mathrm{noise}_{\mathrm{featvar}}(\mathsf{F}) := \mathsf{F} + \sigma_F\, \varepsilon \odot \mathrm{sg}\left[\sqrt{\mathrm{var}_n(\mathsf{F}) + \epsilon}\right] \tag{18}$$

Here $\mathrm{L2}(X)$ is the per-sample norm of Eq. (13), broadcast over d. Both scales are stop-gradiented: without sg the model could evade its own noise by shrinking feature variance, and the ϵ inside the root guards the $0/0$ gradient where a feature's batch variance underflows to exactly zero. Because σ_K enters the softmax as σ_K/t, holding it constant under a falling temperature would grow effective exploration rather than hold it, so models with an annealed temperature anneal σ_K with it. Each model's noise settings are in Appendix B.3.

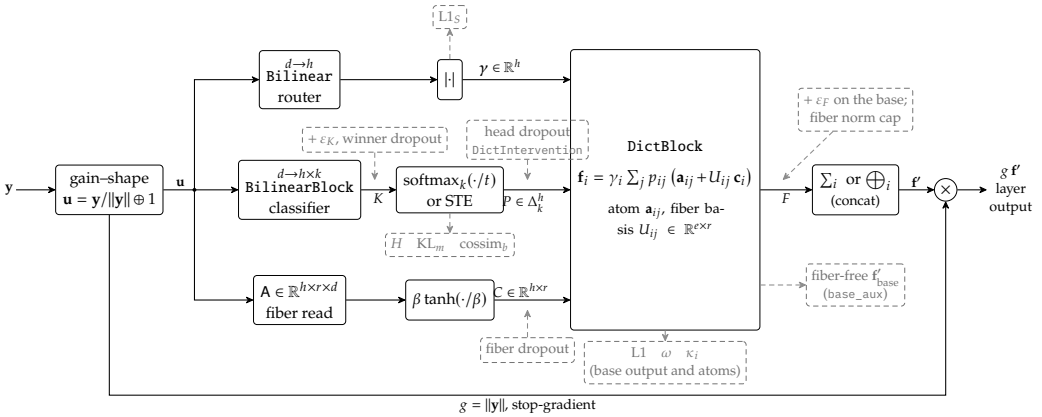


FIGURE 6 · **DictEnc with router and fibers.** The gain–shape split hands the unit input $\mathbf{u}$ to three branches: the classifier (which entry, P), the router (how much, $\boldsymbol{\gamma}$, one non-negative scale per head) and the fiber read (where within the entry's fiber, C, r coordinates per head). They meet in the lookup (Eq. (19)), and the combined heads are scaled by the measured gain g. Dashed boxes above an arrow act on it in training; dashed boxes below are statistics of the fiber-free base output.

Router and fibers. Two optional branches read the same shape $\mathbf{u}$ as the classifier (Eq. (14), with the layer index dropped) and meet it in the dictionary lookup (Figure 6). Both are optional; the models that use them are listed in Appendix B.3.

The *router* (`scaled`) is a second bilinear map, from $\mathbf{u}$ to one scale per head, $\boldsymbol{\gamma} \in \mathbb{R}^h$, made non-negative by an absolute value (`router_signed` drops it). Its scale γ_i multiplies head i's whole output, so the classifier decides *what* a head says and the router *how much*; its own L1, L1$_S$, prices the router alone, where L1 on the output can be paid down by either factor.

Fibers (`fiber_rank` r) give every entry a local basis $U_{ij} \in \mathbb{R}^{e \times r}$ (in its head's slice under `ConcatDictBlock`), and each head reads r coordinates $\mathbf{c}_i = \beta \tanh(A_i \mathbf{u}/\beta)$ from the input, so

$$\mathbf{f}_i = \gamma_i \sum_{j=1}^{k} p_{ij} \left(\mathbf{a}_{ij} + U_{ij}\, \mathbf{c}_i\right). \tag{19}$$

Under a hard code the selected entry is a discrete base point and the fiber a continuous correction within the plane its basis spans (Figure 7). Unconstrained, the correction can take over and leave the base meaningless, so the base is kept honest by fiber dropout, a norm cap ($\|U_{ij}\mathbf{c}_i\| \le \rho$ per head, or summed over heads), a base-only deep-supervision term (`base_aux`, which also decodes the fiber-free output), or training the fibers after freezing the base. A radial fiber (r=1, no basis) only rescales the selected entry. Statistics describe the base output. Head dropout (`p_head_drop`) is the third addition here: in training it replaces a head's classification by the batch mean, its origin, so heads cannot co-adapt and ablating one to its origin stays in distribution.

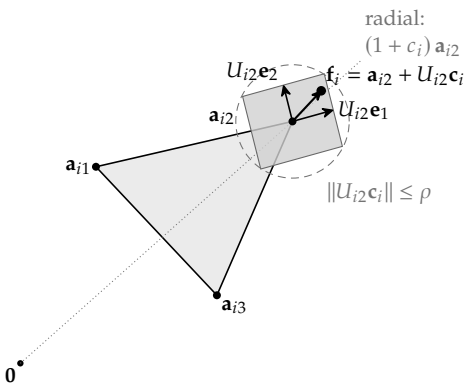


FIGURE 7 · **Fiber geometry** for one head, extending Figure 5. The head selects the entry with atom $\mathbf{a}_{i2}$; its fiber is the plane spanned by U_{i2} at that point, and the coordinates $\mathbf{c}_i$ place the output within it. The dashed circle is the per-head norm cap; the dotted ray is the radial variant, which moves the output only along the atom's own direction.

3.2 Training

Embedding model. We embed texts using SONAR [36], a multimodal and language-agnostic sentence embedding model: a translation-trained encoder whose states are mean-pooled into one 1024-dimensional vector, and a decoder that generates text from that vector alone, in a target language chosen at decoding time. Each embedding here covers the opening of a web document (up to its first 512 tokens), longer than the sentences SONAR is trained on. Translations embed close together, so a reconstructed or steered embedding can be decoded in one fixed language and re-encoded, which is how text fidelity and text-cycle steering are scored (Appendices C.5 and C.8). Large Concept Models [7] generate directly in SONAR space, one sentence embedding per step, which makes it a natural substrate for steering experiments: an edit to a SONAR embedding is an edit to such a model's context or output. The same work finds SONAR decoding fragile, with small perturbations of an embedding able to lose much of its meaning once decoded.

Loss function. Reconstruction fidelity is trained with MSE, where $\mathbf{w} \in \mathbb{R}^d$ is a fixed per-dimension weight vector.

$$\mathrm{MSE}(X, \hat{X}) := \frac{1}{bd} \sum_{n=1}^{b} \sum_{q=1}^{d} w_q \left(\hat{x}_{nq} - x_{nq}\right)^2 \tag{20}$$

The total objective adds the ghost-gradient reconstruction and the auxiliary statistics, each weighted by a scalar. Statistics are *summed* over the l layers rather than averaged.

$$L := \mathrm{MSE}(X, \hat{X}) + s_g \,\mathrm{MSE}(\hat{X}_g,\ X - \hat{X}) + \sum_{\ell=0}^{l-1} \mathbf{s} \cdot \boldsymbol{\psi}_\ell \tag{21}$$

Each layer contributes the statistics (Algorithms 2 and 1)

$$\boldsymbol{\psi}_\ell = \left[\, \mathrm{L1}_K \;\; \mathrm{L1} \;\; H \;\; \mathrm{cossim}_b \;\; \mathbb{E}_i \kappa_i \;\; \max_i \kappa_i \;\; \mathrm{KL}_m \;\; \mathrm{KL}_{\mathrm{pwak}} \;\; \mathrm{L2}_{\mathrm{pwak}} \;\; \mathrm{L1}_S \;\; \omega \,\right]_\ell^\top \tag{22}$$

where $\mathrm{L1}_K$ is the L1 of the classifier's gate, κ_i the mean within-head cosine between head i's atoms, $\mathrm{KL}_{\mathrm{pwak}}$ and $\mathrm{L2}_{\mathrm{pwak}}$ the two partition-consensus statistics, and $\mathrm{L1}_S$ the router's L1. The max of κ_i is logged for a setpoint controller and never itself penalized. Appendix B.4 gives the implementation's row layout.

Each model's weights are in Appendix B.3. The ghost path is disabled throughout, since $\hat{X}_g$ fires only on features that are exactly zero across the whole batch, which bilinear classifiers with softmax selection and absolute-valued dictionaries never produce.

Reconstruction is trained with deep supervision [45]: $\mathrm{MSE}(X, \mathrm{dec}(R_\ell))$ is computed separately for every prefix $\ell = 1, \ldots, l$ and averaged, so each prefix carries weight $1/l$ and each layer must explain what previous layers left unexplained. Supervising every prefix through one shared decoder is the deterministic counterpart of SoundStream's quantizer dropout, which trains a residual quantizer at a random depth per example [6], and the nested-prefix loss of Matryoshka SAEs [23] applied to a residual stack rather than to prefixes of one dictionary. A stop-gradient variant (`deepsup_sg`, off in every reported model) stops each prefix's gradient at the accumulator it was handed, so that a layer trains only against its own residual. The MSE is whitened: $\mathbf{w}$ in Eq. (20) is the per-dimension inverse variance, normalized to mean one, the same whitening the evaluation applies. The implementation applies $\sqrt{\mathbf{w}}$ to X and $\mathrm{dec}(R_\ell)$ before the squared error, which is equivalent. Four statistics are active as loss terms in the reported models (Figure 3; Appendix B.3): between-sample similarity cossim_b, computed from P through the dictionary Gram so it measures feature-space similarity of the per-head reconstructions, and which in practice also minimizes per-sample entropy H; the batch-mean term KL_m, anchoring each head's mean assignment vector to a uniform distribution so that a uniform code corresponds to a maximally generic feature; an L1 penalty on features; and a between-head disjointness term, the support overlap ω, which measures the extent to which heads use nonoverlapping coordinates of the dictionary space. Most reported models trained before ω existed and used its predecessor (Appendix B.4). L1 tends to fall under MSE alone; including it in the loss speeds up convergence. Classification sharpness is controlled by the temperature. Winner dropout and the noise terms ε_K and ε_F regularize classifications and features respectively.

Algorithm 3: `Ontologizer` in training (`Ontologizer.withStats`) and the loss it feeds (`Hyperparams.loss`), for the residual forward (`forward="resid"`) every configuration here uses. Without an encoder, enc is the identity; with direct atoms, dec is.

Input: batch $X \in \mathbb{R}^{b \times d}$

Data: encoder enc; layers $\underset{0}{\texttt{DictEnc}}, \ldots, \underset{l-1}{\texttt{DictEnc}}$ (Algorithm 1); decoder dec (Eq. (12)); MSE weights $\mathbf{w}$; loss weights $\mathbf{s}$, s_g

Hyperparameters: input noise σ_X; deep supervision; `deepsup_sg`; `resid_norm`, `resid_const`, `resid_gain`

Output: loss L; per-layer statistics $\boldsymbol{\psi}_0, \ldots, \boldsymbol{\psi}_{l-1}$

begin

- $E \leftarrow \mathrm{enc}\big(\mathrm{noise}_X(X)\big)$ `// Eq. (17)`
- $R \leftarrow \mathbf{0} \in \mathbb{R}^{b \times e}$ `// state 0: no layers applied`
- $E \leftarrow E \oplus [1]$ `// if resid_const`
- **for** $\ell \leftarrow 0$ **to** $l-1$ **do** `// layer` ℓ `reads state` ℓ`, writes state` $\ell+1$
  - $R_{\mathrm{prev}} \leftarrow R$
  - $R, \mathsf{P}_\ell, \boldsymbol{\psi}_\ell \leftarrow \underset{\ell}{\texttt{DictEnc}}(R, E)$
  - **if** *deep supervision* **then** `// a decode per prefix`
    - $\hat{X}_{\ell+1} \leftarrow \mathrm{dec}\big(R - R_{\mathrm{prev}} + \mathrm{sg}[R_{\mathrm{prev}}]\big)$ `// dec(R) without deepsup_sg`
  - **end**
  - **if** $\ell < l-1$ **then**
    - $Y \leftarrow \mathrm{sg}\big[X - \mathrm{dec}(R)\big]$ `// Eq. (13)`
    - $E \leftarrow Y \oplus [1]$ `// if resid_const; the next layer splits it into gain and shape, Eq. (14) (normalized here instead under resid_norm alone)`
  - **end**
- **end**
- **if** *not deep supervision* **then** $\hat{X}_l \leftarrow \mathrm{dec}(R)$
- $\mathrm{MSE} \leftarrow \mathbb{E}_\ell \, \mathrm{MSE}_{\mathbf{w}}(X, \hat{X}_\ell)$ `// Eq. (20); over the prefixes computed, each weighted` $1/l$
- $L \leftarrow \mathrm{MSE} + s_g \, \mathrm{MSE}_g + \mathbf{s} \cdot \sum_{\ell=0}^{l-1} \boldsymbol{\psi}_\ell$ `// Eq. (21);` $\mathrm{MSE}_g = 0$ `without ghosts`
- **return** L, $(\boldsymbol{\psi}_\ell)_{\ell=0}^{l-1}$

end

3.3 Evaluation

Evaluation runs against `TopK` SAEs trained on identical cached SONAR embeddings under an identical whitened reconstruction objective. The battery follows SAEBench [37] in judging a dictionary with many instruments rather than by reconstruction at fixed sparsity, and adapts three of its eight metrics: a loss-recovered analogue, as downstream text fidelity through the paired M2M100 decoder; blinded autointerp detection scoring; and sparse probing, of language. The rest is specific to this comparison: a reconstruction-versus-capacity Pareto; categorical-form and semantic-coherence tests of head structure; cross-model feature splitting and seed stability; shrinkage decomposition; and steering fidelity. A structural-ablation ladder of SAE variants, each adding one Ontologizer-style commitment at a time, attributes effects to specific structure. AxBench [40] found that prompting, finetuning and one simple supervised direction, the difference of means, beat SAE latents at steering, while a linear probe's direction did not; the steering comparison here, which includes both, is built to face that result.

3.4 Data, models and evaluation

This section is an overview; Appendix B is the full description, with every evaluation's defining equation or algorithm.

Data. The SONAR models train on a fixed cache of 4,000,000 unit-norm SONAR sentence embeddings (d=1024) from an interleaved 86-language mC4 sample, with a language label per row and text recoverable by row index; 80 of the 86 languages, 93% of rows, were encoded under SONAR's English source tag rather than their own (Appendix B.1). Sections 7.1 and 5.5 also use residual-stream activations of GPT-2 small [53] at the output of block 8 of 12 (d=768), a substrate where sparse dictionary learning is known to work [22]; Section 6's separate GPT-2 setup compares only within that section (Appendix B.1).

Objective and score. Every model minimizes the whitened MSE of Eq. (20), and we report whitened FVU (Eq. (23)) on the final 32,768 cache rows, which SAEs and straight-through arms never train on. The softmax Ontologizers predate that holdout, but their scores move by at most 0.3% on freshly encoded rows that no model saw (Appendix B.1). Every score uses the run's own final temperature.

Models. The reference softmax Ontologizer `resid_nc` has l=5 residual layers of h=32 heads with k=32 entries; a selection sweep varies only its selection rule, and the straight-through arm (Section 5) replaces it with hard codes. The SAEs are `TopK` SAEs after Gao et al. [22], plus a structural-ablation ladder that adds one Ontologizer constraint at a time. Two of its rungs, `g160top1` and `g160softmax` (160 heads of hard or softmax competition over a shared decoder), are each a one-layer

Ontologizer without the bilinear classifier, residual stack or non-negative dictionary, `g160top1` with a router fused into its classifier, and are treated as simplified variants rather than baselines (Appendix B.2; configurations in Appendix B.3).

Evaluations. Each is a standalone script on the shared eval split (Appendix B.5): reconstruction against code capacity and a Gaussian rate–distortion reference (`pareto.py`); whether SAE latents form exhaustive, exclusive head-like groups (`headstruct.py`); the semantic coherence of a head's atoms against size-matched nulls (`headcoh.py`); feature splitting and cross-seed head matching by partition and by contribution (`splitting.py`, `partition.py`, `headcontrib.py`); text fidelity through the M2M100 decoder (`textfid.py`); language probing by latent and by partition (`langprobe.py`, `headlang.py`); shrinkage by least-squares refitting on a frozen support (`refit.py`); steering fidelity in text and in embedding space (`steerfid.py`, `steerembed.py`, `steergeom.py`); additivity of paired interventions (`compose.py`); and autointerp (`autointerp.py`).

4 Results: the softmax Ontologizer against SAEs

4.1 The price of structure: reconstruction–capacity Pareto

The SAE frontier dominates the soft Ontologizer's code at every capacity measured (Figure 8b). Truncating the reference model's code to the ν largest deviations per head from its mean assignment (Appendix B.5), it needs 320 coefficients to reach the $\mathrm{FVU}_w \approx 0.49$ that the k_{SAE}=32 SAEs reach with 32, and its coarsest code, one deviation per head (160 coefficients, FVU_w 0.677), is already worse than the SAE's 32-coefficient point, so the very sparse regime is closed to it. Unstructured reconstruction is better still: the dense control reaches FVU_w 1.1×10^{-4} with 3.4× fewer active coefficients than the soft code, about 50× lower error at a matched budget. The truncations are applied after the fact to a model never trained to truncate, so they may overstate what the structure itself costs: a residual quantizer degrades when truncated without training for it, and one trained at a random depth per example does not [6]. The code's information is spread across heads: keeping whole heads instead of deviations is far weaker at matched coefficients (FVU_w 0.50 for 8 of 32 heads per layer, against 0.094 at ν=8).

The soft model's own discrete reading is not a code. Its argmax reconstructs at FVU_w 17.3, 77× the Gaussian reference for 800 bits (0.226; Figure 8a), the least distortion an 800-bit code could reach if the embeddings were Gaussian with their measured covariance (Eq. (25)). Real embeddings are no harder to compress than that [54, Problem 10.8], so the argmax sits at least 77× above the data's distortion–rate function, and on a selection-sweep run of the same shape the gap is nine orders of magnitude (8.7×10^6 against 0.0007 for the soft code). The entries that interventions on a softmax model act on are therefore not what carries its embedding — the failure VQ-VAE reports for a soft relaxation of quantization

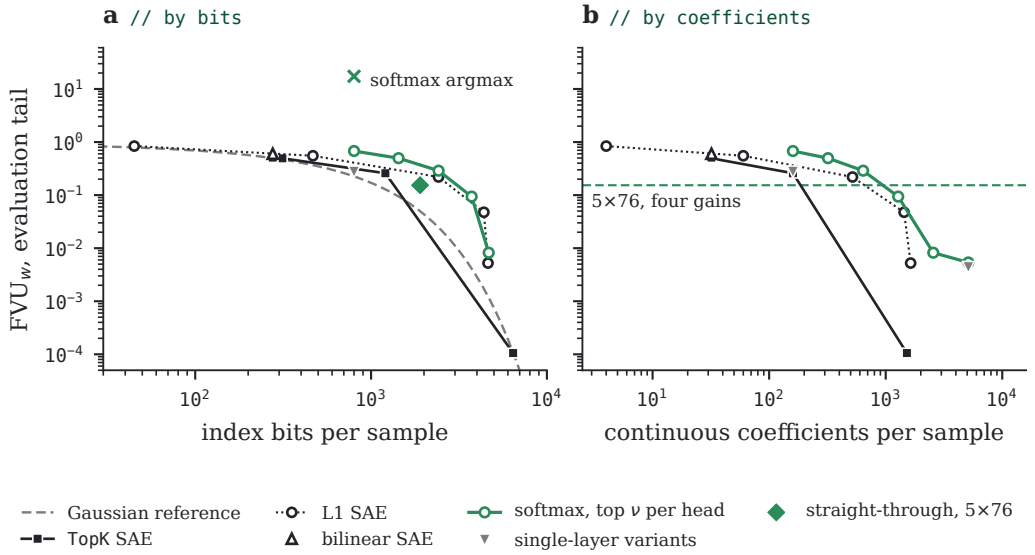


FIGURE 8 · Reconstruction against code size on the shared evaluation tail, by index bits (a) and by continuous coefficients (b): the reference softmax model's truncated codes (`resid_nc`), the SAEs (the L1 SAEs at all five weights of Table 13), the single-layer variants and the straight-through stack (`ste_h76_init01`, Section 5). A code appears only where its size is nonzero, so the soft codes are in (b) alone, the softmax argmax is in (a) alone, and the straight-through stack, whose only continuous numbers are four per-layer gains, is a level in (b). The dashed curve is the Gaussian reference (Eq. (25)). Index bits are the whole rate of a hard code, apart from those gains, but only the support of a code that also sends coefficients, which is how the k_{SAE}=32 and k_{SAE}=5120 SAEs reach the reference in (a); Figure 29 puts every code on one rate axis. The softmax model saw the tail in training and the others hold it out; on rows no model saw, every point moves by at most 0.3%, and the dense control's 1.1×10^{-4} by 0.9% (Appendix B.1). One seed per point, except the k_{SAE}=32 SAE's seed-43 replica.

trained from scratch, whose decoder learns to invert the relaxation [55] — which motivates the straight-through arm of Section 5. The reference does not say that 800 bits are too few, since non-Gaussian data can be compressed below it.

4.2 Structure decomposition (the ladder)

rung	FVU_w	L0	note
k32 baseline	0.499	32	0 dead
g160top1	0.274	160	1469/5120 dead (29%)
g160softmax	**0.0044**	5120	beats `resid_nc` (0.0054)
k32_p5	0.509	32	nested prefixes, ~2% cost
k32_bl	0.602	32	1920/5120 dead (38%)

TABLE 1 · Converged structural rungs, m=5120, 144.3k steps each.

Per-head competition sets most of the gap to unstructured codes: the single-layer soft variant costs about 30× the dense control (0.0044 against 1.1×10^{-4}). Everything the reference Ontologizer adds to it (bilinear classifiers, five residual layers, a shared 2048-dimensional dictionary space) adds nothing to reconstruction, at 2.2× the parameters. A selection-sweep run of the same shape does reach 0.0007, six times below the single-layer variant, and keeps that on rows no model trained on (0.00067 against 0.0044; Appendix B.1). A stack can therefore beat the single layer out of sample, but the reference checkpoint does not, so for the model the rest of this section studies the case has to rest on steering.

Hard per-head selection with a router fused into the classifier (Appendix B.2) costs only about 6% over an unstructured code of equal sparsity (0.274 for the single-layer hard variant against 0.258 for the k_{SAE}=160 SAE). The softmax model's catastrophic argmax is a training artifact, not a property of discrete head codes: one-hot selection inflates its output norm 3.7×, because winning atoms carry about 4× the soft mixture's norm, while a stack trained under the argmax, with no per-head scales at all, reaches FVU_w 0.154 (Section 5). A bilinear encoder costs about 20% in FVU_w and leaves over a third of the dictionary dead, though its factors start random where every linear rung starts tied to its decoder (Appendix B.3), so the cost is not yet the bilinear form's alone.

Categorical structure appears where it is trained in as hard selection. The hard variant's partition is head-like, while soft per-head codes converge to dense, near-uniform mixtures and the SAE's spontaneous groups are co-firing topical clusters (Appendix C.1). The rest of the softmax-versus-SAE battery (feature reliability, text fidelity, language localization and shrinkage) is in Appendix C.

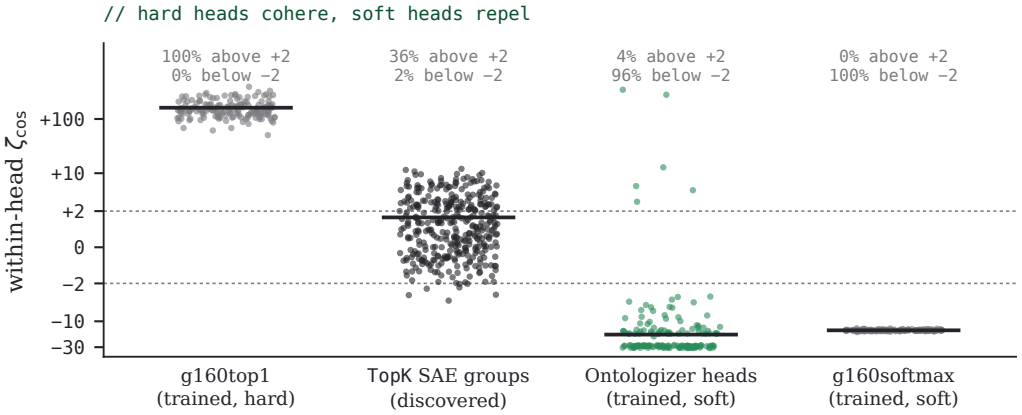


FIGURE 9 · Within-head decode-direction coherence: each head's, or discovered group's, ζ_{cos} against a size-matched null drawn from the same pool (n_{null}=500), on a scale linear between the dotted lines at ±2 and logarithmic beyond them; bars are means. The soft heads are those of `resid_nc` and `g160softmax`, the discovered groups those of the m=11264, k_{SAE}=32 SAE. One seed per model.

4.3 Semantic coherence: heads are contrast sets

Hard-trained heads are coherent, and soft-trained heads are contrast sets (Figure 9). Every head of the single-layer hard variant decodes to directions far more similar than a random set of the same size: winner-take-all with a free magnitude leaves each head one direction at a time, the regime in which covering a multi-dimensional feature takes many near-collinear atoms [27], so each head's atoms tile one region. An SAE's post hoc co-firing groups are mildly coherent, as topical clusters would be. Both soft architectures show the opposite: in 96–100% of heads, the entries' decode directions are more dissimilar than random sets. Softmax competition pushes a head's atoms apart, so its entries are well-separated alternatives, and swapping one for another is a large, clean move along the head's own dimension. The repulsion does not depend on the inner product: under the whitened metric the objective optimizes, it strengthens for the Ontologizer (mean ζ_{cos} −24.4 against −17.6) and is essentially unchanged for the single-layer variants (Appendix C.2).

The hard-code Ontologizer shows the same in its weights. Its within-head atom cosine, decoded into output space, is 0.023 at layer 0 and −0.015 to −0.020 at layers 1–4, so a head's decoded atoms are orthogonal and slightly repelling; read in the dictionary space instead, the statistic is dominated by the decoder's null space (Appendix G.1). Of the two orthogonalities in Section 3.1.4, the within-head one holds in both code types and the between-head one holds in neither: no trained model partitions its dictionary space between heads. The support overlap ω sits at or above a null that shuffles each head's coordinates, and each head's usage covers 52–98% of the coordinates, where a partition into 32 heads needs about 3% (Appendix G.2). Heads are contrast sets that share their coordinates.

4.4 Steering fidelity

Steering is measured in embedding space: a step from the input along a direction, scored by whether the target entry becomes selected (its realization) and by how many other heads change (its collateral, a side effect inside the code rather than in the downstream features that SAE steering studies count [56]; Algorithm 4). The text round trip cannot resolve an intervention, because even an unsteered round trip preserves a head's argmax on only 11–13% of rows (Appendix C.8).

direction	hard code @0.25 / 1.0	softmax @0.25 / 1.0
decode	0.364 / 0.607	**0.824** / 0.569
grad	0.304 / 0.103	0.453 / 0.477
margin	0.158 / 0.275	—
adjoint, oriented	0.001 / 0.001	0.427 / 0.501
adjoint, oriented per sample	—	0.446 / 0.512
random	0.018 / 0.024	0.020 / 0.037
collateral	0.66 / 0.88	0.58 / —

TABLE 2 · Realization rate (target entry newly selected) by direction and strength; 60–64 entries × 32 held-out rows, chance 0.031. Collateral agrees within 0.01 across directions at a given strength, so rows compare at an equal budget. Hard code: the 5×76 stack, whose seed-43 twin agrees within 0.015 in every decode, grad and random condition. The adjoint direction's orientation is in Appendix B.5.

In embedding space steering works. The decode direction realizes the target entry on 82% of rows for the softmax model at strength 0.25. The hard code is the less steerable of the two: it realizes 36% while already disturbing more of the other heads (0.66 against 0.58), because a hard argmax has margins a small step cannot cross. Swept over strength, its decode direction saturates near 0.65 (Figure 10b), so a third of its interventions cannot be realized at any strength, and it passes the soft model only beyond collateral 0.85, where most of the other heads have moved too. Steering by target layer reverses the aggregate ranking of the decode and gradient directions (Appendix E.5). In both stacks the decode step realizes through directions the data barely vary along: at the full step's length, its component in the subspace holding the bottom fifth of the data's variance realizes 95% (hard) and 100% (softmax) of what the whole step does, and its component in the top half 20% and 8% (Appendix E.7).

Against supervised steering vectors. The same measurement scores the two standard supervised steering vectors for each target: the difference of means between rows that select the target and rows that do not (the mean-difference vector of contrastive activation addition [57], which averages over many contrast pairs the single-pair difference of activation addition [58]), and the weight vector of a class-balanced logistic probe, both standard supervised baselines [40], fitted on

8,192 reference rows with the steered rows held out. Unlike contrastive activation addition's pairs, which differ only in the property, these rows are unpaired, so the difference also carries whatever else varies with the target. Their labels are the code's own selections, so they are an oracle baseline: how well a supervised method that already knew the partition would steer into it.

model	own decode	diff. of means	probe	random
softmax stack	**0.75** / 0.57	0.46 / 0.46	0.47 / 0.51	0.02 / 0.56
hard-code stack	0.37 / 0.66	**0.91** / 0.63	**0.94** / 0.64	0.02 / 0.65
hard-code flat	1.00 / 0.33	0.97 / 0.27	1.00 / 0.31	0.01 / 0.37
`g160top1`	0.58 / **0.03**	0.94 / 0.13	0.95 / 0.14	0.04 / 0.23
`g160softmax`	1.00 / 0.30	0.99 / 0.27	1.00 / 0.31	0.01 / 0.34
`TopK` SAE, k_{SAE}=32 (m=11264)	1.00 / 0.21	0.92 / 0.21	0.97 / 0.23	0.00 / 0.25
SAE, bilinear encoder	1.00 / 0.07	0.97 / 0.12	0.99 / 0.11	0.01 / 0.13

TABLE 3 · Each model's own steering direction against supervised vectors fitted to the same targets (realized / collateral), at a step of $0.25\|\mathbf{x}\|$; 64 targets × 32 held-out rows per model, one seed. An SAE's collateral counts its other active latents, so compare within a row.

Only the softmax stack's own direction beats the supervised vectors, reaching its categories on 75% of rows against 46–47%. The hard-code stack's categories are reachable, but its own direction misses them: the supervised vectors reach the same entries on 91–94% of rows, where its decode direction reaches 37%. The single-layer hard variant trades reach for precision (58% realized against 94%, at a fifth of the collateral); every other model's own direction disturbs the other heads as much as a random step does. Knowing the partition buys the supervised vectors little precision (the difference of means sits at most 0.1 below random's collateral), and an SAE's decoder direction is already as good as the oracle. At each method's own step size the difference of means realizes more in most models, partly by taking longer steps (Appendix E.6). This tests reaching the model's own categories; Appendix D.5 steers toward an external concept, language, against supervised vectors fitted to its labels.

A single entry can still steer content: on the reference softmax model, forcing entry 17 of head 8 in layer 1 reliably yields Burmese decodes of otherwise preserved content.

4.5 Description specificity (autointerp)

Individual SAE latents are about twice as describable as individual entries. Under the same plain summarization prompt, 51% of k_{SAE}=32 latents receive a specific description (20% naming a language or script), against 23% of the reference model's entries (8%). Blinded detection over the full campaign agrees:

Describability tracks code density, not architecture, among the learned dictionaries compared here (raw model units score lower at similar sparsity [38]).

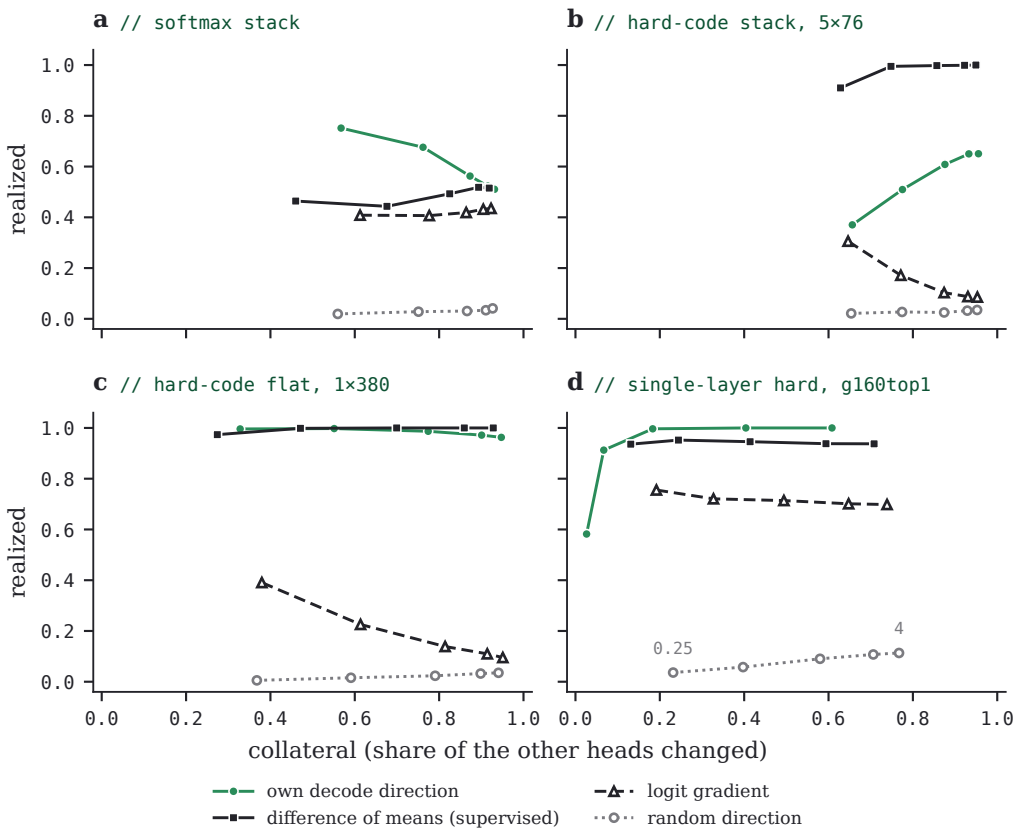


FIGURE 10 · Realization against collateral as the step grows from $0.25\|\mathbf{x}\|$ to $4\|\mathbf{x}\|$, left to right along each curve, on the draw of Table 3. (a) The softmax stack (`sweep_softmax_shm`). (b) The hard-code stack (`ste_h76_init01`). (c) The flat hard code (`ste_l1_h380_i01_hm1e4`). (d) The single-layer hard variant `g160top1`. At each strength the models' own directions disturb as many other heads as the random step (a direction drawn isotropically in the embedding's raw coordinates), to within about 0.05, except `g160top1`'s decode direction, which disturbs far fewer; what a direction buys is its height above the random curve. Against a random step shaped like the data, the softmax stack's own directions disturb 0.14–0.20 more heads at strength 0.25 and the hard codes' at most 0.07 more (Appendix E.7). The logistic probe tracks the difference of means (within 0.03 realized, 0.05 collateral) and is not drawn.

model	density	freq	params	eig	acts	cacts	features
onto (resid_nc)	100%	10.80%	0.039	—	0.473	0.392	225
k32	0.6%	0.61%	0.139	—	0.645	0.603	250
k32_bl	0.6%	2.62%	0.136	0.034	0.578	0.470	75
g160top1	3.1%	5.08%	0.111	—	0.646	0.594	150
g160softmax	100%	0.00%	0.086	—	0.449	0.364	256
k5120 (dense ctrl)	45.5%	17.97%	0.010	—	0.470	0.332	256

TABLE 4 · Mean per-feature detection F1 by description mode (Appendix B.5). The always-match null is about 0.67, and a generic description that matches every item scores the null, so generic descriptions inflate toward it. Density is nominal code density; freq is the firing rate at the $2/k$ threshold. `k32_bl`'s pool is only 75 live features; no null control was recorded for this campaign.

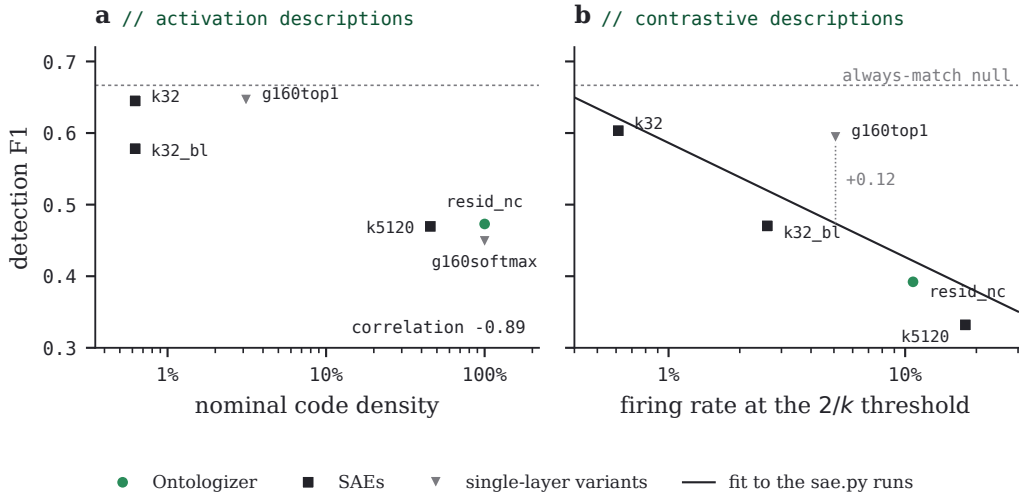


FIGURE 11 · Mean detection F1 per run of Table 4 against code density. (a) Activation descriptions against nominal density, the share of a code's coefficients that are nonzero. (b) Contrastive descriptions against the firing rate at the $2/k$ threshold, with a line fitted in log rate to the four `sae.py` runs that fire; `g160softmax` never crosses the threshold and is not drawn, and the Ontologizer (`resid_nc`) is held out of the fit. Detection sets are half positives, so a description that matches every item scores the always-match null, 2/3 (dotted). One seed per run.

Activation-based descriptions rank the sparse codes (the k_{SAE}=32 SAE and the single-layer hard variant, 0.65) above the bilinear SAE (0.58), the reference model and the dense control (0.47) and the single-layer soft variant (0.45), and density alone carries about 80% of that variance (Figure 11a). Much of it is language. mC4 cycles through 86 languages, and neither the contrastive background nor the detection negatives are matched for language, so a latent whose top rows share one is described by naming it and detected against random rows: 63 of the k_{SAE}=32 SAE's 250 latents and 41 of the hard variant's 150 have their 15 judged rows in one to three languages and are detected at F1 0.90–0.97, while no dense code has such a latent. Across the six runs mean language purity accounts for 93% of the variance in mean F1 (98% for contrastive descriptions), though over six runs it moves with density. Restricted to features whose top rows span 13–15 languages, as random rows do, the gap to the reference Ontologizer in activation descriptions shrinks from 0.17 to 0.10 for the k_{SAE}=32 SAE and to 0.01 for the hard variant. The reference Ontologizer sits on the SAEs' density trend, and the hard variant, 0.12 above it on all features (Figure 11b), is 0.02 above a trend refitted on the multilingual ones (Appendix D.6). Per-latent detection F1 therefore cannot compare architectures at unmatched density or language spread, and every soft Ontologizer is dense by construction. Distribution-based monosemanticity scores show the same confound, rising as an SAE's L0 falls at fixed width [59]. The contrastive descriptions (`cacts`) give the more honest number, though their background is not matched for language either: they score below plain summaries everywhere, least for the sparse codes (by 0.04–0.05) and most for the dense control (by 0.14), so part of every plain-summary score is generic matching.

Parameter-space decodes are not descriptions. Decoding the parameters scores 0.14 at best (the plain and bilinear SAEs) and 0.04 for the reference model, where activation descriptions of the same features score 0.45–0.65. Bilinear eigenfeature decodes score 0.034, but they decode the top-$|\lambda|$ eigenvector, which for most of the bilinear SAE's latents is a direction the latent rejects (Appendix B.2), so that score does not test eigenfeatures. Part of the parameter decodes' failure is how the decode is taken: an atom's cosine to its head's mean atom is 0.849, so a raw decode returns one template sentence with the details shuffled. Subtracting the origin decode makes the decodes legible but not detectable (on one head, 0.079 against a 0.021 null, where contrastive activation descriptions score 0.479 against 0.273). A decoded sentence is an instance of what an entry writes, and detection needs a criterion, so naming requires activation evidence. Detection also asks an input-side question of an output-side description; intervention scoring [38] would be the matched test.

5 A hard-code arm: straight-through selection

The softmax model's discrete reading is not its code (Section 4.1), so its argmax entries are not what interventions act on. Under straight-through selection the forward pass is the argmax, and no such gap can open: at every checkpoint scored, the hard and soft-forward reconstructions agree to four decimals.

5.1 Capacity, and why the head count rises

A hard code transmits exactly $l \cdot h \log_2 k$ index bits. The Gaussian reference (Eq. (25)) gives the bits a code would need to reach a given error if the embeddings were Gaussian; real embeddings may need fewer:

FVU_w target	bits	heads at k=32, l=5
0.226	800	160 (the softmax architecture)
0.05	1,898	380 (this arm, h=76)
0.0023	4,167	833

Heads are the cheap axis: bits and parameters both grow linearly in h, and heads run in parallel where layers run in sequence. The arm has 5×76 heads of 32 entries, 1900 bits and 51.93M parameters. Training a classifier under a hard forward on SONAR takes three settings, all because the argmax is scale-invariant and, in a naive port, nothing pushed the logits up from their small initial scale: logit noise relative to the logit scale, a surrogate temperature matched to that scale, and a rescaled s_{KL_m} (Appendix E.1; every arm's configuration is in Table 10). The GPT-2 arms ran without them, at t=1 with absolute logit noise σ_K=0.02 and s_{KL_m}=10^{-6} (Table 11), the naive port's settings, so any comparison between the substrates also compares two calibrations. Nor did their logits behave alike: in `ste_h76` they grew 452-fold, until the noise changed about 1% of layer-0 assignments, while in the concatenated seed pair they stayed at their initial scale and the noise changed 71–90% of layer-0 assignments at every checkpoint scored (Appendix E.1).

5.2 Initialization: a third of the error

The dictionary is non-negative, so a layer's h rows add coherently and its output grows with h. The per-layer gain compounds that, so at the default initialization the residual grows geometrically with depth: the first arm started at 1.84×10^8 where it should start near 1. Rescaling each dictionary at initialization so that a layer writes about a tenth of its input (`--dict-init-scale 0.1`) divides h out and starts the cascade as a near-identity:

One scalar takes a third off the held-out error and has every head using its entries almost uniformly, the largest single effect in the straight-through work. It also closes the utilization gap that rescaling the entropy pressure and whitening the input each closed in part (Appendix E.1). The hard-code results below use the

arm	shape	s_{KL_m}	realized bits	FVU_w	params
`ste_h76` (shipped init)	5×76	10^{-4}	1714 (90.2%)	0.2293	51.93M
`ste_h76_init01`	5×76	10^{-4}	1895 (99.7%)	**0.1535**	51.93M
`ste_h76_i01_s43` (seed 43)	5×76	10^{-4}	1895 (99.7%)	**0.1535**	51.93M
`ste_k2_h380`	5×380, k=2	5×10^{-4}	1887 (99.3%)	0.1844	17.67M
`ste_l1_h380_i01_hm1e4`	1×380	10^{-4}	1898 (99.9%)	0.2750	51.93M
`ste_l1_h380_i01`	1×380	5×10^{-4}	1898 (99.9%)	0.2870	51.93M
`ste_l1_h380_i01_s43`	1×380	5×10^{-4}	1898 (99.9%)	0.2871	51.93M

TABLE 5 · Straight-through arms at the fixed initialization (except the first row), held-out tail. Realized bits are the summed entropy of each head's argmax usage over the tail, against the nominal $l\,h \log_2 k$; the sum bounds the information the code carries per sample from above, with equality only when heads are independent (Eq. (36)).

fixed initialization, except the steering-by-layer and head-health measurements in Appendix E.5.

A discrete code beats `TopK` SAEs, but not every sparse code. On one frontier of converged SAEs and the single-layer hard variant, all trained for about 144k steps (Figure 8a), the stack's 1900 index bits reach FVU_w 0.154. Its only continuous numbers are the gains of layers 1–4 (layer 0's is identically 1), and with those pinned to their means it is a pure 1900-bit code at 0.158 (Appendix E.4). A code that also sends coefficients reaches that total only at some precision for them, and no coding of its coefficients takes an SAE below the least-squares fit on its own support (Eq. (33)). The `TopK` SAE with 160 active latents names them in 1,206 bits, so its total is 1900 bits at 4.3 bits per coefficient, where its error is at least its fit's, 0.236 (0.258 as trained): at matched total rate the pure code is at least 33% below it, and more precise coefficients only lengthen the SAE's code. The single-layer hard variant, 800 index bits plus 160 coefficients, matches the rate at 6.9 bits per coefficient, where it is at least 0.273, so the pure code is at least 42% below it; it shares the stack's hard per-head selection but has one layer of linear classifiers and keeps a ReLU magnitude on each winner. An SAE whose sparsity is not fixed is another matter. With everything each code sends quantized and entropy-coded (Appendix E.3), an L1 SAE with a weak penalty (s_{L1}=3×10^{-6}), quantized coarsely, reaches 0.128 at 1900 bits, 18% below the stack: its latents fire unevenly, so naming the active ones costs a fifth less than a fixed-length code would, while the stack uses its entries almost uniformly and has nothing to save. With every support charged its fixed-length cost the stack leads every SAE (0.158 against 0.179). For a hard model the capacity axis is $l \cdot h \log_2 k$ itself: truncating toward the origin only adds mass the model was never trained to decode (0.259 at v=1). The stack is less efficient per bit than a single layer, 3.08× the Gaussian reference at its rate against 1.28× for one layer's 380-bit prefix, and bits are not the whole capacity: 380 binary heads, at the same bits, reach 0.1844 (Appendix E.2).

5.3 Depth at fixed capacity

Five layers of 76 heads and one layer of 380 have the same nominal capacity (1900 bits) and exactly the same parameter count (51.93M; the dictionary hke and the bilinear classifier $2hkd$ depend on l and h only through $l \cdot h$). At the fixed initialization both use their entries almost uniformly (summed head entropy 99.7% and 99.9% of nominal), and depth is worth 1.79× the reconstruction error (0.1535 against 0.2750). Seed does not explain the gap: replicas agree to four digits. Summed head entropy bounds what each code carries from above, and whether the two codes carry different amounts jointly is not measured. Neither arm has converged, and the ratio is narrowing slowly, but the stack leads at every checkpoint (Appendix E.4).

What depth buys is not usage, input conditioning or the per-layer gain. A flat arm with a whitened input uses every entry uniformly and still reconstructs 1.7× worse, and pinning the gains costs the stack only 3.0% (Appendix E.4). Blending each layer's subtracted reconstruction toward another row's, without retraining, shows instead that what matters is pairing each residual with its own sample: the error rises well before the pairing is fully severed. A flat layer decodes like an additive quantizer [3] but does not encode like one: additive quantization's beam search codes each codebook against what the codewords already chosen leave, while the layer's heads choose in parallel, so no head sees another's error — harmless in a product quantizer, whose heads own disjoint coordinates, but not when atoms overlap. The stack is a residual quantizer, each stage coding what earlier stages missed [6]. Every stack also trains each of its prefixes to reconstruct (deep supervision), which a single layer cannot, so whether the gain needs that supervision or only the residual forward is not yet separated.

5.4 Entanglement: the cost of the cascade

A single-head intervention in embedding space realizes its target on almost every row of the flat model (99.6%) while disturbing a third of the other heads; in the stack it realizes 37% while disturbing 66%. Restricting the stack to its layer 0, so that the target is as close to the input as in the flat model, widens the gap (realized / collateral at strength 0.25, and at each model's native step for the last row; 12 layer-0 targets in the stack, 64 in the flat arm):

direction	stack, layer 0	flat
decode	0.049 / 0.680	**0.996** / 0.328
grad	0.281 / 0.656	0.391 / 0.380
margin	0.034 / 0.661	0.076 / 0.379
random	0.008 / 0.655	0.005 / 0.368
native decode	0.044 / 0.678	0.169 / 0.043

Collateral is not a property of the intervention. In both models every direction

disturbs as many other heads as a random step of the same length, to within 0.05 (Figure 10b, c), and as many as a random step shaped like the data to within 0.025 in the stack and 0.07 in the flat arm (Appendix E.7), because the heads sit almost on their decision boundaries: the median margin between winner and runner-up logits is 1.5×10^{-4}, at layer 0 and in the flat arm alike, so a step of a quarter of the input's norm crosses many of them. Linearizing each head's logits with the upstream codes held fixed separates two sources. The stack's layer-0 heads flip directly about as often as the flat arm's (0.22 against 0.24), so what the stack adds is downstream layers reclassifying a changed residual: the cascade roughly doubles collateral but is not its only source. The same reclassification answers a removed head by describing the input anew rather than restoring the old description (Appendix E.8). At the model's own step size (the forced entry's output change, unscaled) the flat arm's step is small (a median 0.028 of $\|\mathbf{x}\|$) and moves 4% of the other heads, while the stack's (0.13) realizes 8% of targets across layers and moves 54%, as a random step of that length does. This is the steering counterpart of the composition result, in which non-additivity decays with the number of layers below an intervention (Section 8). Steering by target layer, the head-health levers tried against entanglement (only a row-collinearity setpoint bought steerability), and a head-independence penalty that turned out to measure estimator bias are in Appendices E.5 and E.9.

5.5 Seed reproducibility: the error is determined, the heads are not

Two runs differing only in seed, both at the fixed initialization, reach identical held-out error: 0.1535 and 0.1535 for the stack, 0.2870 and 0.2871 for the flat arm (Table 5). Nothing that identifies an individual head agrees:

what is compared	stack	flat	null
head partitions (mean matched NMI)	0.0121	0.0177	0.0048
head contributions (centered cosine)	0.0074	—	0.0015
per-layer subspace, rank 8	= data's	—	see text
atom Grams, entries matched	1.05–1.19× chance	—	within-layer pairing
reconstruction (residual correlation)	0.30	0.34	0.53 (same run)

Partitions agree barely above the null, where a run scores 0.38 against its own earlier checkpoint. Contributions agree no better. Per-layer subspaces agree exactly as much as each model agrees with the data's own principal directions, and per-head Grams match at chance across seeds even with entries matched exactly, while within one run each head's Gram matches its own earlier checkpoint at 8.5–10.7× chance (Appendix E.10): heads persist within a run and do not replicate across seeds. Nor is the reconstruction one function. The two seeds' whitened residuals correlate at 0.30 (the flat arm's at 0.34), against 0.53 for the stack against its own checkpoint 19,500 steps earlier and 0.83 for the k_{SAE}=32 SAE's seed pair,

whose two seeds compute nearly the same reconstruction (`reconseeds.py`). What reproduces is aggregate (the error) or the data's own (its principal directions), and what identifies an individual head, with one exception, does not. That is the pattern of weak rather than strong universality, in which networks trained on the same task share an algorithm while differing, even between seeds, in the features that implement it [60], though here equal error comes from different reconstructions, in the flat arm as in the stack, so the cascade does not cause the disagreement. Claims about individual heads, including the Burmese entry and the per-head steering tables, therefore describe one seed. Non-negativity and the layer-dependent gain put structure into the floor of every such measure, and several of them, among them uncentered contributions and subspace overlap, read as agreement until compared against a null of the same shape.

Layer 0 is the exception in every head-level measure, as the only layer whose input does not depend on upstream choices, and the one head that reproduces is there: the topic head of Section 5.6, the strongest layer-0 partition in both seeds at different indices, matched across them at NMI 0.41 against a 0.01 median for other heads. The softmax Ontologizer fares no better: a seed replica at matched training matches at most 0.1% of entries, where the k_{SAE}=32 SAE pair matches 3–31% (Figure 25), on a thresholded activation measure stricter than the decoder-cosine [41] and activation-correlation [42] measures on which language-model SAEs agree across seeds at about 0.8.

Replication on GPT-2. A GPT-2 pair differing only in seed (5×20 heads, k=256, concatenated heads at d_{head}=128, 371,900 steps) gives the same picture. Each measure is run across seeds and, as a positive control, against the first run's own checkpoint at step 350,000 (Figure 12).

Only layer 0 reproduces, at under a fifth of the control, and each of its heads matches into the other seed's layer 0 on both measures. Layer 1 keeps its identity (19 and 20 of its 20 heads land in the other seed's layer 1) while agreeing on almost nothing it separates or writes; layers 2–4 lose even that. The partition null is high, because plug-in NMI between two 256-way partitions of 32,768 rows is dominated by bias, so it cannot resolve the deep layers; against the tight contribution null they sit two to four times above it, at about 1% of the control, which is real and negligible. The excess over the null is larger than on SONAR (24× against 5–11×), almost all of it in layer 0, though the GPT-2 arms also ran without the SONAR arm's calibration, and in this pair the training noise, not the classifier, chose most layer-0 assignments throughout training (Table 11; Appendix E.1).

What layer 0's lead is made of. Three findings, detailed in Appendices E.11 and E.12. Layer 0's larger contributions are all gain: it receives the raw input, while its atoms are no larger than layer 1's (Table 19), and at equal atom spread it still reproduces 14× better on GPT-2 and 23× on SONAR. The stack costs reproducibility at every

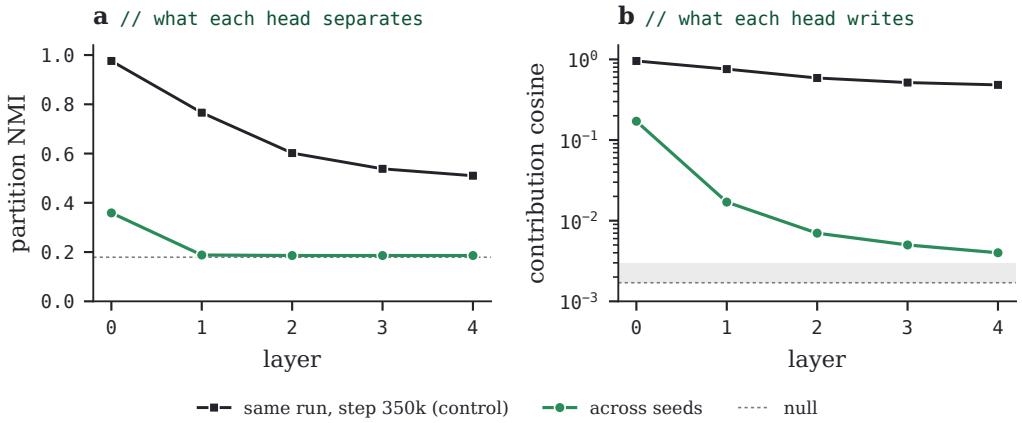


FIGURE 12 · The GPT-2 concatenated-head seed pair by layer: what each head separates (a, partition NMI on 32,768 rows) and what it writes (b, centered contribution cosine on 8,192 rows, log scale), across the two seeds and against the first run's own checkpoint at step 350,000. Dotted lines are the nulls; the band in (b) runs from the null's mean to its maximum. One seed pair.

layer: against flat-arm heads of the same size, which see the same input with nothing downstream, the stack's layer 0 reproduces about 8× worse and its deeper layers 19–49× worse, the reproducibility side of entanglement. Within layer 0 the atoms decide: layer-0 entries are unequal by atom norm rather than usage (Table 20). On SONAR one head (54) holds 8 of the layer's 11 atoms above three times the median, reproduces at 0.844 against 0.249 for the next, and drives the within-layer correlation between agreement and largest atom (Table 21); without it the layer-0 mean falls only from 0.080 to 0.070. GPT-2's layer 0 has no such head: every head carries a graded tail of large atoms, and agreement is narrowly spread.

5.6 Heads against a known factor: language, and what they encode instead

Language is the corpus's one factor with free labels, which is why it is measured, not because a head was expected to encode it: SONAR maps translations to nearby points [36], so language is a weak, residual factor of the embedding. No head encodes it outright. The best reaches a partition NMI of 0.18 against an attainable 0.875 (0.165 on the second seed), and no head-health lever changes that (Table 22). Partitions do carry the factor well above chance, and the hard code carries more of it: 3.5–3.9 times the softmax reference on null-corrected best-head NMI, and 0.42 against 0.27 in a 32-head probe. Single cells and conjunctions of cells detect language neighborhoods rather than identities (Finnish with Estonian, Japanese with Korean), which matches an embedding in which languages are weakly separated regions rather than directions (Appendix E.13).

What the strongest head holds: topic and genre. A semantic, language-agnostic embedding should be organized by meaning, and its strongest head is. Ranking layer-0 heads by η^2 and matching partitions across runs (65,536 held-out rows):

run	top head	η^2	next head	match across runs
`ste_h76_init01` (seed 42)	h54	0.117	0.062	—
`ste_h76_i01_s43` (seed 43)	h14	0.122	0.068	NMI 0.413 (others 0.010)
`ste_h76` (unfixed init, seed 42)	h54	0.111	0.080	NMI 0.355 (others 0.024)

The topic head is the one head found in every run, the strongest partition in each by nearly a factor of two. A different seed grows it at a different index, and the same seed keeps its index across the initialization fix. It is also the head that reproduces on contributions (Appendix E.12), the exception to Section 5.5's finding that heads do not reproduce, though at NMI 0.41 against a 0.87 same-run control it is the same variable carved differently, not the same partition. It also carries more language than almost any other head: in seed 43 it is the most language-informative of all 380 heads (NMI 0.165, Table 22) and in seed 42 the third (0.161), though against an attainable 0.875 each of its entries still mixes many languages. Decoded as deviations from the head's mean atom, its entries read as topic and genre: crime news, geopolitics and parliament, encyclopedic place description, manufacturing, marketing, legislative text and entertainment. Contrastive autointerp agrees, with every category of description above its null (Appendix E.13). Testing whether heads are natural variables therefore needs topic or genre labels, which this corpus does not have.

6 Hard codes on GPT-2 small: atoms in activation space

A second line of straight-through runs, from the `headline` branch, takes the hard code to GPT-2 small and adds what an embedding substrate lacks: the fraction of GPT-2's loss recovered when the reconstruction is spliced back into the model, $(\mathrm{CE}_{\mathrm{ablate}} - \mathrm{CE}_{\mathrm{recon}})/(\mathrm{CE}_{\mathrm{ablate}} - \mathrm{CE}_{\mathrm{clean}})$, with the residual mean-ablated as the floor — a lenient scale against an ablation floor [22], so Table 6's caption also gives the cross-entropies themselves. These runs model the residual stream entering block 8, one block earlier than the GPT-2 runs of Section 5.5, centered and scaled to unit mean square, on FineWeb, and their FVU_w is unwhitened; they compare only with each other and with SAEs trained on the same cache. Held-out evaluation uses documents disjoint from training (520k tokens for the 40M-token runs, 65k for one-epoch runs).

The design. Relative to Section 5, three things change. Atoms are direct: signed vectors in activation space, with no latent dictionary space and no decoder, so every atom is literally a steering direction. Every layer, layer 0 included, classifies the unit

direction of what it corrects, and each layer's contribution is scaled by the observed residual norm, clipped at $\|\mathbf{x}\|$, which makes the model positively homogeneous ($f(\xi\mathbf{x}) = \xi f(\mathbf{x})$ for $\xi > 0$). The straight-through temperature is annealed early in training (Table 12). With 5×32 heads of 32 entries the code is 800 bits per token, and its only continuous numbers are the five layers' gains.

model	atoms	code per token	FVU_w	loss recovered
direct + gain + STE	signed, in $\mathbb{R}^{768}$	800 bits + 5 gains	0.195	**97.6%**
private heads + gain + STE	48-dimensional blocks, shared decoder	800 bits + 5 gains	0.186	97.5%
private heads, k=64	48-dimensional blocks, shared decoder	960 bits + 5 gains	**0.164**	**97.9%**
`TopK` SAE, k_{SAE}=32, m=5120	—	32 coeffs + ~276 bits	0.189	96.2%
`TopK` SAE, k_{SAE}=128, m=5120	—	128 coeffs + ~859 bits	0.116	98.6%
soft Ontologizer (1 epoch)	—	4960 mixture weights	0.016	99.85%

TABLE 6 · GPT-2 small, residual entering block 8. Ontologizers: one pass over 40M tokens, one seed. SAEs: 10 epochs over 10M tokens with `sae.py`'s unwhitened objective. Clean cross-entropy 3.806, mean-ablated 8.283; the direct code reaches 3.914.

A discrete 800-bit code recovers more of GPT-2's loss than a k_{SAE}=32 SAE at nearly the same FVU_w: 97.6% against 96.2%, at 0.195 against 0.189. Across architectures, then, FVU_w does not order downstream loss, while within the Ontologizer family on SONAR, text fidelity was monotone in FVU_w (Appendix C.5). Part of such a gap can be mismatch rather than lost information: adapting the downstream model to a frozen SAE closes 30–55% of its cross-entropy gap [61]. With 64 entries per head (960 bits) the code passes the SAE on both measures. Every layer contributes, entry usage stays near uniform, and the observed gain is close to the best per-sample scale for each layer; one-epoch ablations show that the gain-shape split and direct atoms each help, direct atoms most, and that annealing matters (Appendix F.1).

Fibers. A rank-r local basis per entry, with coordinates read linearly from the unit residual, turns the code into a discrete base point plus r numbers per head. "Base only" is the same model with the fibers removed, which measures whether the discrete code still means what it did:

variant (1 epoch, direct + gain + STE)	extra numbers	FVU_w (base only)	loss recovered
base	0	0.217	96.95%
rank-4 fiber, fiber dropout 0.5	640	0.059 (0.274)	99.4%
rank-8 fiber, fiber dropout 0.5	1280	0.014 (0.304)	99.9%
rank-4 fiber, joint norm cap 0.2	640	0.180 (0.227)	97.6%
rank-1 radial fiber (gain per entry)	160	0.148 (0.669)	98.5%
frozen base, rank-4 fibers trained after	640	**0.016 (0.215)**	99.9%
unconstrained fibers	—	— (~10)	—

Fibers recover far more than the SAE's advantage in FVU_w, but they carry hundreds of continuous numbers per token, more than the SAE's 32 coefficients, so this is not a matched-capacity comparison. The question they raise is what keeps the discrete base honest. Unconstrained, the fibers take over (base-only FVU_w about 10); fiber dropout holds the base at 0.23–0.30 whatever the rank; and a joint norm cap guarantees it at the cost of most of the fibers' value. Training the fibers after freezing the base is best on both counts: the base is untouched by construction (0.215), and the fibers reach the reconstruction of unconstrained joint fibers. The mechanisms used in joint training suppress the fibers rather than protect the base.

Seed stability, replicated. Matching heads across two seeds by partition NMI, the partitions agree only in layer 0 (NMI 0.09 against 0.05 for unrelated heads within one run) and not at all in layers 1–4. With direct signed atoms, the gain-shape split from layer 0, a different site and a different objective, this repeats Section 5.5: which entries share a head depends on the seed past the first layer.

7 Dictionary geometry and the non-negativity constraint

Two geometric facts frame this section (Appendix G). Row collinearity measured in the dictionary space is mostly gauge: 74–78% of each atom's energy lies in the decoder's null space, so dictionary statistics should be computed after decoding (Appendix G.1). And the selection rule shapes the dictionary less than training duration does, while the last layer becomes inert once the error before it is small (Appendix G.4).

7.1 The orthant forces a collinear dictionary

Non-negativity binds through the Gram, not the null space. Two non-negative vectors have a non-negative inner product exactly, so for independent $|\mathcal{N}(0,1)|$ coordinates the chance pairwise cosine is $2/\pi = 0.637$ (minimum +0.58 over 4000 draws), where signed draws give 0.000. Dropping the absolute value (`--signed`) on GPT-2 layer 8, with the straight-through arm's shape, crossed with dictionary width and trained to convergence (371,900 steps):

arm	$\bar{\kappa}$ (row cosine)	eff. rank	FVU_w	at step 90k
signed, e=1536	+0.0009	27.37	**0.1540**	0.2910
signed, e=768	+0.0019	26.95	0.1543	0.2942
abs, e=1536	+0.3038	9.15	0.1934	0.3962
abs, e=768	+0.4543	5.41	0.2927	0.4631
concat, d_{head}=32	+0.5353	3.30	0.4155	0.8055

TABLE 7 · Non-negativity × width on GPT-2 layer-8 residuals, held-out tail: geometry from the weights, FVU_w scored from each checkpoint. No collinearity penalty is active in any arm.

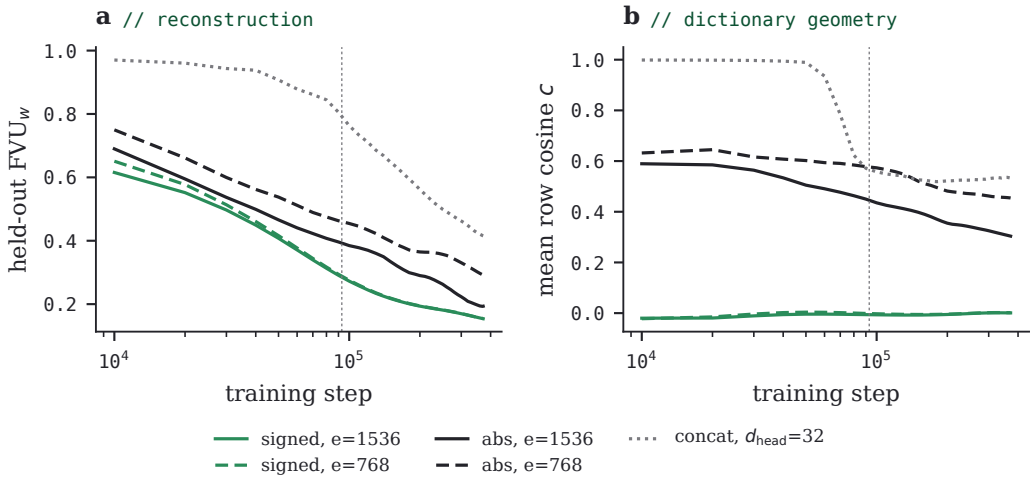


FIGURE 13 · The orthant arms over training: held-out FVU_w scored from each checkpoint (a) and mean within-head row cosine (b). The dotted line marks step 90k. The signed arms coincide at both widths; `abs` is behind at both, and further behind at the square decoder.

Forced collinearity costs reconstruction. At e=1536 the non-negative dictionary runs at a third of the signed effective rank and 26% more held-out error; at e=768, 90% more. Across the five arms $\bar{\kappa}$ orders the error exactly at convergence (Spearman +1.00; +0.90 at step 90k, where concat is the single inversion). The cost is an interaction: narrowing e costs 0.2% with signed atoms and 51% with non-negative ones, so the wide dictionary buys room for the constraint rather than capacity (Appendix G.1). That undercuts Section 3.1.4's claim that head-disjointness is what $e > d$ buys, and no model attains head-disjointness in any case (Appendix G.2). The non-negative arm's gap at e=1536 grew from 8% at step 20k to 50% at 200k and has since narrowed to 26% (Figure 13), so it may be partly slower, but it has not caught up by the end of a full run.

Removing the absolute value buys a decorrelated, near-full-rank dictionary and lower error, roughly what the collinearity setpoint exists to buy. Signed atoms give up two things: a subtractive feature is harder to read as "this concept is

present", and the $\|F\|_1 \geq h$ floor under `norm_rows`, which rests on non-cancellation, disappears. Concatenated heads, which face a collapsed tradeoff rather than a raised floor, are the worst arm (Appendix G.3).

8 Formal reading: heads as causal variables

The steering frame has a natural formalization in the causal-abstraction vocabulary [62, 63, 64, 65]. An abstraction relates two causal models through a map between their states and a map between their interventions, so stating the frame means naming both models, and it shows what each instrument in the battery does and does not estimate.

Definitions. The high-level model is the Ontologizer's forward pass, read as a causal model over the input and the heads: each head (ℓ, i) is a candidate *variable* with value space $\{1, \ldots, k\}$, $\mathbf{x}$ is a parent of every head, and with residual forwarding every earlier head is a parent of every later one, since layer $\ell{+}1$'s classifier reads the residual written by layers $\leq \ell$. These edges belong to the code: unlike those of transcoder circuits [66], which pass through a model's own sublayers, they describe how the code refines one representation, not how the encoder computes it. `withArgs` set-interventions are this model's hard interventions: $\mathrm{do}[(\ell, i){=}j']$ forces the head's assignment while leaving all non-descendants untouched by construction, and marginalizing the continuous intermediates out of the forward pass is an exact transformation by construction [62, Thm. 10]. A soft head's state is a point of its simplex, so reading it as one of k values is a lossy value merge [65], and a set on a soft head is a steering intervention rather than an interchange. The low-level model is the encoder whose embeddings the code reads, and the map between their states is assign, which sends an embedding $\mathbf{x}$ to its per-head assignments. Whether the head model abstracts the encoder under assign is an approximate-abstraction question [65, Def. 41], and the abstraction cannot be a constructive one [63], since every head reads the same embedding.

What the instruments estimate. Input-space steps are not interventions such an abstraction maps; they score control of the head model, not whether the encoder computes with the heads [65, Sec. 3.7]. `steerembed.py`'s realization rate checks one coordinate of the condition that a stepped input's code equal the code under $\mathrm{do}[(\ell, i){=}j']$, so a rate below 1 already rules out an exact transformation that maps the step to that do. Its collateral counts the other heads that change against the *unsteered* code, which in a stack includes the downstream reclassification the do itself would cause. `steerfid.py`'s cycle-consistency gain asks the same of a text-level implementation, through the closed loop assign $\circ$ encode $\circ$ decode: it is a normalized activation gain net of a random control, not an interchange-intervention accuracy [67], since it patches nothing in from a source input, and the loop's ~13%

unsteered round-trip survival (Section 4.4) limits how often an exact intervention could be observed through it. Seed-to-seed matching asks whether two runs' assign maps agree up to relabeling. What the softmax leaves unidentified in principle is the per-head entry order (permutation) and any logit gauge the temperature absorbs; what must be reproducible for the heads to be properties of the data rather than of one run is therefore the partition itself and the decode directions up to relabeling. Section 5.5 measured exactly that on the hard code, on SONAR and GPT-2. Past layer 0 no head reproduces, and nothing measured there shows that the layer does either: per-layer subspaces agree only as much as each model agrees with the data's principal directions. Whether a layer reproduces as a variable merge of its heads [65, Def. 37], of which the per-head variables are a seed-dependent factorization, needs a whole-layer comparison that translates one seed's layer code into the other's. Layer 0 partly reproduces, on SONAR almost entirely through a single large-atom head.

Why the cascade costs reproducibility. The stack's reproducibility deficit at matched head size (Section E.11) and its intervention collateral (Section 5.4) are one fact under this reading. A head's value is a function of its own input, and every head past layer 0 takes as input the residual that upstream heads leave; a different upstream solution changes what any downstream variable is *about*. Even layer 0, whose input is the data, is pinned only partly: its heads are trained jointly with everything downstream of them, so which partition of the raw input is best depends on what the later layers will do with the error. A flat layer has neither dependency: its heads are more reproducible at equal size and realize interventions far more reliably, with half the collateral. They are still not independently addressable — the hard code's heads sit almost on their decision boundaries in either shape — so the cascade compounds an entanglement that the code already has.

Metric hygiene. Claims about decode-direction geometry should not depend on an arbitrary choice of inner product, a point Park et al. make for language-model representations, which they read under an estimated causal inner product [46]. The contrast-set result does not depend on it: under the training objective's inverse-variance metric (a whitening of the data, not their causal inner product) the Ontologizer's within-head repulsion strengthens (−24.4 vs. −17.6 raw) and both rivals' signatures are unchanged. Two further rules emerged. Measure in output space, not in the dictionary space, whose null-space component is unconstrained (Section G.1) and is exactly the kind of causally disconnected direction that makes subspace patching misleading [68]. And separate a layer's gain from a head's atoms before attributing anything to head size: across layers, size is the gain, and the gain is the layer (Section E.11).

Composition. A variable reading earns little if interventions on two heads interact uncontrollably. Measured additivity of decode deltas on `resid_nc` (`compose.py`, whitened metric, medians over pairs):

stratum	pairs	cos	rel. err	later layers
same-layer L0	28	0.959	0.350	4
same-layer L1	28	0.947	0.354	3
same-layer L2	28	0.967	0.259	2
same-layer L3	28	0.996	0.092	1
same-layer L4	28	1.000	0.019	0 (numerical floor)
cross-layer (earlier = L0/L1)	42	0.97–0.98	0.20–0.23	
cross-layer (earlier = L2/L3)	18	1.00	0.02–0.04	

Three observations. (i) Joint interventions point where the summed singles point everywhere (median cosine ≥ 0.95 in every stratum): `withArgs` interventions compose approximately additively. (ii) The non-additive component is entirely attributable to downstream reclassification: it decays monotonically with the number of layers below the intervention and hits the numerical floor when there are none (the measured 0.019 is float32 rounding amplified by the decoder cancellation, $\sim 3\times10^{-6}$ in float64). (iii) Cross-layer pairs compose *better* than same-layer pairs at matched depth: a set-intervention is immune to upstream perturbation, so cross-layer interaction is one-way. Note the contrast with Section 5.4: `withArgs` sets a head directly and composes well, while an *input-space* push reaches every head and disturbs most of them. The downstream reclassification that is a 5–35% correction to a set-intervention is the dominant effect of an input-space one.

9 Preliminary conclusions

1. **Contrast-set structure is real, and it is what the architecture buys.** Trained softmax heads' decode directions repel (96% of heads below null), and hard-code heads' decoded atoms are orthogonal to slightly repelling: each head holds well-separated alternatives. This is invisible to reconstruction metrics and to per-latent probes.

2. **The other designed orthogonality did not emerge: heads share their coordinates.** No trained model partitions its dictionary space between heads; the support overlap sits at or above a coordinate-shuffle null in every model, each head's usage covers 52–98% of the coordinates, and this holds for the part of each atom the decoder sees as well as for its null-space part. Only very long training (8.27M steps) moved it, by sparsifying each head's profile rather than separating heads. A static reading of the dictionary cannot assume disjoint heads; concatenated heads are the only route to them, by construction.

3. **A discrete ontology is possible, but not with the softmax architecture's readout.** The softmax model's argmax is not its code (at least 77× from the best 800-bit code for `resid_nc`, nine orders of magnitude on `softmax_shm`). A straight-through arm with 2.4× the heads makes the discrete code the model: with its four per-layer gains pinned it is a pure 1900-bit code at 0.158. At the same total rate the converged `TopK` SAE and the single-layer hard variant cannot get below 0.236–0.273 however their coefficients are coded, but a coarsely quantized L1 SAE, entropy-coded, reaches 0.128; the hard code leads every SAE only when each support is charged its fixed-length cost (0.158 against 0.179), because its own entries are used almost uniformly and an L1 SAE's latents are not. It sits 3.1× above the Gaussian reference for its rate, so at least that far from the data's distortion–rate function, a limit over long blocks of samples that a code describing each sample alone need not be able to reach [54]. It needed a temperature matched to the logit scale, relative logit noise, a rescaled entropy bonus, and an initialization fix worth a third of its error.

4. **The discrete code holds up downstream.** On GPT-2 small, an 800-bit hard code with signed atoms in activation space and gain-shape on every layer recovers 97.6% of GPT-2's loss, more than a k_{SAE}=32 SAE at nearly the same FVU_w (96.2%), and 64 entries per head pass it on both measures. Across architectures FVU_w does not order downstream loss. Fibers on top of a frozen discrete base recover almost all the rest (99.9%) while leaving the base exactly as it was, so a discrete ontology and a near-lossless reconstruction need not trade against each other — at the price of hundreds of continuous numbers per token.

5. **Reconstruction is the price sheet, not the verdict, for soft codes.** Unstructured codes reconstruct ~50× better at matched coefficient budgets, and the per-head simplex constraint alone costs ~30×; text fidelity is a monotone function of FVU_w and adds no independent axis.

6. **The residual cascade is measured on both sides of the ledger.** For: 1.79× lower error than a flat layer at equal bits, parameters and near-uniform head usage on both sides, sourced in the pairing of each stage's input with the sample's own error (not utilization, conditioning, or the gain side-channel; deep supervision, which only the stack has, is not yet ruled out). Against: an input-space intervention disturbs 66% of the stack's other heads against 33% of a flat layer's, and realizes its target on 37% of rows against 99.6% — in both, collateral equals a random step's, because the hard code's heads sit almost on their decision boundaries; and at equal head size the stack's heads reproduce across seeds several times worse than flat heads at every layer, layer 0 included. The single-layer softmax variant `g160softmax` matches `resid_nc` on reconstruction, but a config-matched softmax Ontologizer reaches 6×

lower error, on rows no model trained on as on the tail, so the reconstruction verdict against the stack was checkpoint-specific.

7. **Steering works in embedding space and the text cycle hides it.** The decode direction realizes a target entry on 82% of rows for the softmax model at strength 0.25, against a 3% chance rate, while the text round trip can observe at most ~13% of perfect interventions. The soft code is the more steerable at low collateral; steerability does not track reconstruction, and the classifier's own gradient is the right direction only where the input reaches the classifier directly.

8. **Individual heads past layer 0 are not reproducible, and layer 0 only partly.** Seeds agree on the error to four digits but not on the reconstruction: their residuals correlate at 0.30, where an SAE's seeds reach 0.83. Past layer 0 they agree on nothing that identifies a head — partitions, contributions, subspaces beyond the data's, or Grams — each tested against a same-shape null, on SONAR and on GPT-2, and again under the direct-atom design of Section 6, though within one run every head persists. Layer 0 reproduces at under a fifth of a same-run control. Its lead over the deeper layers is not atom magnitude: its atoms are no larger, and its larger contributions are its gain, which is a property of the layer. Within layer 0, heads with outsized atoms are the reproducible ones — on SONAR almost entirely one head. This remains the most consequential result for the steering frame: a head that is not the same head on the next run cannot be named, and every per-head claim here inherits that.

9. **Entries contribute near-equally except where the input is raw.** In layers 1–3 every entry carries close to its share (Gini 0.06–0.15, no dead entries); layer 0's entries are unequal (Gini 0.33–0.59) because its atom norms vary widely inside heads, while usage stays near-uniform. The flat arm shows the same pattern, so it belongs to classifying the raw input.

10. **Heads follow what the embedding encodes: meaning before language.** The strongest head encodes topic and genre, as a language-agnostic semantic embedding should be organized, and it is the one head that recurs across seeds. Language, the one factor with free labels, was a probe rather than an expected variable: it is carried as neighborhoods across heads, never by one head, consistent with languages being weakly separated regions of SONAR space. On that probe the hard code carries 3.5–3.9 times what the softmax reference does, so partitions do align with known factors; a test of whether heads are natural variables needs topic or genre labels.

11. **Per-latent lenses mis-measure classifier codes.** Sparse probing ranks the Ontologizer worst while partition-level analysis finds verified causal language

heads; detection F1 is ~80% explained by code density and more by how many languages a latent's top rows span, which moves with it, and on features spanning as many languages as random rows do, the sparse codes' lead over the Ontologizer falls to 0.10 or less. Parameter-space decodes are instances, not descriptions, and must be taken as deviations from the head mean to be legible at all, as the geometry of categorical concepts predicts: a category's value vectors share their parent's component and separate only as differences from it [46].

12. **Non-negativity is expensive in geometry and in error.** It forces a collinear, low-effective-rank dictionary (a third of the signed rank) and costs 26% of held-out FVU_w at the wide dictionary, 90% at a square decoder; row collinearity orders all five GPT-2 arms by error at convergence. Signed atoms decorrelate the dictionary and reconstruct better at either width. Dictionary statistics computed in the dictionary space rather than after decoding are mostly gauge.

13. **Every measure in this architecture needs a null of the same shape.** Non-negativity and the layer-dependent gain put structure into every floor. Three of four seed-reproducibility measures, the HSIC head dependence, the null-space readings, and the apparent size effect on reproducibility each reversed or dissolved under the right control.

10 Follow-up investigations

Ordered by relevance to the steering goal.

- **Head identifiability first.** Before naming heads, make them reproducible or establish which are. The softmax Ontologizer's partitions are now compared across seeds too, and at 370k steps match essentially nothing (Appendix D.4). A k=128 pair sharpens the problem. Two architectures trained from the same seed, one decoding its atoms and one with direct atoms, share their partitions head for head (matched NMI 0.18, every head matched to the head at its own index), while each architecture across seeds reaches 0.064 against a 0.059 null. Seed replicas see the same data in the same order, so the seed acts only through the initialization and the training noise; it, rather than the data or anything after the classifier, largely fixes what a head carves, as in toy networks, where seeds trained on identical data implement one algorithm through different features [60]. Candidates: replicas of `resid_nc_hm` to its full 4.4M steps and of `g160softmax`; whether large-atom heads' reproducibility can be extended by training pressure; whether the k=2 arm's one-bit heads reproduce better than 32-way ones; what lets the topic head recur when no other head does (at k=128 a dominant layer-0 head recurs with a third to a

half of its agreement); whether a per-head router's gain is what reproduces, since on GPT-2 the router doubles contribution agreement while switching off the last two layers; and a GPT-2 flat arm, so the matched-size comparison of Section E.11 has a second dataset.

- **Bring the rest of the GPT-2 hard code onto this branch.** Direct atoms are ported (Appendix F.1), but `resid_first` and `gain_clip` are not, so Section 6's runs still reproduce only on the `headline` branch. On SONAR, 5×54 heads at k=128 (1,890 bits, matching the k=32 stack's 1,900) converge to twice its error (FVU_w 0.31–0.32 against 0.1535, two seeds each), with direct atoms 0.009 ahead of decoded ones. Both codes are full, so the gap is not utilization; but the temperature, learning rate and mean-entropy weight were tuned at k=32 and carried over, and need retuning before the gap reads as a property of k. Direct atoms also make steering literal, a set-intervention adding $g_\ell(\mathbf{a}_{ij'} - \mathbf{a}_{ij})$ at one layer, and that has not been measured.

- **Fibers on a frozen base.** The two-stage recipe kept the base exact and matched unconstrained fibers' reconstruction on GPT-2. Fibers are ported, but no SONAR run has used them; the recipe should be the default there, scored at matched continuous capacity against SAEs (rank-1 radial fibers already sit at 160 numbers per token).

- **Remeasure on the fixed initialization.** Entanglement and the gain and blend ablations ran on the shipped initialization, which was 1.84×10^8 off at step 0 and left 10% of the code unused.

- **The steering overlay at scale.** Supervised vectors are now scored on the model's own categories (Section 4.4) and, for `resid_nc_hm`, on an external concept, language (Appendix D.5). Still open: the language overlay on the hard code and the flat arm; bilinear eigenfeatures; per-entry effect consistency across contexts, which for SAE features is partly predictable before intervening [56]; and the flat arm's near-perfect realization, halved collateral and better reproducibility weighed against its 1.79× reconstruction cost. Collateral needs a lever more than a better direction. It is direction-blind even with the cascade excluded: every Ontologizer's decode direction, linearized within its own layer, flips other heads at a random step's rate. In decoded text too, removing what a head wrote costs no more than a random step of the same length. The hard code's heads sit close to their decision boundaries: linearized within its layer, a random step of $0.25\|\mathbf{x}\|$ flips 30% of the other heads in that layer, and distance from the boundary looks like the lever. `ste_k2_h380`, five layers of 380 binary heads, is the hardest Ontologizer to disturb. The same step flips 13% of its other heads to first order, and measured through the stack it moves 17–19% of them in every direction, random included, against 66% for the 32-way stack. Its logit margins are

34 times the 32-way stack's, but that ratio is not the evidence: they doubled between steps 370k and 739k while the first-order share held, so the logits' scale grew with them. The step count explains neither gap: at step 370k, the 32-way stack's whole run, the first-order share and the measured collateral were both the same. Nor does the mean-entropy weight, which matches the 32-way stack's per head (KL_m averages over heads, and 5×10^{-4} over 380 heads is 10^{-4} over 76). But the arm trained on one seed and at half the batch size, so on a noisier gradient at every step, and the comparison does not isolate the number of entries.

- **Task reconstruction.** Recover what each head classifies. Head-level autointerp, which describes a head from its entries' texts side by side and scores blinded assignment against sibling entries, finds no above-chance signal at 20 heads per condition, on heads not screened for identifiability (Appendix D.6). Scored by modularity on a nearest-neighbor graph of the residual each layer classifies, the softmax Ontologizer's layers 1–4 are two to three times as modular as size-matched regroupings of their entries; neither the discovered groups nor the trained heads of a Matryoshka SAE [23] are, on matched prefix-residual graphs (one checkpoint each; Appendix C.3). Still open: partition-vs-label-library MI, led by topic and genre (from a labeled corpus or unsupervised topic clusters) since those are what the embedding is organized by, with length, formatting/noise and script as secondary factors; head-level autointerp on heads that pass the identifiability bar, at a sample large enough to separate from its null; and FLORES parallel-text invariance to split content heads from form heads. Run each identically on Ontologizer heads and `g160softmax`'s heads.

- **Hierarchy as MoE routing.** Conditional MI between successive layers' partitions tests whether depth builds compositional tasks; `forward="labels"` at the fixed initialization tests the pairing account from the other side. Pairwise effective information [69] bounds the edges of Section 8's causal graph: on the hard code, forcing a head to one of its live entries moves any single downstream head's choice by about 0.01 of a possible 5 bits, at every depth, though 11–16 times more than the pair co-varies naturally. Whether that influence is spread thinly or joint across sources, which pairwise measures cannot see, is open.

- **The collinearity setpoint's mechanism.** The one lever that buys steerability acts on a statistic that is mostly gauge; repeat it on the decoded statistic, and on signed dictionaries.

- **Penalize head overlap where a decoder cannot absorb it.** Trained as a loss term, ω reaches its target in the dictionary space and nowhere else: the partition moves into the decoder's null space, or a near-null one when the

decoder is square (Appendix G.2). The penalty has to act on decoded atoms, or on direct atoms, which have no decoder. The k=128 direct runs trained only at s_ω=10^{-6}, too weak to move ω, where on GPT-2 it takes hold from 10^{-5}. Output coordinates are not privileged axes, so a disjoint-coordinate partition there is a strong and somewhat arbitrary constraint, but it cannot be faked. Concatenated heads remain the only route to disjoint support.

- **Logit-scale control for straight-through training.** Normalize logits per head, or control t against a spread/t setpoint, in place of the hand-matched temperature, or add a term that pushes the logits to commit, the counterpart of the commitment loss by which VQ-VAE ties an encoder's scale to its code [55].

- **Codes trained for their rate.** The operational rates price codes that were never trained for a codec: the hard code's entries are used almost uniformly, the most expensive usage to entropy-code, and no SAE was trained with quantized coefficients. Training each against its entropy-coded rate — an entropy penalty on the hard code's usage, quantization-aware SAEs swept through the 1,900-bit region — would compare the architectures rather than these models.

- **Winner dropout for the one-layer variants.** `g160top1`'s 29% dead latents are within-head winner-take-all starvation. They survived Gao et al.'s auxiliary loss on dead latents [22] (`aux_k` 512, on throughout training), which gives a dead latent gradient without making it win its head. Two older remedies are the alternatives to test beside winner dropout: k-sparse autoencoders starve the same way at low k_{SAE} and avoid it with a budget that falls from 100 to its target over the first half of training [21], here a top-k_{sel} per head annealed to top-1; and neural codecs re-seed a codeword whose running usage falls too low at an input from the current batch [6], as `sae.py`'s neuron resampling already can.

- **Trained truncatability.** A deviation-sparsity penalty at training time would show whether the soft Pareto curve can shift left, as training against several sparsities at once makes a `TopK` SAE's code progressive [22].

- **Rung composition and frontier fill.** `groups+prefixes` (2a+3) is now trained, as `m5120_g160top1_p5` (FVU_w 0.294), for the modularity control above, but is not yet on the ladder. Still to run: the one-layer variants at m=11264, the mid-range `TopK` sweep ($k_{\text{SAE}} \in \{512, 2048\}$), and random-support and nonnegative-LS refit controls, the second being the positivity-constrained refit of Gao et al. [22].

- **Closed.** HSIC head independence (heads are already independent), the HSIC penalty's freeze (real, and the biased estimator's: only a frozen head

escapes its floor; Appendix D.1), decoder sparsification and bias (no structure; $< 1\%$), code-forwarding alongside the residual, input whitening (subsumed by initialization), entropy-pressure rescaling across shapes (likewise), and per-layer decoders (on GPT-2 every layer adds what it added with the shared decoder, to within 0.007 of FVU_w, so a basis fitted to layer 0's error is not what limits depth). The `resid_nc` layer-4 freeze is explained by saturation, and the probe correction leaves it unchanged; the two anomalously coherent heads (L2h25, L3h28) remain to inspect.

Reproducibility

All code is available at `https://github.com/WakaruLabs/ontologize`. The evaluation protocols are root-level scripts (`sae.py`, `pareto.py`, `headstruct.py`, `headcoh.py`, `splitting.py`, `textfid.py`, `langprobe.py`, `refit.py`, `steerfid.py`, `compose.py`, `autointerp.py`), driven by the `sae_ladder.sh`, `sae_evals.sh`, and `autointerp_run.sh` runners (idempotent, crash-resumable systemd user units; judge calls rate-metered). The straight-through arm and its diagnostics live in `experiments/ste-arm/` (`train_ste.py`, `steerembed.py`, `blendablate.py`, `gainablate.py`, `inputgeom.py`, `dictgeom.py`, `lastlayer.py`, `partition.py`, `headcontrib.py`, `entryshare.py`, `bigatoms.py`, `headlang.py`, `headscript.py`, `headeta.py`, `embedgeom.py`, `decodehead.py`, `make_zca.py`); its notebook `experiments/ste-arm/notes.md` records every run's command and the superseded readings. The GPT-2 hard-code runs of Section 6 live in `experiments/gpt2/` (`train_onto.py`, `diagnose.py`, `seedstab.py`; notes in `DIRECT_GAIN_STE.md` and `results/README.md`, with the two headline runs' configs, diagnostics and loss curves) and need the `headline` branch's code. The curated test suite pins each protocol's statistics against planted-structure synthetics. Artifacts live under `data/out/sonar/` and `data/out/gpt2_l8/`. At most three straight-through runs fit concurrently on one 24 GB device.

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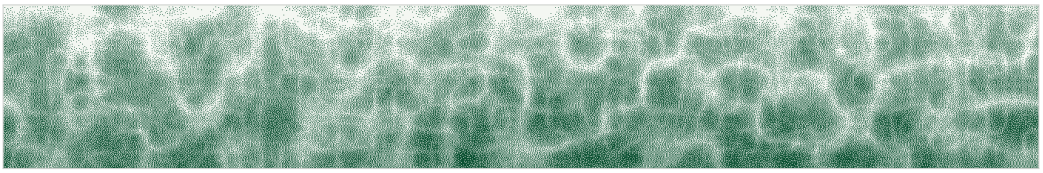
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RIDGE · SEED 32 · THE SAME PIPELINE, CLOSING THE FILE

A Notation

This appendix collects the symbols used throughout, with where each is defined.

Conventions. Lowercase italic letters are scalars (t, g_ℓ), bold lowercase letters vectors ($\mathbf{x}$, $\mathbf{a}_{ij}$), uppercase italic letters matrices (W_{dec}, D_i), and sans-serif letters tensors of rank three or more (D, P). A subscripted object takes the font of what it is: p_{nij} is an entry of P, $\mathbf{p}_{ni}$ a vector slice of it, and $D_i := \mathsf{D}_{i\cdot\cdot}$ a matrix slice; where a lowercase entry would read as a count, it is bracketed, as in $[\mathsf{K}]_{nij}$. Batching adds a rank: one sample's assignments are $P \in \mathbb{R}^{h\times k}$, a batch's $\mathsf{P} \in \mathbb{R}^{b\times h\times k}$. Vectors are columns, so a matrix acts on the left ($W_{\text{dec}}\mathbf{r}_\ell$, with $W_{\text{dec}} \in \mathbb{R}^{d\times e}$ mapping $\mathbb{R}^e$ to $\mathbb{R}^d$); a batch stacks its samples as rows ($X \in \mathbb{R}^{b\times d}$), so a batched product carries a transpose, $\hat{X} = R\,W_{\text{dec}}^\top$, or is written as function application, $\text{dec}(R)$. Sets are calligraphic ($\mathcal{S}_n$, $\mathcal{G}$). Named statistics, losses and operators are roman (FVU_w, MSE, KL_m, dec, sg), as are H, entropy, and L, the loss; the paragraph around Eq. (36) follows information theory in writing random variables in uppercase. A hat marks a reconstruction ($\hat{\mathbf{x}}_\ell$), a tilde a noised quantity ($\tilde{\mathsf{F}}$, $\tilde{\mathbf{f}}_i$), a bar a mean ($\bar{\mathbf{p}}_i$, $\bar{\kappa}$), and a prime on an index a second index of the same kind (i', j'). $\odot$ is the elementwise product and $\oplus$ concatenation. The cosine similarity of two vectors is the cosine of the angle between them, $\cos\angle(\mathbf{a},\mathbf{b}) := \langle\mathbf{a},\mathbf{b}\rangle/(\|\mathbf{a}\|\|\mathbf{b}\|)$, so that cos keeps one argument; $|\cos\angle|$ alone is its absolute value for the pair in context. $\mathbb{E}$ is an average over what its subscript names: $\mathbb{E}_n$ over the samples in context (a batch

in a training statistic, the scored rows in an evaluation), $\mathbb{E}_i$ over heads, $\mathbb{E}_{i \neq i'}$ over pairs of distinct heads, $\mathbb{E}_{j \sim \bar{\mathbf{p}}_i}$ under a distribution. Without a subscript it is over the data. Pr is likewise a fraction of samples, $\Pr_n(\cdot)$ the fraction on which its condition holds. $\mathbf{1}[\cdot]$ is the indicator, and sd and var are the standard deviation and variance, subscripted as $\mathbb{E}$ is.

Layers and states. A layer is indexed by its position, $\ell = 0, \ldots, l-1$, and a state of the stack by the number of layers applied, $0, \ldots, l$: layer ℓ reads state ℓ and writes state $\ell + 1$. So $\mathbf{r}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{x}$, the final reconstruction is $\hat{\mathbf{x}}_l$, and the deep-supervision prefixes are $\hat{\mathbf{x}}_1, \ldots, \hat{\mathbf{x}}_l$. Tables and prose name layers ("layer 0"), so a table's row for layer ℓ reports the prefix through it, $\hat{\mathbf{x}}_{\ell+1}$.

Indices and sizes.

index	ranges over	size	
n, n'	samples	b	batch size
ℓ, ℓ'	layers, $0, \ldots, l-1$	l	layers
i, i'	heads of a layer	h	heads per layer
j, j'	entries of a head	k	entries per head
q	coordinates	d, e	input dimension, dictionary width
		r	fiber rank
		d_{head}	a head's slice of e under `ConcatDictBlock`
		k_{sel}	entries kept by the `top`k_{sel} selection rule
		h'	heads kept, probed or conjoined, out of h
		n_{const}	constant coordinates appended to a layer's input
		m, k_{SAE}	SAE width, and active latents under `TopK`

k alone always counts Ontologizer entries.

Symbols. Grouped by the part of the paper that introduces them.

symbol	meaning	defined
The model		
$\mathbf{x}$, X; $\bar{\mathbf{x}}$	an input (d), a batch ($b \times d$); the data mean	Eqs. (20), (23)
enc	the encoder, the identity unless stated	Alg. 3
$\mathbf{u}_\ell$, $\mathbf{u}$	layer ℓ's shape: its input's unit direction, with the constant coordinate appended; what the classifier, router and fibers read ($\mathbf{u}$ within a layer)	Eq. (14)
W, V	the bilinear classifier's weight tensors	Eq. (1)
K, K	logits, one sample's ($h \times k$) and a batch's	Eq. (1)
t	classifier temperature, $P = \text{softmax}_k(K/t)$	§3.1.3
P, P; $\mathbf{p}_{ni}$, p_{nij}	assignments, one sample's and a batch's; head i's distribution, one entry's probability	§3.1.3

symbol	meaning	defined
$\bar{\mathbf{p}}_i$	head i's mean assignment over the data, its usage	Eq. (4)
p_{drop}	winner-dropout probability	§3.1.2
D, D_i	the dictionary ($h \times k \times e$), and head i's slice ($k \times e$)	Eq. (5)
$\mathbf{a}_{ij}$	an atom: $\lvert\mathsf{D}_{ij\cdot}\rvert$, or $\mathsf{D}_{ij\cdot}$ when signed	Eq. (5)
G_i	head i's dictionary Gram	Eq. (3)
$\mathbf{f}_i$; F, F	head i's output (e); stacked over heads, and over heads and samples	Eq. (5)
$\mathbf{f}'_\ell$, F'	layer ℓ's pooled output, and a batch's	Eq. (10)
$\boldsymbol{\gamma}$, γ_i, Γ	router scales: per head, one head's, and a batch's	Eq. (19)
U_{ij}, $\mathbf{c}_i$; C, C	fiber basis ($e \times r$) and coordinates; one sample's coordinates, and a batch's	Eq. (19)
A, A_i; β, ρ	fiber readout, and head i's; coordinate bound, norm cap	§3.1.7
g_ℓ, g_n	gain: the norm of layer ℓ's input, or of one sample's	Eq. (11)
$\mathbf{r}_\ell$, R	accumulator after ℓ layers, and batched	Eq. (11)
W_{dec}, dec	the decoder shared by all layers	Eq. (12)
$\hat{\mathbf{x}}_\ell$, $\hat{X}$	reconstruction after ℓ layers; the final one, batched	Eq. (12)
$\mathbf{y}_\ell$	residual after ℓ layers, which is layer ℓ's input	Eq. (13)
$\text{sg}[\cdot]$	stop-gradient	Eq. (13)
W_{flat}	the code-to-output map of a model's flattened code, $\hat{\mathbf{x}} = W_{\text{flat}}\mathbf{p}_{\text{flat}}$; its columns are the entries' decode directions	§B.5
Noise, statistics and loss		
ε; ε_K, ε_F	a standard normal draw; the logit and feature noise terms	§3.1.6
σ_X, σ_K, σ_F	noise scales for inputs, logits and features	§3.1.6
$\tilde{\mathsf{F}}$, $\tilde{\mathbf{f}}_i$	noised per-head outputs	Eq. (7)
ϵ	floating-point epsilon of the working dtype	Eq. (14)
τ_{rev}	dead-entry revival threshold	Alg. 2
entropy_k, H	entropy of a head's assignment in bits, and its mean over samples and heads	Eq. (2)
cossim_b	between-sample similarity of per-head outputs	Eq. (3)
KL_m	divergence of each head's batch-mean assignment from uniform, averaged over heads	Eq. (4)
κ_i, $\bar{\kappa}$	mean cosine between head i's atoms, and its mean over heads (row collinearity)	Eq. (22); §G.1
$\boldsymbol{\pi}_i$, ω	head i's usage-weighted coordinate profile; the support overlap between heads	Eq. (6)
L1, L1_K, L1_S	feature L1; the classifier gate's; the router's	Eqs. (7), (22)
KL_{pwak}, L2_{pwak}	partition-consensus statistics	Eq. (22)

symbol	meaning	defined
$\boldsymbol{\psi}_\ell$	layer ℓ's statistics row	Eq. (22)
$\mathbf{w}$	per-dimension MSE weights: inverse variance, normalized to mean one	Eq. (20)
$\mathbf{s}$; s_g, s_ω, s_{L1}, ...	loss weights: the vector paired with $\boldsymbol{\psi}_\ell$, and individual weights	Eq. (21)
MSE, L	whitened reconstruction error; the total loss	Eqs. (20), (21)
Evaluation and baselines		
FVU_w	whitened fraction of variance unexplained	Eq. (23)
v, $\mathcal{S}_v$; $\mathbf{o}$	deviations kept per head, and their entries; the origin, the mean assignment	Eq. (24)
$\text{FVU}_{w,\text{G}}$, n_{bits}; λ_j, θ	the Gaussian reference at a bit budget; covariance eigenvalues, the water level	Eq. (25)
H_{real}	realized bits: the summed per-head entropy of usage	Eq. (36)
Δ, Δ_j; H_{op}	the quantizer step, shared and coefficient j's; the operational rate in bits per sample	Eq. (26)
$\mathcal{G}$; exh, exc	a head or latent group; its exhaustiveness and exclusivity	Eq. (27)
$\kappa(\mathcal{G})$, $\zeta_{\cos}$; $\mathbf{v}_j$	a group's mean decode-direction cosine, and its standardized score; a decode direction	Eq. (28)
$(\cdot)^{(1)}$, $(\cdot)^{(2)}$	the two models in a pairwise comparison	Eq. (29)
$\mathbf{q}_i^{(1)}$; $[M^{\text{part}}]_{ii'}$, $[M^{\text{contr}}]_{ii'}$	head i's entry choices over the rows; the head-matching matrices	Eq. (29)
$\boldsymbol{\phi}_{ni}$, size_i	head i's output contribution on sample n; its share of the variance	Eqs. (29), (30)
$\mathbf{a}_{ij}^{\text{dec}}$, $\bar{\mathbf{a}}_i^{\text{dec}}$	a decoded atom; head i's usage-weighted mean	Eq. (30)
recovered	fraction of the next-token loss recovered	Eq. (31)
η^2; C, n_{rows}	variance explained by a partition; its cells, the rows scored	Eq. (32)
k_{probe}	the sparse-probing budget	§B.5
$\mathbf{z}_n^\star$, $\mathcal{S}_n$; W, $\mathbf{b}$	refitted coefficients on a frozen support; a linear code's decode directions (the columns of W) and offset	Eq. (33)
μ_{ni}; j_1, j_2; J_i; α	a head's logit margin between two entries; the Jacobian of head i's logits; step strength	Eq. (34)
$j^\star$, dir, $\boldsymbol{\xi}_n$	a steering target, a direction, the random control	Alg. 4
assign	the abstraction map from an input to its code	§8
$\mathbf{z}$, z_{nj}	SAE latent activations	Eq. (27)
W_{enc}, $\mathbf{b}_{\text{enc}}$; W_{dec}, $\mathbf{b}_{\text{dec}}$	SAE encoder weights and bias; its decoder, whose columns are the latents' directions, and the pre-subtracted decoder bias	§B.2
L0, s_{L1}, `aux_k`	active SAE latents per sample; an L1 SAE's weight; dead-latent revival	§B.2; Table 13

B Supplementary methods

B.1 Data and objective

Embedding space and data. All SONAR models are trained on a fixed cache of 4,000,000 SONAR sentence embeddings (d=1024, unit-norm) computed from an interleaved 86-language mC4 sample, with a language label sidecar. The stream order is deterministic, so raw text is recoverable by row index (verified against the language sidecar per row). Only six of the 86 configurations (English twice, French, Spanish, German and Chinese) were encoded under their own SONAR source-language tags; the other 80, 93% of rows, were encoded under the English tag, which SONAR reads as the text's language. Re-encoding 400 rows per language under the intended tags (`experiments/lang-tags/retag.py`) moves the changed rows by a whitened FVU of 0.305, about twice the hard code's reconstruction error and almost all of it specific to each row, while language explains 6.7% of the embedding's variance under the English tag and 7.4% under the intended ones; the trained models' error rises by 4–12% on the correctly tagged versions, the hard code and the SAEs alike. "SONAR embeddings" here means SONAR's embeddings under these tags, and the language results inherit the difference (Appendix C.6). The cloud is unit-normed but not centered: $\|\bar{\mathbf{x}}\| = 0.564$, so it is a spherical cap and 0.318 of every squared norm is one shared direction. It is dense without being isotropic — effective dimension 108 of 1024 (participation ratio 268 on the whitened eval tail), yet 943 principal components for 99% of variance — and samples average a pairwise cosine of 0.31. Sections 7.1 and 5.5 additionally use residual-stream activations of GPT-2 small [53], the residual leaving block 8 of 12 (`blocks.8.resid_post`; d=768, harvested by `encode_acts.py`), as a substrate where sparse dictionary learning is known to work [22]. Section 6 uses a separate GPT-2 setup from the `headline` branch — the residual stream entering block 8 on FineWeb, unwhitened, scored also by GPT-2 loss recovered — whose numbers compare only within that section.

Evaluation. All models minimize the whitened MSE of Eq. (20). We report FVU_w, whitened MSE divided by the whitened variance of a constant predictor,

$$\mathrm{FVU}_w := \frac{\sum_n \sum_q w_q \, (\hat{x}_{n,q} - x_{n,q})^2}{\sum_n \sum_q w_q \, (x_{n,q} - \bar{x}_q)^2}, \qquad \bar{\mathbf{x}} := \mathbb{E}_n[\mathbf{x}_n], \tag{23}$$

with $\mathbf{w}$ the per-dimension inverse variance normalized to mean one, on the final 32,768 cache rows. This tail is excluded from SAE training and from the training of every straight-through arm. The softmax Ontologizers (`resid_nc` and the selection sweep) predate the holdout and trained on the full cache including the tail. To measure what that is worth, the 131,072 stream rows that follow the training corpus were encoded the same way (`encode_fresh.py`), rows no model has seen. On them every `pareto.py` point of `resid_nc`, the selection sweep's softmax run,

the straight-through stack and the SAEs moves by at most 0.3% (one SAE at FVU_w 10^{-4} by 0.9%), and every operational rate of Appendix E.3 by at most 0.0004 in FVU_w, so the softmax models' scores on the tail are not inflated by having seen it. Where activations are needed, Ontologizer features are the flattened per-head classification probabilities $P \in \Delta^{k\,l\cdot h}$ at the run's final temperature (entry j of head i in layer ℓ is feature $(\ell h + i)k + j$; fire threshold $2/k$), and SAE features are latent activations (fire threshold 0; runs with softmax heads use the above-uniform threshold $2/(m/h)$ for h heads, matching the entry convention). Scoring a softmax run at the default temperature 1.0 rather than its annealed 0.03 puts a healthy checkpoint at $\mathrm{FVU}_w \approx 0.91$, which is indistinguishable from a broken one; every number here uses the run's own temperature.

B.2 Models and baselines

Every model's configuration is tabulated in Appendix B.3.

Softmax Ontologizers. The reference Ontologizer (`resid_nc`) is the architecture of Section 3.1 with softmax selection: l=5 residual layers of h=32 heads with k=32 entries each. Layers 0–3 trained healthily (prefix FVU_w 0.129 → 0.0051); layer 4 froze 22/32 heads late in temperature annealing and is mildly harmful (Section G.4 explains why). A later *selection sweep* holds that shape fixed and varies only the `DictBlock` selection rule — `top1`, `top2`, `top4`, `top8`, `top16`, `softmax` — each with and without a raised mean-entropy bonus (`_shm`).

The straight-through arm. `train_ste.py` imports `sonar.py`'s configuration and overrides `select="ste"`: the forward pass selects the argmax one-hot, and the backward pass uses softmax(K/t) as its surrogate. Its shape, calibration and initialization are described with its results in Section 5.

SAEs. SAEs follow the architecture of Gao et al. [22]: pre-activation relu$(W_{\mathrm{enc}}(\mathbf{x} - \mathbf{b}_{\mathrm{dec}}) + \mathbf{b}_{\mathrm{enc}})$, reconstruction $\hat{\mathbf{x}} = W_{\mathrm{dec}}\mathbf{z} + \mathbf{b}_{\mathrm{dec}}$, with `TopK` selection of k_{SAE} latents, decoder directions (the columns of W_{dec}) renormalized to unit norm after every step, a linear encoder initialized to the decoder's transpose, and `aux_k` dead-latent revival (512 dead latents at weight 1/32, a latent counting as dead after 1000 steps without firing), trained to convergence. Unlike their recipe, the `TopK` pre-activation keeps the encoder bias $\mathbf{b}_{\mathrm{enc}}$, which their `TopK` encoder omits, $\mathbf{b}_{\mathrm{dec}}$ starts at the data mean rather than the geometric median, the objective is whitened, Adam runs with its default ϵ and without weight averaging or decoder-gradient projection, and training makes many passes over a fixed cache. ReLU+L1 SAEs replace `TopK` with an L1 penalty and neuron resampling.

The structural-ablation ladder. To attribute differences to specific architectural commitments rather than to whole architectures, we train SAE variants that each add one Ontologizer-style constraint (all m=5120, same data and objective):

rung	run	added structure	analogue
1	`k32`	none (`TopK`, k_{SAE}=32)	baseline
2a	`g160top1`	160 heads, hard winner per head	hard heads, router
2b	`g160softmax`	160 heads, softmax per head	heads (soft)
3	`k32_p5`	5 nested prefix losses (matryoshka)	deepsup layers
—	`k32_bl`	bilinear encoder, $[\mathbf{x}-\mathbf{b}_{\text{dec}}; 1]$ factors	bilinear classifier

Rungs 2a and 2b divide the m=5120 latents into 160 heads of 32, and we present them as simplified Ontologizer variants, not as baselines. Per-head competition over a shared decoder is a single layer of linear classifiers selecting dictionary entries, without the bilinear classifier, the residual stack or the non-negative dictionary. `g160softmax` is the one-layer softmax Ontologizer: a head's coefficients sum to one, so it has no magnitude of its own. `g160top1` is the one-layer hard-code Ontologizer with a router fused into its classifier: each head selects its entry by argmax, and the winning logit's ReLU value scales that entry's unit-norm atom. That value is the router's per-head scale (Section 3.1.7), computed by the classifier itself rather than by a separate map, so the code's 160 coefficients are router scales beside its 800 index bits. The fused router is also what trains the hard selection without a straight-through estimator: the winning logit receives reconstruction gradient through its role as a scale, and the losing logits receive none. The same function is an SAE with a rectified local winner-take-all activation [70] over the heads. In quantization terms both rungs have the form of one additive quantizer, 160 codebooks of 32 atoms that each span the whole output [3]: `g160softmax` mixes each codebook's atoms, and `g160top1` takes one atom per codebook with the router scale as its gain, so that every head is a gain–shape vector quantizer [5]. Switch SAEs [13] are the transposed design: they route each input to one expert with several latents active inside it, where `g160top1` keeps every head active with one latent inside each. `k32_bl` computes each latent as the product of two linear projections of the input, which is how the bilinear autoencoder of Dooms et al. implements its atomic latents, though it defines them as a single squared projection [71]. `k32_bl` centers the input and appends a constant coordinate so that its latents are not sign-blind, where that autoencoder reads the raw input with no bias. Where it has no activation, a Hoyer sparsity penalty and a tied decoder that reconstructs $\mathbf{x}\mathbf{x}^\top$, `k32_bl` feeds a `TopK` activation and an ordinary linear decoder of $\mathbf{x}$: it is a `TopK` SAE with a bilinear encoder. Rung 3 is a Matryoshka SAE [23] with five equal blocks of 1024 latents where the original's blocks roughly double in size.

SAE baselines at width m=11,264 (parameter-matched to the Ontologizer): `k32`, `k160` (coefficient-matched to one deviation per head), and `k5120` (coefficient-

matched to the full soft code; the `TopK` constraint never binds — measured L0 ≈ 1520 — so it is effectively a dense ReLU autoencoder, the regime in which `TopK` latents go dense and PCA-like [22]). A seed replica `k32_s43` supports stability analysis. The bilinear encoder's latents admit closed-form "eigenfeature" analysis: each latent's interaction matrix $\mathrm{sym}(\mathbf{w}_1\mathbf{w}_2^\top)$ (Pearce et al. [52]) has rank at most 2, and its top-$|\lambda|$ eigenvector is $\mathbf{w}_1/\|\mathbf{w}_1\| \pm \mathbf{w}_2/\|\mathbf{w}_2\|$, with the sign of $\mathbf{w}_1 \cdot \mathbf{w}_2$. When the two factors anticorrelate, as they do for 90% of `k32_bl`'s latents, that eigenvector's eigenvalue is negative: the form falls along it, so it is a direction the latent rejects, and the positive-eigenvalue eigenvector $\mathbf{w}_1/\|\mathbf{w}_1\| + \mathbf{w}_2/\|\mathbf{w}_2\|$ is the one that excites it. The `eig` autointerp mode (`sae.eigenfeatures`) takes the top-$|\lambda|$ eigenvector. Under the steering frame the simplified variants are the load-bearing comparison: `g160softmax` isolates the "series of softmax classifiers" idea from the residual stack, bilinear classifiers and shared dictionary, and `g160top1` does the same for hard selection with a fused router.

B.3 Model configurations

Tables 9–13 list every reported model's configuration as its run recorded it: training hyperparameters from each run's final configuration record, any value that changed when a run was resumed, and the architecture from its checkpoint.

B.4 The statistics row and its retired entries

The implementation logs each layer's statistics as a row whose layout is append-only: a new statistic goes on the end and a retired one keeps its slot, so a shorter row from an older run still aligns column for column and the positional loss weights keep pairing with the right quantity. The row of Eq. (22) is therefore stored with two empty slots,

$$\begin{bmatrix} \mathrm{L1}_K & \mathrm{L1} & H & \mathrm{cossim}_b & \mathrm{cossim}_h & \mathbb{E}_i \kappa_i & \max_i \kappa_i \\ \mathrm{KL}_m & \mathrm{KL}_{\mathrm{pwak}} & \mathrm{L2}_{\mathrm{pwak}} & \mathrm{cossim}_{\mathrm{flat}} & \mathrm{L1}_S & \omega & \end{bmatrix},$$

where the two retired statistics are logged as NaN and zeroed before the weighted sum.

Between-head cosine cossim_h was the mean absolute cosine over ordered pairs of heads within a sample,

$$\mathrm{cossim}_h(\mathsf{F}) := \mathbb{E}_n \, \mathbb{E}_{i \neq i'} \left| \frac{\langle \mathbf{f}_{n,i}, \mathbf{f}_{n,i'} \rangle}{\|\mathbf{f}_{n,i}\| \|\mathbf{f}_{n,i'}\|} \right|,$$

computed on the clean features. It stood in for disjoint support, which on non-negative features its zero implies (Section 3.1.4), but it measured that indirectly and at $O(bh^2e)$, and it says nothing about support under a signed dictionary, where orthogonal outputs can be cancellation rather than disjointness. It saw only the

shared by every softmax run	
input	SONAR mC4 embeddings, d=1024, unit norm
shape	l=5 layers × h=32 heads, k=32 entries per head
classifier	bilinear (order 2), no gate
forward	residual; `resid_norm`, `resid_const`, `resid_gain`; joint deep supervision
optimizer	Adam, lr 5×10^{-5}, gradient clip 1.0, batch 256, 24 epochs
temperature	t=1 → 0.03, geometric over 50k steps, then held
noise	logits normal, σ_K=0.02 → 0.006; features featvar, σ_F=0.1; input off
winner dropout	p_{drop}=0 → 0.1 over the anneal
loss	whitened MSE; s_{cossim_b}=10^{-5}; ghost path off
seed	42

run	selection	e	s_{KL_m}	other	steps
`resid_nc`	softmax	4096	$10^{-6\dagger}$	s_{L1}=10^{-9}, s_{cossim_h}=0 → $10^{-5\dagger}$	372.6k
`resid_nc_hm`	softmax	2048	10^{-6}	s_{L1}=10^{-9}, s_{cossim_h}=10^{-5} → $10^{-6\dagger}$	8.27M‡
`sweep_softmax`	softmax	2048	10^{-6}	s_{cossim_h}=10^{-6}	369.5k
`sweep_top`k_{sel}	top-k_{sel}, $k_{\text{sel}} \in \{1, 2, 8, 16\}$	2048	10^{-6}	s_{cossim_h}=10^{-6}	369.5k
`sweep_top8_relu`	top-8	2048	10^{-6}	ReLU gate on the classifier	369.5k
`sweep_*_shm`	softmax, top-k_{sel}, $k_{\text{sel}} \in \{2, 4, 8, 16\}$	2048	3×10^{-5}	s_{cossim_h}=10^{-6}	369.5k

TABLE 9 · Softmax Ontologizer configurations. †Changed at a resume: `resid_nc` acquired s_{KL_m} and its s_{cossim_h} mid-run, and `resid_nc_hm` lowered s_{cossim_h}. ‡Latest checkpoint; results attributed to `resid_nc_hm` in Appendix D use its 4.38M-step checkpoint. cossim$_h$ is the retired between-head cosine (Appendix B.4).

shared by every SONAR straight-through arm	
input	SONAR mC4 embeddings, d=1024; eval tail held out
selection	straight-through (`ste`): argmax forward, softmax backward
classifier	bilinear (order 2), no gate
dictionary	non-negative, e=2048, shared linear decoder
forward	residual; `resid_norm`, `resid_const`, `resid_gain`; joint deep supervision
optimizer	Adam, lr 10^{-5}, gradient clip 1.0, batch 256, 24 epochs
temperature	t=1.5×10^{-4}, constant
noise	logits batchnorm, σ_K=0.3; features featvar, σ_F=0.1; input off
winner dropout	p_{drop}=0.1
loss	whitened MSE; s_{L1}=10^{-9}, s_{cossim_b}=10^{-5}, s_{cossim_h}=10^{-6}; ghost path off
seed	42 unless noted

arm	shape	k	bits	s_{KL_m}	init scale	other	steps
`ste_h76`	5×76	32	1900	10^{-4}	—		369.5k
`ste_h76_init01`	5×76	32	1900	10^{-4}	0.1		369.5k
`ste_h76_i01_s43`	5×76	32	1900	10^{-4}	0.1	seed 43	369.5k
`ste_h76_hm1e6`	5×76	32	1900	10^{-6}	—		369.5k
`ste_h76_kcos`	5×76	32	1900	10^{-4}	—	row-cosine setpoint 0.5	377.3k
`ste_h76_hsic3`	5×76	32	1900	10^{-4}	—	head CKA 10^{-3} (biased)	380.2k
`ste_k2_h380`	5×380	2	1900	5×10^{-4}	0.1	batch 128	739.0k
`ste_l1_h380`	1×380	32	1900	10^{-4}	—		369.5k
`ste_l1_h380_hm5`	1×380	32	1900	5×10^{-4}	—		369.5k
`ste_l1_h380_zca`	1×380	32	1900	5×10^{-4}	—	frozen ZCA encoder	353.3k
`ste_l1_h380_enc`	1×380	32	1900	5×10^{-4}	—	trained linear encoder	369.5k
`ste_l1_h380_kcos`	1×380	32	1900	10^{-4}	—	row-cosine setpoint 0.5; batch 256 → 128†	929.4k
`ste_l1_h380_i01`	1×380	32	1900	5×10^{-4}	0.1		369.5k
`ste_l1_h380_i01_s43`	1×380	32	1900	5×10^{-4}	0.1	seed 43	369.5k
`ste_l1_h380_i01_hm1e4`	1×380	32	1900	10^{-4}	0.1		369.5k
`ste_k128_h54`	5×54	128	1890	10^{-4}	0.1	s_ω=10^{-6}‡	369.5k
`ste_k128_h54_direct`	5×54	128	1890	10^{-4}	0.1	signed direct atoms, s_ω=10^{-6}‡	369.5k

TABLE 10 · SONAR straight-through configurations. Init scale is `--dict-init-scale`: each dictionary rescaled at initialization so a layer writes that fraction of its input (— leaves the default initialization). †Changed at a resume. ‡Trained after cossim_h was retired, so s_ω takes its place.

shared by every GPT-2 straight-through arm on this branch	
input	GPT-2 small, residual leaving block 8, d=768; document-level holdout
selection	straight-through (`ste`)
classifier	bilinear (order 2)
forward	residual; `resid_norm`, `resid_const`, `resid_gain`; joint deep supervision
optimizer	Adam, lr 5×10^{-5}, gradient clip 1.0, batch 256, 24 epochs
temperature	t=1.0, constant (the naive port's setting, Appendix E.1)
noise	logits normal, σ_K=0.02 (absolute, not relative to the logit scale); features featvar, σ_F=0.1; input off
winner dropout	p_{drop}=0.1
loss	MSE weighted by `gpt2_l8.mse_weights_matched`; s_{L1}=10^{-9}, s_{cossim_b}=10^{-5}, s_{cossim_h}=10^{-6}, s_{KL_m}=10^{-6}

arm	shape	k	dictionary	gate	init scale, seed	steps
`ste_h76`	5×76	32	non-negative, e=1536	none	1.0, 42	371.9k
`ste_h76_e768`	5×76	32	non-negative, e=768	none	1.0, 42	371.9k
`ste_h76_sgn`	5×76	32	signed, e=1536	none	1.0, 42	371.9k
`ste_h76_sgn768`	5×76	32	signed, e=768	none	1.0, 42	371.9k
`ste_h76_cat32`	5×76	32	non-negative, concat, d_{head}=32	none	1.0, 42	371.9k
`ste_h20_cat128`	5×20	256	signed, concat, d_{head}=128	sigmoid	—, 42	371.9k
`ste_h20_cat128-43`	5×20	256	signed, concat, d_{head}=128	sigmoid	—, 43	371.9k
`ste_h20_cat128-sc`	5×20	256	as above, plus router	sigmoid	—, 42	371.75k
`ste_h20_cat128-sc43`	5×20	256	as above, plus router	sigmoid	—, 43	371.75k

TABLE 11 · GPT-2 straight-through configurations on this branch: the orthant arms of Table 7 and the concatenated seed pairs of Section 5.5. The router (`scaled`) is sigmoid-gated and non-negative.

shared by the headline-branch runs of Section 6	
input	GPT-2 small, residual entering block 8, d=768; mean-centered, scaled to unit mean square; FineWeb
shape	l=5 layers × h=32 heads, k=32 entries per head (800 bits) unless noted
selection	straight-through, temperature t=1 → 0.03 over the first 60k steps
classifier	bilinear (order 2)
forward	residual; `resid_first`, `resid_gain`, `gain_clip`, `resid_norm`, `resid_const`; joint deep supervision
optimizer	Adam, lr 2×10^{-4}, batch 256
noise	logits normal, σ_K=0.02 → 0.006; features featvar, σ_F=0.1
winner dropout	p_{drop}=0 → 0.1
loss	unwhitened MSE; s_{L1}=10^{-9}, s_{cossim_b}=10^{-5}, s_{cossim_h}=10^{-6}, s_{KL_m}=10^{-6}
seed	42 unless noted

run	dictionary	data
direct + gain + STE	signed atoms in $\mathbb{R}^{768}$, no decoder	40M tokens, one pass (156k steps)
private heads + gain + STE	non-negative, e=1536 in 48-dimensional blocks, shared decoder	40M tokens, one pass
one-epoch ablations	as in Appendix F.1	10M tokens

TABLE 12 · Configurations of the headline-branch GPT-2 runs (`experiments/gpt2/results/D40_*/config.json`). Fiber variants add `fiber_rank` r with the dropout, cap or freezing given in Section 6's table.

shared by every `sae.py` *run*	
input	the SONAR cache and eval tail of the Ontologizers, d=1024
objective	whitened MSE, as the Ontologizers
architecture	pre-subtracted decoder bias, encoder initialized tied to the decoder, except `m5120_k32_bl`'s two factors, which are random and scaled to the tied encoder's pre-activation size
decoder	unit-norm directions, the columns of W_{dec}, except `g160softmax`'s, which carry magnitude because its coefficients are bounded
dead latents	`aux_k` 512 (`TopK` and grouped runs) or neuron resampling every 5000 steps (L1 runs)
optimizer	Adam, batch 4096

run	m	sparsity	encoder	structure	lr	steps
`m5120_k32`	5120	top-32	linear	—	4×10^{-4}	144.3k
`m5120_k32_s43`	5120	top-32	linear	seed 43	4×10^{-4}	144.3k
`m5120_k32_bl`	5120	top-32	bilinear	—	4×10^{-4}	144.3k
`m5120_k32_p5`	5120	top-32	linear	5 nested prefixes (Matryoshka)	4×10^{-4}	144.3k
`m5120_l1*`	5120	L1, five weights s_{L1} (caption)	linear	—	4×10^{-4}	144.3k
`m11264_k32`	11264	top-32	linear	—	10^{-4}	336k
`m11264_k160`	11264	top-160	linear	—	$10^{-4\dagger}$	336k
`m11264_k5120`	11264	top-5120	linear	— (dense, L0 ≈ 1520)	10^{-4}	22k
`m5120_g160top1`	5120	160 heads, top-1	linear	one-layer hard variant	4×10^{-4}	144.3k
`m5120_g160softmax`	5120	160 heads, softmax	linear	one-layer soft variant	4×10^{-4}	144.3k

TABLE 13 · SAE baselines (above the rule) and the single-layer variants (below it; Section B.2). The L1 sweep's five weights s_{L1} are 3×10^{-6}, 10^{-5}, 3×10^{-5}, 10^{-4} and 3×10^{-4}; $*$ is the weight as the run name spells it (`m5120_l13e-05`). †Not recorded; inferred from the run sharing `m11264_k32`'s step count.

entries a sample selected, so a hard code's unselected entries were unconstrained. The support overlap ω measures the property directly, for every used entry, under either sign, at $O(hke + h^2e)$.

Flattened row cosine $\text{cossim}_{\text{flat}}$ was the mean off-diagonal cosine over every pair of atoms in the layer, between heads as well as within. It was added to catch heads whose atoms are individually diverse but parallel to another head's, which the within-head κ_i cannot see. Its mean is dominated by between-head pairs ($k-1$ of a row's $hk-1$ partners share its head), and on non-negative atoms a between-head cosine of zero is again disjoint support, so ω, weighted by usage and computed per head pair, replaced it too.

B.5 Evaluation protocols

Each protocol is a standalone script sharing the eval split and activation conventions; the root-level ones are unit-tested against planted-structure synthetic data. Figure 14 shows what each script reads and the question it answers.

Capacity Pareto (`pareto.py`). Capacity is continuous coefficients transmitted per sample, plus index bits. Ontologizer code points are *top-ν deviation truncations*: per head, keep the ν entries with largest $|p_j - o_j|$, pin the rest to the origin distribution $\mathbf{o}$, renormalize,

$$\mathcal{S}_\nu := \text{top-}\nu\, j\, |p_j - o_j|, \qquad p'_j := \frac{\mathbf{1}[j \in \mathcal{S}_\nu]\, p_j + \mathbf{1}[j \notin \mathcal{S}_\nu]\, o_j}{\sum_{j' \in \mathcal{S}_\nu} p_{j'} + \sum_{j' \notin \mathcal{S}_\nu} o_{j'}}, \qquad \mathbf{o} := \mathbb{E}_n[\mathbf{p}_n], \quad (24)$$

applied to every head, so the code transmits ν coefficients per head. The origin is the corpus-mean classification $\mathbf{o}$ — measured, not uniform: the model output is jointly *linear* in the flattened classifications (zero residual seed, linear decoder), so decode($\mathbf{o}$) = $\mathbb{E}[\hat{\mathbf{x}}]$, which decodes to the corpus-mean generic text, while the uniform code lands off-manifold (cosine 0.067 to $\bar{\mathbf{x}}$). `resid_nc` trains without KL_m, the term meant to tie the uniform code to the generic one (Section 3.1.3); this probe is what motivated the term, and whether it achieves that in the models that train with it is not measured. Endpoints: ν=0 (input-independent origin), ν=k (exact soft model), and end-to-end argmax ("hard"). An alternative head-level code mean-ablates all but the h' lowest-entropy heads per layer. SAE points use measured eval L0. Index bits are the length of the code that names which coefficients are sent. The coefficients follow in a fixed order, so naming r of n costs $\log_2 \binom{n}{r}$ bits rather than $r \log_2 n$: $l \cdot h \log_2 \binom{k}{\nu}$ for the deviation code, $l \log_2 \binom{h}{h'}$ for the head-level code, and $\log_2 \binom{m}{\text{L0}}$ at the mean L0 for an SAE (for the single-layer hard variant, which heads fire and each one's winner). Index bits are not a rate for a code that also sends coefficients. We compare such a code with a hard one at matched total rate, index bits plus coefficients at whatever precision brings the two totals level, and bound its error there by the least-squares fit on its own support (Eq. (33)), which no coding of its coefficients can beat. A hard code's index

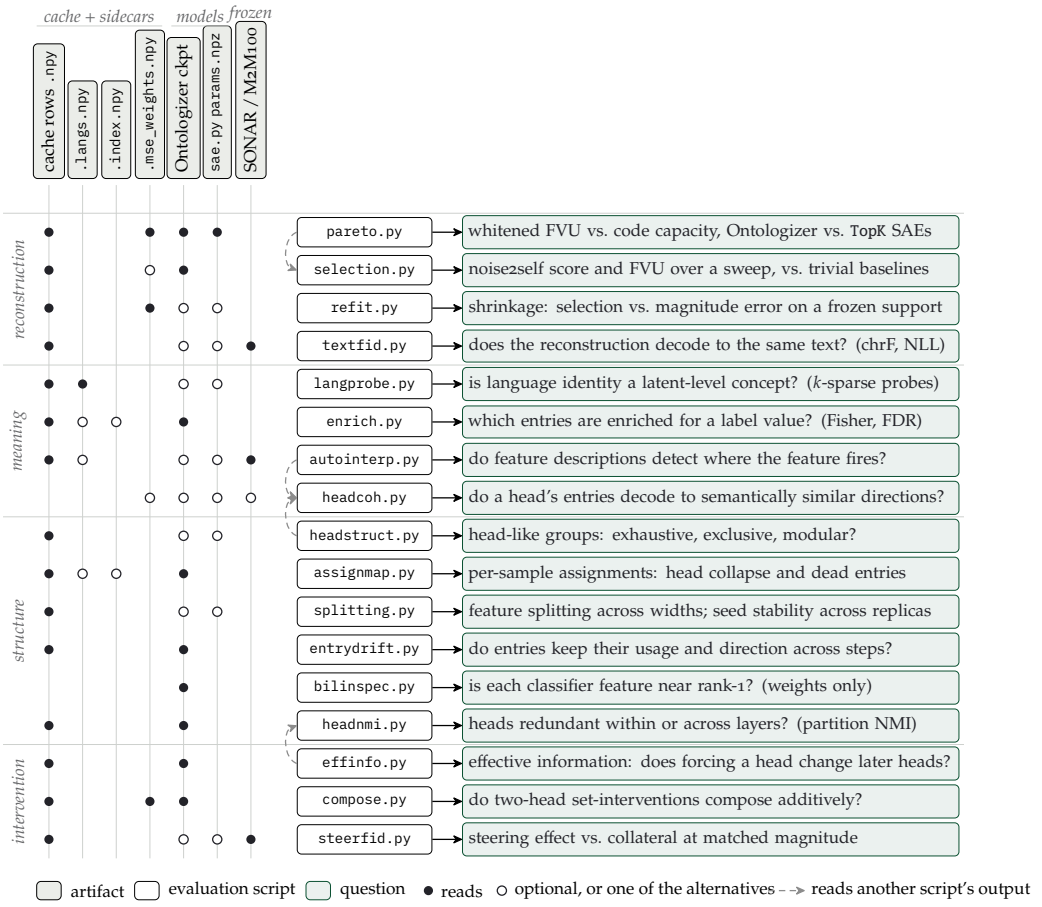


FIGURE 14 · **What validates what.** Columns are the artifacts each evaluation script reads: the embedding or activation cache, whose tail rows are the eval split `sae.py` excludes from training, with its sidecars; Ontologizer checkpoints; `sae.py` baselines; and the frozen SONAR encoder and M2M100 decoder. A filled mark is a required input, an open mark an optional input or one of several alternatives (most scripts take either model type; `splitting.py` takes two runs, such as seed replicas). Dashed arrows are one script's output read by another. `effinfo.py` (in `experiments/effective-information/`) is described with `headnmi.py`, which reads it.

bits are $l \cdot h \log_2 k$, its whole rate apart from any per-layer gains (Appendix E.4); we compare each against a *Gaussian reference*: the least distortion a code of that rate can achieve on a Gaussian source with the eval tail's whitened covariance, given by reverse water-filling [54, Thm. 10.3.3]. With $\lambda_1, \ldots, \lambda_d$ that covariance's eigenvalues, a rate of n_{bits} bits per sample gives

$$\mathrm{FVU}_{w,\mathrm{G}}(n_{\text{bits}}) = \frac{\sum_j \min(\theta, \lambda_j)}{\sum_j \lambda_j}, \qquad n_{\text{bits}} = \sum_j \max\Big(0,\ \tfrac{1}{2} \log_2 \frac{\lambda_j}{\theta}\Big), \tag{25}$$

with the water level θ solved from the rate. It is a reference, not a floor, for two reasons. Among sources of a given covariance the Gaussian has the largest rate–distortion function under squared error [54, Problem 10.8], so for the real embeddings the distortion–rate function at n_{bits} is *at most* $\mathrm{FVU}_{w,\mathrm{G}}(n_{\text{bits}})$ and may be well below it. And that function is a limit over block length: it is approached by fixed-rate codes that index long blocks of samples jointly [54, Sec. 10.5], whereas our codes describe each sample alone, and its converse holds at every block length [54, Sec. 10.4]. The reference is therefore safe in one direction only: a code *above* it is at least that far above the data's distortion–rate function, though not necessarily that far from the best per-sample code, and nothing we report can establish that a given rate is too small.

Operational rate (`quantrate.py`). To put codes with and without coefficients on one rate axis, everything a code sends is quantized and entropy-coded. A coefficient z_{nj} with origin o_j and decode direction $\mathbf{v}_j$ is sent as an integer symbol and decoded as

$$\hat{z}_{nj} = o_j + \Delta_j \Big\lfloor \frac{z_{nj} - o_j}{\Delta_j} \Big\rceil, \qquad \Delta_j = \frac{\Delta}{\|\mathbf{w}^{1/2} \odot \mathbf{v}_j\|}, \qquad H_{\text{op}} = \sum_j H\Big(\Big\lfloor \frac{z_{\cdot j} - o_j}{\Delta_j} \Big\rceil\Big), \tag{26}$$

with $\lfloor \cdot \rceil$ rounding to the nearest integer, so a coefficient within half a step of its origin is not sent and every coefficient costs about the same whitened error, $\Delta^2/12$ at fine steps. The origin is 0 for a sparse code and the training-row mean for a dense one (the $\mathbf{o}$ of Eq. (24) for a soft Ontologizer). H_{op}, in bits per sample, codes each slot's symbol with its own distribution: plug-in entropy on the evaluation tail with the Miller–Madow correction, which tables fit on held-out training rows reproduce within half a bit. A grouped top-1 SAE codes one symbol per head, its winner and level or silence. Ontologizers are quantized inside their forward pass, so each layer's residual is that of the quantized code; a hard code's entries cost their realized entropy (Eq. (36)) and its gains are quantized like coefficients. Nothing is refit after quantization, and sweeping Δ traces each code's curve. A second rate charges the support its fixed-length cost instead, $\log_2 \binom{m}{\mathrm{L0}}$ per sample (the nominal $l \cdot h \log_2 k$ for a hard code), with the values still entropy-coded.

Categorical form (`headstruct.py`). An Ontologizer head is exhaustive (always fires) and exclusive (one winner). We test whether SAE latents form such units: discovery clusters latents by *paradigmatic* affinity — similar co-fire profiles with the rest of the dictionary × anti-correlation with each other (PMI < 1) — under a group-size cap, on one half of the rows; scoring on the held-out half computes per-group exhaustiveness, exclusivity and the coefficient of variation of the group's summed activation,

$$\begin{aligned} \mathrm{exh}(\mathcal{G}) &:= \Pr_n\big(\textstyle\sum_{j\in\mathcal{G}} \mathbf{1}[z_{nj} > 0] \geq 1\big), \\ \mathrm{exc}(\mathcal{G}) &:= \mathbb{E}_n\big[\textstyle\sum_{j\in\mathcal{G}} \mathbf{1}[z_{nj} > 0] \;\big|\; \sum_{j\in\mathcal{G}} \mathbf{1}[z_{nj} > 0] \geq 1\big], \end{aligned} \tag{27}$$

against size-matched random partitions of the live-latent pool. A head scores exh = exc = 1. Head-like structure requires exhaustiveness *above* null with exclusivity *below* null; topical co-firing clusters show the opposite signature. `--trained-groups` scores a run's own trained heads instead.

Semantic coherence vs. contrast (`headcoh.py`). The complement to categorical form: do a head's entries *mean* similar things, or do they form a contrast set? Every entry has an exact decode direction in SONAR output space — for the Ontologizer, the columns of the linear map W_{flat} with $\hat{\mathbf{x}} = W_{\text{flat}}\mathbf{p}_{\text{flat}}$; for SAEs, decoder directions — and SONAR space is a semantic sentence space, so within-head direction similarity is semantic similarity of what entries decode to. Per head or group $\mathcal{G}$ we report mean pairwise cosine and top-singular-value energy fraction, z-scored as ζ against size-matched random subsets of the same pool (same layer for Ontologizer heads),

$$\kappa(\mathcal{G}) := \mathbb{E}_{j\neq j'\in\mathcal{G}} \cos\angle(\mathbf{v}_j, \mathbf{v}_{j'}), \qquad \zeta_{\cos}(\mathcal{G}) := \frac{\kappa(\mathcal{G}) - \mathbb{E}_{\mathcal{G}'}[\kappa(\mathcal{G}')]}{\mathrm{sd}_{\mathcal{G}'}[\kappa(\mathcal{G}')]}, \tag{28}$$

with $\mathbf{v}_j$ entry j's decode direction and $\mathcal{G}'$ ranging over 500 random same-size subsets; as a metric-hygiene control the same statistics are recomputed under the training objective's inverse-variance inner product (`--metric whitened`). $\zeta_{\cos} \gg 0$ means a head clusters semantically; $\zeta_{\cos} \ll 0$ means its atoms repel — the signature of a *contrastive classifier*, whose entries' atoms are pushed apart. A second view SONAR-encodes autointerp description texts and tests pooled within-head similarity against a label-permutation null. `--assignment` scores discovered (post-hoc) groups.

Between-head support (`headsupport.py`). The complement on the other axis: do distinct heads occupy distinct coordinates, as Section 3.1.4 argues non-negative heads should? On a trained checkpoint, each head's usage-weighted coordinate profile and their mean pairwise cosine are those of Eq. (6), with $\bar{\mathbf{p}}_i$ the head's mean assignment over the evaluation tail at the run's final temperature, from

the model's own forward. The *decoded* variant replaces each atom by $|\mathbf{a}_{ij}^{\text{dec}}|$, with $\mathbf{a}_{ij}^{\text{dec}} := \text{dec}(\mathbf{a}_{ij}) - \text{dec}(\mathbf{0})$ the output-space direction steering acts in; for an `sae.py` run, which has no dictionary space, the profile of a head or group $\mathcal{G}$ is $\boldsymbol{\pi}_{\mathcal{G}} = \sum_{j \in \mathcal{G}} \mathbb{E}[z_j]\,|\mathbf{w}_j^{\text{dec}}|$. Non-negative profiles overlap heavily by chance, so ω is scored against a null that permutes each head's profile coordinates independently (50 draws), keeping every head's own distribution and destroying only the alignment between heads. A head's *spread* is its profile's participation ratio as a fraction of the dimension, $(\sum_q \pi_{iq})^2/(e \sum_q \pi_{iq}^2)$, with d in place of e when decoded; a partition into h blocks needs it near $1/h$.

Feature splitting and seed stability (`splitting.py`, `partition.py`, `headcontrib.py`). For a coarse model (1) and a fine model (2) on shared rows, each latent j' of model (2) is classified per parent latent j of model (1) by two-way firing containment at threshold τ: a *child* satisfies $\Pr(j \mid j') \geq \tau$ and $\Pr(j' \mid j) < \tau$, with $\Pr(j \mid j')$ the fraction of the rows where j' fires on which j fires too; a *match* satisfies both. Splits (≥ 2 children) are scored by union coverage of the parent's firing set and pairwise child overlap. Containment is the activation-coverage criterion, which admits dense latents as false parents [26]. Decoder geometry adds *geometric twins*, latents of model (2) at decoder cosine ≥ 0.6 to a parent of model (1) that fire where it rarely does ($\Pr(j \mid j') < 0.2$), and each model's nearest-neighbor decoder-cosine distribution vs. a random baseline. The code calls twins "absorption candidates", but they are not absorption in the sense of Chanin et al. [25], a within-model failure scored against a ground-truth probe, in which the absorbing latent's decoder carries only a small component of the general direction. For two same-architecture runs differing only in seed, the match fractions are the reproducible-feature fractions. For hard-code Ontologizers, `partition.py` matches heads across runs by the NMI of their partitions (what a head separates), and `headcontrib.py` by the cosine of their contributions (what a head writes): each head's decoded output contribution in the whitened frame, less $\text{dec}(\mathbf{0})$, centered over rows — uncentered, the non-negative dictionary's shared offset reads as agreement. With $q_{ni}^{(1)}$ the entry head i of run (1) selects on row n, $\mathbf{q}_i^{(1)}$ those selections over all rows, and $\boldsymbol{\phi}_{ni}^{(1)}$ the head's whitened decoded contribution on row n,

$$\begin{aligned} [M^{\text{part}}]_{ii'} &:= \text{NMI}(\mathbf{q}_i^{(1)}, \mathbf{q}_{i'}^{(2)}), \\ [M^{\text{contr}}]_{ii'} &:= \cos \angle\big(\text{vec}_n[\boldsymbol{\phi}_{ni}^{(1)} - \bar{\boldsymbol{\phi}}_i^{(1)}],\ \text{vec}_n[\boldsymbol{\phi}_{ni'}^{(2)} - \bar{\boldsymbol{\phi}}_{i'}^{(2)}]\big), \end{aligned} \tag{29}$$

where $\text{NMI}(\mathbf{q}, \mathbf{q}') := 2I(\mathbf{q}; \mathbf{q}')/(H(\mathbf{q}) + H(\mathbf{q}'))$ over the rows' empirical distribution, $\bar{\boldsymbol{\phi}}_i$ is the mean over rows, and each cosine runs over the contributions concatenated across rows. Each matrix is matched one-to-one (Hungarian) by the pairing of heads that maximizes its summed entries, and the reported agreement is the mean over matched pairs. The null repeats this with run (2)'s rows shuffled, and the positive control matches the run against its own earlier checkpoint. `headcontrib.py`

also splits each head's contribution size into its layer's input gain and its own decoded atoms. Since a head's contribution on row n is its layer's gain g_n times the decoded atom $\mathbf{a}^{\text{dec}}_{i,q_{ni}}$ of the entry it selected,

$$\text{size}_i := \frac{\sum_n \|\boldsymbol{\phi}_{ni} - \bar{\boldsymbol{\phi}}_i\|^2}{\sum_n \|\mathbf{w}^{1/2} \odot (\mathbf{x}_n - \bar{\mathbf{x}})\|^2} \approx \frac{\mathbb{E}_n[g_n^2]\, \mathbb{E}_{j\sim\bar{\mathbf{p}}_i} \|\mathbf{a}^{\text{dec}}_{ij} - \bar{\mathbf{a}}^{\text{dec}}_i\|^2}{\mathbb{E}_n \|\mathbf{w}^{1/2} \odot (\mathbf{x}_n - \bar{\mathbf{x}})\|^2}, \tag{30}$$

with $\bar{\mathbf{p}}_i$ head i's usage over entries (its mean assignment) and $\bar{\mathbf{a}}^{\text{dec}}_i$ its usage-weighted mean decoded atom: gain squared times atom spread squared, up to the gain's small variation. With a reference seed pair (a flat arm) it compares agreement at matched head size.

Text fidelity (`textfid.py`). The "loss recovered" analogue: decode $\mathbf{x}$ and recon($\mathbf{x}$) through M2M100 (forced `eng_Latn`, both rescaled to SONAR's median mean-pooled norm over the corpus, 0.307, so the metric sees direction error; SONAR's own vectors are far from unit norm, and its decoder is sensitive to their scale [72]; every decode in this document uses that scale, near which decode-then-re-encode fidelity peaks) and report chrF2 [73, 74] (character 1–6-grams, whitespace removed, recall weighted twice precision; sentence level, averaged over rows on a 0–1 scale, equal to sacreBLEU's sentence chrF2 [75] divided by 100) and exact-match between the decodes, plus the decoder's teacher-forced per-token NLL of the *original's* decode under the original (ceiling), the reconstruction, and the corpus-mean embedding (floor — which decodes to the generic text). Loss recovered places the reconstruction between the two,

$$\text{recovered} := \frac{\text{NLL}_{\text{floor}} - \text{NLL}_{\text{recon}}}{\text{NLL}_{\text{floor}} - \text{NLL}_{\text{ceil}}}, \tag{31}$$

the form GPT-2's loss recovered takes with cross-entropy in place of NLL (Section 6). Under the steering frame this doubles as the collateral ceiling: a code that cannot preserve text cannot steer text with low collateral.

Language probing (`langprobe.py`, `headlang.py`, `headscript.py`). k_{probe}-sparse probing after SAEBench [37], with the free 86-language labels in place of its balanced tasks: top latents per language by two-sample t-statistic on training rows; held-out one-vs-rest F1 at k_{probe}=1 (single latent, fitted threshold), k_{probe}=4, 16 (logistic probes), and a dense probe on the raw embedding as ceiling. Macro-F1 over languages. *Frame caveat: this measures whether a factor lives in individual latent magnitudes. A classifier head carries its factor in its assignment (partition), which single-latent probes cannot see. Distribution-based monosemanticity scores [59], which credit a unit for responding to many concepts if its responses separate them, are the per-unit measure that transfers to heads.* `headlang.py` is the matching instrument: per head, the NMI between its partition and the label, null-corrected by a label shuffle, plus

a held-out probe on the best h' heads' one-hot codes against a dense ridge ceiling. NMI here normalizes by the arithmetic mean of the two entropies, so its attainable maximum is $2\min(H_{\text{head}}, H_{\text{lang}})/(H_{\text{head}} + H_{\text{lang}})$ — 0.875 for a 5-bit head against the 6.43-bit language label. `headscript.py` repeats this for script and scores per-label cells and cross-head conjunctions; `headeta.py` and `topichead.py` measure each partition's share of embedding variance,

$$\eta^2(C) := \frac{\sum_{C_c \in C} |C_c| \, \|\bar{\mathbf{x}}_{C_c} - \bar{\mathbf{x}}\|^2}{\sum_n \|\mathbf{x}_n - \bar{\mathbf{x}}\|^2}, \qquad \mathbb{E}_{C'}\left[\eta^2(C')\right] \approx \frac{|C| - 1}{n_{\text{rows}} - 1}, \tag{32}$$

for a partition C of n_{rows} rows into cells C_c, with $\bar{\mathbf{x}}_{C_c}$ the mean of the rows in cell C_c, and report it raw and against the baseline, its mean over random partitions C' into $|C|$ cells.

Shrinkage (`refit.py`). The refinement of Gao et al. [22] without its non-negativity constraint: freeze each sample's support $\mathcal{S}_n$ and solve the whitened least-squares coefficients on it,

$$\mathbf{z}_n^\star := \operatorname*{arg\,min}_{\mathbf{z}:\, z_j = 0 \;\forall j \notin \mathcal{S}_n} \left\| \mathbf{w}^{1/2} \odot \left(\mathbf{x}_n - \mathbf{b} - W\mathbf{z}\right) \right\|^2, \tag{33}$$

where the columns of W are the code's decode directions and $\mathbf{b}$ its offset, by masked normal equations (silent head slots forced to zero), and compare FVU_w. Raw−refit is magnitude (shrinkage) error; refit is selection error. The Ontologizer shares the machinery via its exact linearity $\hat{\mathbf{x}} = W_{\text{flat}}\mathbf{p}_{\text{flat}}$, with W_{flat} measured by decoding one-hot constant codes; its supports are the top-ν deviation entries. Points with support $\geq d$ are excluded (underdetermined). Note the raw convention: `refit.py` truncates the soft forward's P (non-adaptive), whereas `pareto.py` truncates inside the forward, letting later layers reclassify against the truncated prefix residual.

Steering fidelity in text (`steerfid.py`). Unit-normalized native steering directions — SAE decoder directions, bilinear eigenfeatures, or the Ontologizer's `withArgs` set-intervention delta — applied at matched embedding magnitudes. Effect is cycle consistency: the steered decode is re-encoded through SONAR and the target feature's activation gain is normalized by its firing-conditioned quantiles, with the baseline taken from the *cycled* base decode so both sides share roundtrip degradation. Collateral is $1 - \text{chrF}$ against the base decode. Random unit directions give the control curve, and *net* effect is feature effect minus that control.

Steering fidelity in embedding space (`steerembed.py`). Drops the text round trip and asks the causal question directly: move the input along a direction, re-run the model, and record whether the intended entry is now selected (*realization*) and what fraction of the *other* heads changed their selection (*collateral*). Directions: `decode` (the decode delta of forcing the entry), `grad` (the gradient of the target logit), `margin`

(the gradient of the target's margin over the head's current winner), `adjoint` (the bilinear classifier's top eigenvector via `BilinearBlock.rev`), and `random`. Strength is a fraction of each row's norm. The *native* step is the model's own intervention size: the output change of forcing the entry, or for an SAE latent its decoder direction at the latent's mean nonzero activation. Algorithm 4 gives the procedure.

Eigenvector orientation. The adjoint direction of a bilinear classifier is an eigenvector of its interaction matrix, which `eigh` returns only up to sign, since a bilinear eigenvector contributes through the square of its projection [52]. Orienting it by the sign of its eigenvalue, with a leading-coordinate tie-break (`nlinear.orient`), realizes 0.427 of targets on the softmax model at strength 0.25, against 0.224 with the sign `eigh` happens to return and 0.446 when each sample is oriented separately. A weight-only convention should not be able to do that: a step of α along $\mathbf{v}$ changes $\mathbf{x}^\top B\mathbf{x}$ by $2\alpha\lambda(\mathbf{x}\cdot\mathbf{v})$, a property of the sample. It can on SONAR because the embedding is anisotropic, so $\mathbf{x}\cdot\mathbf{v}$ is dominated by $\bar{\mathbf{x}}\cdot\mathbf{v}$, which has a consistent sign; on isotropic Gaussian inputs a weight-only direction raises the logit on about half the samples. On the hard code the oriented eigendirection is below random at every strength.

Linearized collateral (`steergeom.py`). Separates the direct effect of a step on the other heads of the target's layer from the cascade through later layers, by linearizing that layer's logits with every upstream code held fixed. Under residual forwarding an input step $\mathbf{v}$ moves the layer's residual input by exactly $\mathbf{v}$, so with J_i the Jacobian of head i's logits at the layer input $\mathbf{e}_n$, head i's winner–runner-up margin and its first-order change along a step of $\alpha\|\mathbf{x}_n\|$ are

$$\mu_{ni} := [\mathsf{K}]_{nij_1} - [\mathsf{K}]_{nij_2}, \qquad \Delta\mu_{ni} := \alpha\,\|\mathbf{x}_n\|\left(\left(J_i\mathbf{v}_n\right)_{j_1} - \left(J_i\mathbf{v}_n\right)_{j_2}\right), \tag{34}$$

with K the logits and j_1, j_2 the head's top two entries on row n. With $i^\star$ the target's head, the linearized flip share is $\mathbb{E}_{n,\,i\neq i^\star}\,\mathbf{1}[\mu_{ni} + \Delta\mu_{ni} < 0]$ at α=0.25, reported for the `decode`, `grad` and `random` directions with the median margin μ. Two further measures use the same Jacobians: *off*, the mean $\|\mathbf{x}_n\|\,|J_i\mathbf{v}_n|$ over other heads and entries as a ratio to the random direction's, which is below 1 when a direction avoids the other heads' detectors; and *own*, the median first-order gain of the target over the incumbent per unit of $\|\mathbf{x}_n\|$, divided by the gap it has to close.

Intervention composition (`compose.py`). Treating heads as candidate causal variables: for pairs of forced assignments $(\ell_1, i_1)\to j_1$, $(\ell_2, i_2)\to j_2$ (8 random heads per layer, one random entry each; all same-layer pairs plus 60 cross-layer pairs), compare the joint intervention's decode delta $\boldsymbol{\delta}_{12}$ against the sum $\boldsymbol{\delta}_1+\boldsymbol{\delta}_2$ of the single-intervention deltas on 512 held-out rows: median per-row cosine and relative error $\|\boldsymbol{\delta}_{12} - (\boldsymbol{\delta}_1+\boldsymbol{\delta}_2)\|/\|\boldsymbol{\delta}_1+\boldsymbol{\delta}_2\|$, raw and whitened. Because the decode

Algorithm 4: Steering scored in embedding space (`steerembed.py`). For an Ontologizer a target $j^\star$ is an entry, of some head i in some layer ℓ, and a choice is a head's argmax; for an SAE with heads a target is a latent and a choice is its head's winner (a top-1 head that abstains counts as its own outcome); for an SAE without heads a target is a latent, it is selected when active, and collateral is the share of the row's other active latents that drop out.

Input: model; held-out rows X; strengths $\alpha \in \{0.25, 0.5, 1, 2, 4\}$; 64 targets per model, stratified by layer, each selected on at least 0.5% of reference rows

Output: realization and collateral per direction and strength, and at the native step

begin

  foreach *target* $j^\star$ **do**

    $X_{j^\star} \leftarrow$ 32 rows on which $j^\star$ is *not* already selected

    record every head's choice on $X_{j^\star}$

    foreach *direction* dir $\in$ {`decode`, `grad`, `margin`, `adjoint`, `random`} **do**

      $\mathbf{v}_n \leftarrow \mathrm{dir}(\mathbf{x}_n, j^\star)/\|\mathrm{dir}(\mathbf{x}_n, j^\star)\|$ `// unit direction per row`

      foreach *strength* α **do**

        $\mathbf{x}'_n \leftarrow \mathbf{x}_n + \alpha\,\|\mathbf{x}_n\|\,\mathbf{v}_n$; $\mathbf{x}'_n \leftarrow \mathbf{x}'_n\,\|\mathbf{x}_n\|/\|\mathbf{x}'_n\|$ `// rescale to` $\|\mathbf{x}\|$

        recompute every head's choice on the steered rows

        realization $\leftarrow \mathbb{E}_n\,\mathbf{1}[j^\star$ selected on $\mathbf{x}'_n]$

        collateral $\leftarrow$ share of (row, other head) pairs whose choice changed

      end

    end

    $\boldsymbol{\delta}_n \leftarrow \hat{\mathbf{x}}_n(j^\star$ forced$) - \hat{\mathbf{x}}_n$, or, for an SAE, the latent's decoder direction times $\bar{z}_{j^\star}$, its mean activation when it fires `// native step`

    score $\mathbf{x}_n + \boldsymbol{\delta}_n$, unrescaled, and the control $\mathbf{x}_n + \|\boldsymbol{\delta}_n\|\,\boldsymbol{\xi}_n$ for a random unit $\boldsymbol{\xi}_n$

  end

  return *the means over targets*

end

is exactly linear in the flattened classification, final-layer pairs compose exactly by construction; that stratum doubles as the measured numerical floor (the $\|\mathbf{r}\|\approx 31 \to \|\hat{\mathbf{x}}\|\approx 1$ decoder cancellation amplifies float32 rounding to ~0.02 relative error between differently-compiled variants — verified exact in float64). All other non-additivity is downstream reclassification: later layers re-reading the intervened residual.

Autointerp (`autointerp.py`). Four stages: harvest (top-activating rows, negatives, frequency-stratified feature selection, parameter-based decode embeddings), texts (deterministic mC4 stream replay recovers text by row index, verified against the language sidecar), describe, and score. Description modes: `params`, the M2M100 decode of a parameter-derived embedding (for entries, the origin $\mathbf{o}$ with one head forced one-hot; for SAE latents, $\mathbf{b}_{\text{dec}} + \alpha\, \mathbf{w}_j^{\text{dec}}$); `pdev`, the same decode with the pure-origin decode subtracted; `eig`, the bilinear eigenfeature decode; `acts`, an LLM summary of top-activating snippets; and `cacts`, a *contrastive* summary — the judge sees top snippets next to random corpus draws and must name what distinguishes them, blocking the degenerate corpus-level answer that plain summaries invite on noisy multilingual web text. Scoring is blinded detection over held-out positives and negatives, identical for all modes; per-feature F1, to be read against the always-match null of F1 ≈ 0.67 on balanced items. The judge is Haiku (`claude-haiku-4-5`), which also writes the `acts` and `cacts` descriptions, so describer and judge are one model and no second judge or human rater checks its decisions. It runs via the Anthropic Messages API, rate-metered, with bounded reply length; failed judgements are never recorded (an earlier outage mode scored empty replies as F1 = 0). Six models: the Ontologizer, rungs `k32`/`k32_bl`/`g160top1`/`g160softmax`, and the dense control `k5120`.

Modularity (`headstruct.py --modularity`). A head's assignments split the samples into soft cells, and soft modularity — Newman's Q [76] with the co-membership indicator replaced by the dot product of membership vectors, the fuzzified modularity of [77, Eq. 14] — scores how much more affinity those cells hold than random placement with the same degrees would put there. Per batch of 4096 rows, with edge weights $\chi_{nn'}$ (the heat kernel $\exp((\cos\angle(\mathbf{y}_n, \mathbf{y}_{n'}) - 1)/0.2)$ between the samples' inputs $\mathbf{y}_n$, self-edges removed, kept for each sample's 15 nearest neighbors and symmetrized by union), degrees $\chi_n = \sum_{n'} \chi_{nn'}$ and total $\chi = \sum_n \chi_n$, head i scores

$$Q_i = \frac{1}{\chi} \sum_{n,n'} \left(\chi_{nn'} - \frac{\chi_n \chi_{n'}}{\chi} \right) \mathbf{p}_{ni} \cdot \mathbf{p}_{n'i}, \tag{35}$$

with the graph and degrees restricted to the samples the head fires on, so a sparse group is not penalized for its coverage. That scores cells as communities of the induced subgraph, where Newman's subdivision keeps the parent graph's degrees [76]. An Ontologizer layer is scored on its own input: $\mathbf{x}$ at layer 0, the

residual it classifies above that. A Matryoshka SAE's prefix block ℓ is scored on its prefix residual, $\mathbf{x}$ less the decode of the code on blocks before ℓ (so block 0 on $\mathbf{x} - \mathbf{b}_{\text{dec}}$), and a flat model on $\mathbf{x}$. An SAE group's cells are its members' activations normalized within the group. A layer's mean Q_i is set against 20 size-matched nulls that regroup the same entries within the layer or block, over cache rows 524,288–1,048,575 (training rows; SAE groups are discovered on the rows before them).

Classifier geometry (`bilinspec.py`). With the classifier as trained (order 2, no gate), each logit is one bilinear pair over the layer's shape, $(\mathbf{w} \cdot \mathbf{u})(\mathbf{v} \cdot \mathbf{u}) = \mathbf{u}^\top B \mathbf{u}$ with $B = (\mathbf{w}\mathbf{v}^\top + \mathbf{v}\mathbf{w}^\top)/2$ the symmetric interaction matrix of a bilinear unit [52]. B has rank at most 2, with eigenvalues $\lambda_\pm = (\mathbf{w} \cdot \mathbf{v} \pm \|\mathbf{w}\|\|\mathbf{v}\|)/2$ on eigenvectors $\mathbf{w}/\|\mathbf{w}\| \pm \mathbf{v}/\|\mathbf{v}\|$, so every scale-free statistic of a feature's spectrum is a function of $\cos\angle(\mathbf{w}, \mathbf{v})$. The script reads them from the weights at retained checkpoints, on the semantic block of $\mathbf{u}$ (the constant coordinate excluded), and reports per layer the median and quartiles of $|\cos\angle|$, the share of features whose dominant eigenvalue is negative, and the constant coordinate's weight relative to the rest of $\mathbf{w}$. The closed-form spectra agree with a dense eigendecomposition to 10^{-14}. No data enter.

Assignment maps (`assignmap.py`). A stratified draw of eval-tail rows, a fixed number per label value (language from the cache's sidecar; for GPT-2 the coarse class of the current token: word, capitalized word, continuation, punctuation, digit, newline or space), is drawn against each head's k entries: rows grouped by label and ordered within a group by their winning entries, entries by mean usage. Each head is summarized by its "use" and "row" (Figure 4). With an SAE, the same rows in the same order are drawn against its latents, each cell a latent's activation as a share of the row's total, latents by usage; a trained group is drawn as a head, and a flat code as one unit whose use and row are computed the same way.

Entry drift (`entrydrift.py`). Across a run's retained checkpoints, each entry's usage $\bar{p}_{ij}$ on the eval tail, at that checkpoint's training temperature, and the cosine between its decoded atom $\mathbf{a}_{ij}^{\text{dec}}$ (the one-hot code decoded, less the decode of the zero code) at that checkpoint and at the last one. An entry with $\bar{p}_{ij} < 1/4k$ is out of use. Such an entry receives no gradient, so its direction freezes and its cosine reads near 1 without meaning anything; drift is summarized over the entries in use at the last checkpoint.

Label enrichment (`enrich.py`). Every eval-tail row is assigned, per head, to its winning entry. For each (entry, label value) cell, the number of rows taking the entry and carrying the label follows a hypergeometric distribution given the two margins if entry and label are independent. The test is two-sided (the tail on the

observed count's side of its expectation, doubled), a cell is tested only if its count or expectation reaches 5, and p-values are Benjamini–Hochberg corrected across every tested cell of every head. The effect size is the cell's 2×2 $\log_2$ odds ratio with 0.5 added to each count. Per head, the NMI between the winning entry and the label (mutual information over the mean of the two entropies), less its mean over five label-shuffled draws, measures how much of the partition the label explains.

Head-partition NMI and effective information (`headnmi.py`, `effinfo.py`). Every pair of heads' winning-entry partitions of the eval tail is compared by NMI, adjusted for chance as $(I - I_0)/(\bar{H} - I_0)$, with I the mutual information, $\bar{H}$ the mean of the two entropies and I_0 the Miller–Madow approximation to the information two independent partitions with the same numbers of used entries share; a permutation null agrees with I_0 to within 10%. 0 is chance and 1 the same partition. A head using fewer than 1.1 effective entries carries no partition and is left out. Effective information [69] from a source head to a later effect head is the mutual information between an intervention that forces the source to one of its entries, uniformly over the entries it uses on the eval tail, and the effect head's resulting choice, over 4,096 eval-tail contexts. On a hard code those entries are the source's natural states, so the intervention stays on distribution.

Selection surfaces (`selection.py`). Each head's assignments P_i on a batch of 2,048 eval-tail rows define the neighbor graph of Figure 4b on the layer's input Y_ℓ: $P_i P_i^\top$ times the heat kernel of the inputs, self-edges removed, rows normalized, N_i. The head is scored by how well its co-classified neighbors predict each row, $\|Y_\ell - N_i Y_\ell\|^2 / \|Y_\ell - \bar{Y}_\ell\|^2$ with $\bar{Y}_\ell$ the batch mean, the leave-one-out score that $\mathrm{L2}_{\mathrm{pwak}}$ trains (Eq. (22)), on the weighted affinity kernel of DEWAKSS [78]. It follows noise2self [79] without that loss's guarantee: removing self-edges keeps a row's value out of its own prediction, but the row's input still sets its weights, through its assignment and through the kernel, so the score is not $\mathcal{J}$-invariant in noise2self's sense. 0 means the neighbors predict a row exactly, 1 that the graph does no better than the mean. Two baselines share the run and rows: every row in one cell (the plain heat-kernel smoother) and the rows permuted, a random partition with matched cell sizes that is independent of the input (three draws per batch). Scores are averaged over heads and batches. A $\mathcal{J}$-invariant split form (`--jinv`) centers each batch's input, rotates it at random and cuts it in half: one half builds every gate and kernel, the other is predicted, so no row's predicted coordinates set its own weights. Under it the bandwidth is chosen as the one whose plain smoother scores lowest, as DEWAKSS chooses its own [78], and the null for a partition is random-projection heads, each putting a row in the cell of its largest projection on k Gaussian directions: singly, as the joint gate of eight, and as the joint gate of 64, which estimates their collision kernel.

Diagnostic scripts (`experiments/ste-arm/`). The straight-through and geometry results rest on further single-purpose instruments, described where used: `inputgeom.py` (geometry of the vector each classifier sees), `blendablate.py` (severing the pairing between a layer's input and the sample's own earlier reconstruction), `gainablate.py` (the norm side-channel), `dictgeom.py` (row collinearity and effective rank read from the weights), `lastlayer.py` (final-layer ablations against the deep-supervision prefixes), `decodehead.py` (decoding a head's atoms as deviations from their mean), `embedgeom.py` (SONAR cloud geometry), `entryshare.py` (how evenly a layer's entries contribute to the output, and whether usage, gain or atom norm drives the inequality), and `bigatoms.py` (how a layer's large atoms are distributed among its heads, and whether the heads holding them reproduce).

C Supplementary results: the softmax Ontologizer against SAEs

C.1 Categorical form

At 1M rows with split-half scoring and 20 size-matched nulls: `g160top1`'s trained partition is *perfectly* head-like (exhaustiveness 1.0000 vs. null 0.64; exclusivity 1.0000 vs. 1.51; sum-CV 0.25 vs. 1.03). Its heads have the form of the feature groups Huang and Hahn hypothesize, whose members exclude one another [29]. Their toy groups are also absent part of the time, which `g160top1` can represent (a head whose winning logit is not positive gets a router scale of zero) and a head without a router cannot, but at exhaustiveness 1.0000 it practically never does. Discovery on every unstructured run (`m11264_k32`, `m5120_k32`, `k32_p5`, `m11264_k160`) finds the *opposite* signature — exhaustiveness below null (ζ from −30 to −50), exclusivity above (ζ +13 to +31): spontaneous SAE group structure consists of co-firing topical clusters, not categorical slots, even when the affinity explicitly selects for anti-correlated same-context pairs. And `g160softmax` never fires distinctly at all: no coefficient ever exceeds 2/32 on any test row — soft per-head codes converge to dense near-uniform mixtures, exactly as the Ontologizer's soft code does (**o** sits 0.06–0.16 bits from uniform in healthy layers; matching soft reconstruction requires ∼16 of 32 deviations per head). **Categorical structure exists precisely when trained in as hard selection, never emerges from softmax competition under a reconstruction objective, and costs ∼60× in FVU_w (0.274 vs. 0.0044) at the SAE's 160 heads.** For steering this means the *soft* architectures' classifier reading must be cashed out in deviations around **o**, not in discrete winners — or the selection must be made hard during training, which Section 5 does at a far smaller cost once the head count is raised. Figure 15 shows one such soft head of `resid_nc`.

Trained in as hard selection, the categorical form shows row by row (Figure 16, `assignmap.py`). On both substrates the straight-through stack's layer-0 heads put each row on one entry (row 1.0) and spread the rows over 14–30 effective entries (use), and rows that share a label tend to share entries: most Cebuano rows sit on one entry of each of the three SONAR heads shown, and most GPT-2 newline and

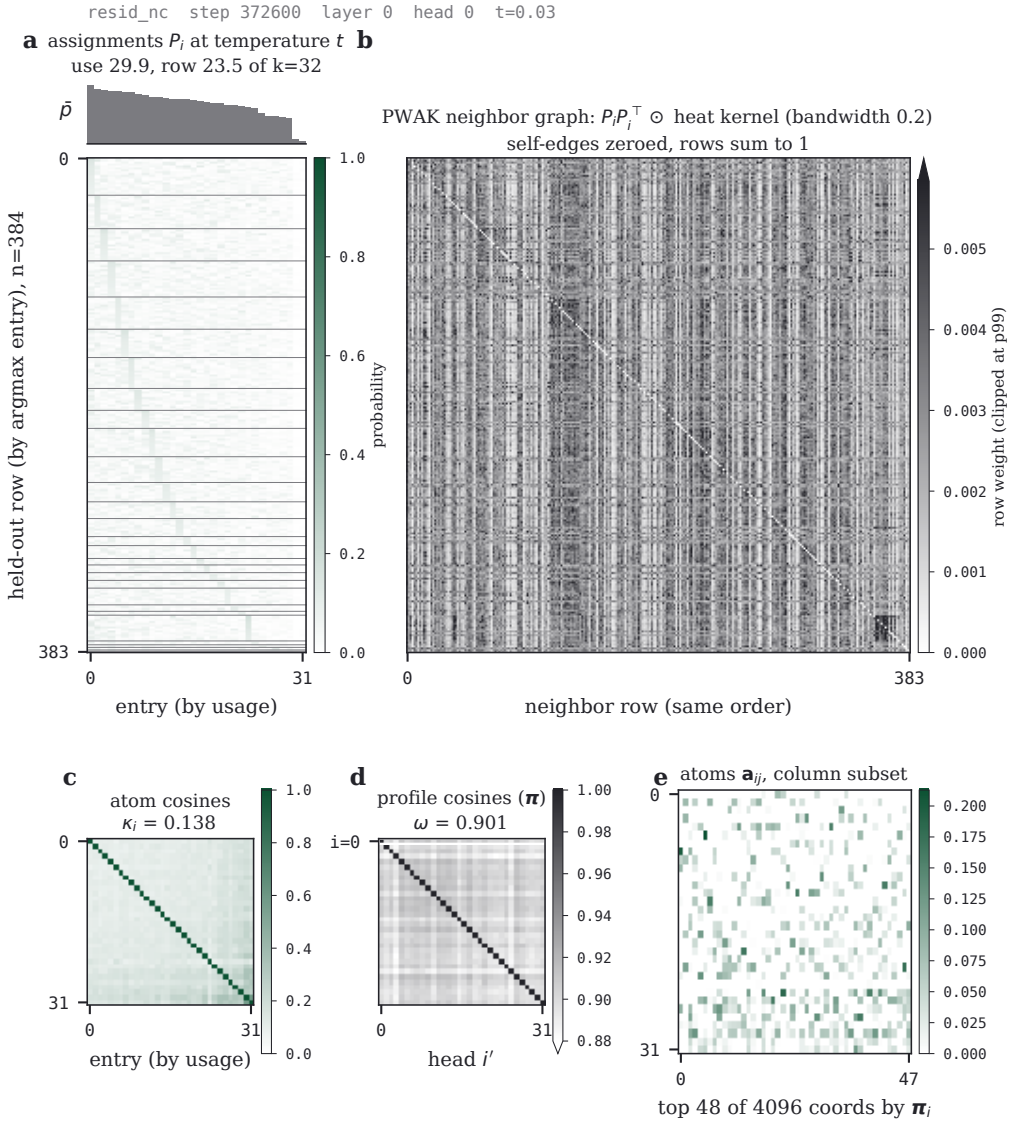


FIGURE 15 · **One softmax head, block by block.** The blocks of Figure 4 for head 0 of layer 0 of `resid_nc` (step 372,600, t=0.03) on n=384 eval-tail rows; one seed. Usage is spread across entries (use 29.9 of k=32), but each row is a near-uniform mixture (row 23.5; a), so the neighbor graph (b) is close to the heat kernel alone, with hub columns rather than blocks. κ_i=0.138, ω=0.901, e=4096. As in Figure 4, the model was not trained with the statistics that use the graph.

space tokens on one entry per head. An SAE on the same rows spreads each over 25–29 effective latents of its k_{SAE}=32 active and the shown rows over 1,066–1,533; a label shows there as a set of co-firing latents (GPT-2's newlines), not as one.

C.2 Semantic coherence in detail

Under the whitened metric the training objective optimizes, the Ontologizer's mean $\zeta_{\cos}$ is −24.4 (−17.6 raw; the fraction of heads below −2 is unchanged at 0.96), the single-layer soft variant's −14.7 (unchanged) and the hard variant's +179.3 (+161.3 raw).

Two heads of the reference softmax model cluster instead of contrasting: layer 2's head 25 ($\zeta_{\cos}$=+346) and layer 3's head 28 (+280), with the next candidates in the partially frozen layer 4. Their sampled entries also decode to generic text, which suggests partial head collapse (entries that never differentiated) rather than semantic coherence; they have not been inspected further.

The description-text view of coherence is uninformative on plain summaries, which are generic for about 77% of entries and 49% of k_{SAE}=32 latents and so inflate the permutation null (Ontologizer heads ζ=+0.14). On the contrastive descriptions it acquires signal. Within-head description similarity clears the head-membership permutation null for Ontologizer heads (ζ=+2.03, 155 within-head pairs), is marginal for the single-layer hard variant (+1.55, 62 pairs), and is weaker still for the single-layer soft variant (+1.09, 217 pairs). Against the geometry, this is the pattern that alternative values of one shared dimension predict: Ontologizer entries' decode directions repel while their described contents weakly cohere. It is also the first measure to separate the Ontologizer from its single-layer soft rival, though only suggestively at this coverage (one global ζ per model over the campaign's sampled features).

C.3 Heads as communities of their input

A head partitions the samples it reads. Whether the partition is a community structure of that input, with more affinity inside its cells than a degree-preserving null graph puts there, is soft modularity (Eq. (35)), compared with size-matched regroupings of the same entries (Figure 17). The softmax Ontologizer is `bl_ctl_full`; the SAEs are the Matryoshka rungs `m5120_g160top1_p5` (trained heads) and `m5120_k32_p5` (discovered groups) and the flat `m5120_g160top1` and `m5120_k32`.

On the input itself nothing is a community: the Ontologizer's layer 0 is 1.25 times its null and every SAE 0.76–1.06. On each layer's own input the Ontologizer's layers 1–4 are 2.1–2.9 times as modular as regroupings of their entries. A Matryoshka SAE scored on matched prefix-residual graphs is not, whether its heads are trained (0.96–1.06) or discovered (0.88–1.03 in the blocks they cover). So the structure is not a property of residual graphs, nor of trained heads in a coarse-to-fine dictionary.

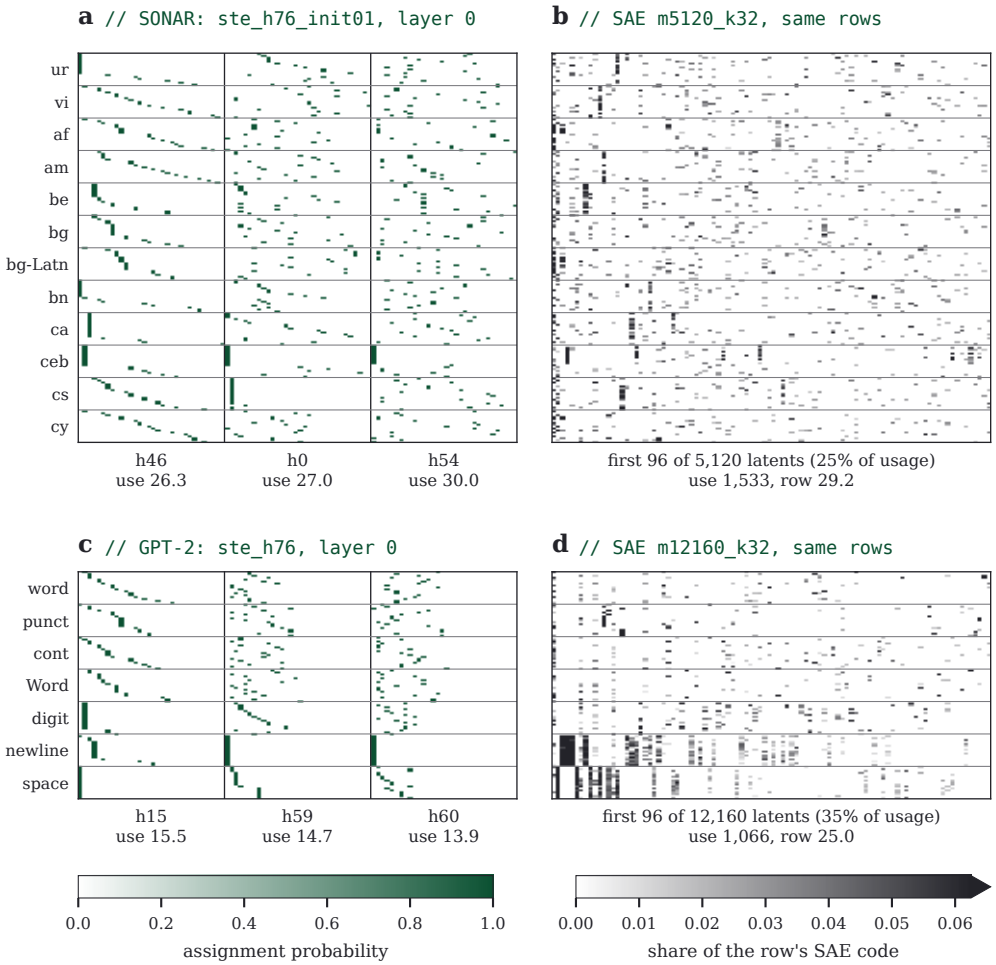


FIGURE 16 · **Hard codes against SAE codes, row by row.** Layer-0 assignments of the straight-through stacks on eval-tail rows, 24 per label value: (a) SONAR `ste_h76_init01` by language, (c) GPT-2 `ste_h76` by token class, for the three heads whose partitions carry the most NMI with the label above its shuffled null. Entries are sorted by usage, rows grouped by label and ordered by their winning entries; each head's use is under it, and its row is 1.0. (b, d) The same rows in the same order against an SAE trained on the same cache (`m5120_k32`, `m12160_k32`), each cell a latent's share of the row's code, for the 96 latents most used on these rows; the whole code's use and row are under it. One seed per model.

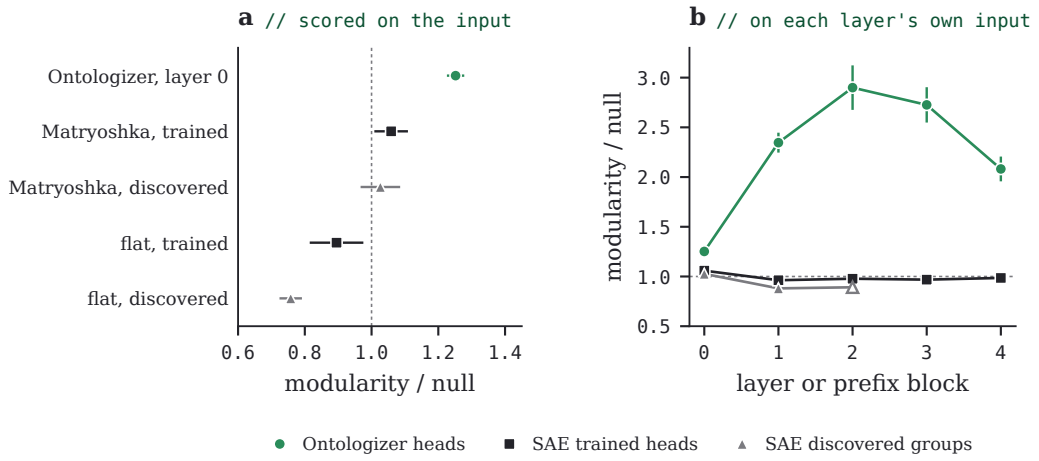


FIGURE 17 · **Ontologizer heads are communities of the residual they classify; SAE heads and groups are not.** Soft modularity over its size-matched null (`headstruct.py --modularity`, Eq. (35)); bars are ±2 null standard deviations. (a) Every model scored on the input itself: the softmax Ontologizer's layer 0, block 0 of the two Matryoshka SAEs (scored on $\mathbf{x} - \mathbf{b}_{\text{dec}}$, the SAE encoder's input), and the two flat SAEs. (b) By depth, each layer or prefix block on the graph of its own input. Hollow: a block whose groups cover under 10% of samples. The discovered groups' blocks 3 and 4 are not drawn: block 3's null is within two of its standard deviations of 0 (3% coverage), and block 4 holds one group, which its null reproduces exactly. One checkpoint per model.

What differs is that an Ontologizer layer classifies the residual it is scored on, while a Matryoshka block is encoded from the input and only trained to reconstruct that residual. Raw modularity rises with depth in real and null alike (layer 4: 0.218 against 0.105), as assignments sharpen and the residual graphs cluster, which is why the ratio is the comparable number. Low values on the input are what a factorial code predicts: a head that encodes one of many independent factors cuts across the input's neighborhoods, since neighbors share most factors and not necessarily this one. One checkpoint per model.

C.4 Feature reliability

Seed stability (`k32` vs. `k32_s43`; identical architecture and data, FVU_w 0.4994 vs. 0.4967): matched-feature fractions are **0.31 / 0.24 / 0.15 / 0.03** at τ=0.3/0.5/0.7/0.9. Roughly three quarters of individual features are not reproducible across initializations at moderate containment. Cross-width splitting (`k32` → `m11264_k32`): 190 of 5114 live parents split at τ=0.7 (median 3 children; union coverage 0.37; child overlap 0.42), with 441 geometric twins and a heavy nearest-neighbor decoder-cosine tail (p50 0.234 vs. random 0.119): splitting is real but is not the dominant organization; topical co-firing is. For a steering generator the unit that must reproduce is the head's partition rather than the individual feature vector; Section 5.5

measures that on the hard-code Ontologizer and finds it reproduces far *worse* than SAE features do.

C.5 Text-space fidelity (collateral ceiling)

model	chrF	exact	NLL (ceiling 1.560, floor 3.999)
corpus-mean floor	0.176	0	3.999
k32 tier (incl. p5, m11264)	0.23	0.00	3.16–3.18
k32_bl	0.204	0.00	3.365
k160 / g160top1	0.27–0.28	0.00	2.41–2.43
onto soft	0.552	0.070	1.585 (gap 0.025)
g160softmax	0.591	0.094	1.568 (gap 0.008)
k5120	0.841	0.562	1.560 (gap 0.000)

TABLE 14 · Downstream text fidelity, 512 tail rows, forced English decodes. The NLL ceiling is the same for every model, the original's decode scored under the original.

The FVU–text mapping is sharply nonlinear: $\mathrm{FVU}_w \approx 0.5$ decodes to same-topic-different-facts text barely above the generic floor, while the Ontologizer's soft code and `g160softmax` are near the likelihood ceiling — almost all decodable content survives. Across the eleven Ontologizer runs with both numbers (the selection sweep and the straight-through arms; Table 16), "loss recovered" is a monotone function of FVU_w (rank correlation -1), so text fidelity is a check that the whitened objective transports to text, not an independent axis — and it does. This matters doubly under the steering frame: a steering primitive that routes through a code which cannot preserve text (the entire $k_{\mathrm{SAE}} \leq 160$ sparse tier) starts from ~0.75 collateral before any intervention, whereas the dense-mixture classifier codes start near zero.

C.6 Concept (language) localization

Language remains distributed in every model (best single-latent macro-F1 0.121 against a dense ceiling of 0.676; script-distinctive languages such as Chinese reach ~0.98 from one latent, same-script families smear; Table 15). The softmax Ontologizer's entry probabilities localize nothing: at one and four entries their F1 is the always-match floor. The hard codes' entries, which are their assignments, localize at one entry as well as the best SAE (0.121 and 0.120 against 0.119), though less at 4 and 16. Read on SAE terms the soft code's floor is an indictment; read on classifier terms it is expected: a soft head encodes a factor in its *assignment*, not in any single entry's probability, and partition-level analysis of `resid_nc` found layer-1 heads whose partitions carry script/family mutual information, one of which ((L1,h8,k17)) was causally verified to force Burmese output. The partition-level instrument (`headlang.py`) was later run on a config-matched softmax reference and on the hard-code arms (Section 5.6): the softmax reference's best head reaches a

	86 languages			five languages, F1@1		
model	F1@1	F1@4	F1@16	own tags	it pt nl ja	ca ro sv ko
k160	0.119	**0.393**	**0.473**	0.684	0.841	0.841
k32 (m=5120)	0.089	0.347	0.465	0.624	0.868	0.876
m11264_k32	0.078	0.364	0.471	0.551	0.860	0.871
k32_bl	0.084	0.228	0.368	0.695	0.840	0.786
k32_p5	0.061	0.322	0.456	0.625	0.862	0.870
g160top1	0.066	0.207	0.302	0.592	0.542	0.700
g160softmax	0.067	0.162	0.328	0.432	0.437	0.433
k5120	0.064	0.130	0.237	0.439	0.448	0.447
softmax Ontologizer (entries)	0.022	0.022	0.135	0.333	0.333	0.333
hard-code stack (entries)	**0.121**	0.153	0.362	0.685	0.804	0.806
hard-code flat (entries)	0.120	0.141	0.319	0.609	0.798	0.703

TABLE 15 · Language localization (`langprobe.py`): one-vs-rest F1 of the single best latent and of logistic probes on the best 4 and 16, on held-out tail rows. Left, over all 86 languages, where always-match scores 0.023 and a dense probe on the embedding 0.676. Right, single-latent macro F1 within three five-language populations, where always-match scores 0.333: English, French, Spanish, German and Chinese, the languages encoded under their own SONAR tags (Appendix B.1), and two populations of the same shape encoded under the English tag. The softmax Ontologizer is `resid_nc`, the hard codes `ste_h76_init01` and `ste_l1_h380_i01`.

null-corrected NMI of only 0.043, a quarter of the hard code's. Language factors are carried weakly and across many heads' partitions, not by a few: in both hard-code seeds the 76 most language-informative of the 380 heads are exactly the 76 layer-0 heads (`headlang.py`), the best of them reaching about a fifth of the attainable NMI, every deeper head carries less than any of them, and without the top 32 the rest of the code still reads most of what they do (Appendix E.13).

What the English tag does to localization. Most of the cache was encoded under SONAR's English source tag (Appendix B.1), and the tag shapes this table. Each `TopK` SAE has one latent that fires on French, Spanish and German, the three Latin-script languages encoded under their own tags, and so separates them from the English-tagged rest without separating them from each other: it gives each F1@1 near 0.49, against 0.03–0.15 for Italian, Portuguese, Dutch, Catalan, Romanian and Swedish. Restricted to the correctly tagged languages, French and German get no latent or entry of their own in any model (best F1@1 0.39–0.50 against the 0.33 floor), while English (0.64–0.96) and Chinese (0.86–0.997 in the sparse and hard codes) do. Populations of the same shape under the English tag localize at macro F1@1 0.70–0.88 against 0.55–0.70 for the correctly tagged one in every model but `g160top1` and the dense codes, although a dense probe separates the correctly tagged five at least as well (0.95–1.00 against 0.92–0.98): each English-tagged Latin-script language carries a signature of its own. So most single-latent localization of Latin-script languages here is the tag's signature, not the language's. Whether a dictionary trained on correctly tagged embeddings would give those languages a latent each is untested.

Entry-level enrichment asks the same question cell by cell (Figure 18, `enrich.py`). In `resid_nc`, 3,698 of 226,763 tested (entry, language) cells are significant at FDR 0.05, and the label structure is a layer-0 property: the best head's NMI above its shuffled null is 0.050 at layer 0 and 0.006, 0.003, 0.002 and 0.001 above it. GPT-2's hard code has the same depth profile for token class (0.149 at layer 0, at most 0.029 above) and for position (0.105, then at most 0.011), with 11,777 of 53,412 (entry, token class) cells significant.

C.7 Shrinkage decomposition

`TopK` SAEs have small magnitude error (share of raw FVU_w removed by support-frozen least squares: 2.2–2.4% for k32 variants, 8.5% for k160, 0.6% bilinear, 1.9% p5); `g160top1` is essentially perfectly calibrated (0.3%) — its reconstruction deficit is pure selection. For the Ontologizer's truncated deviation codes the refit removes 16% at ν=1 and 33% at ν=2, about the share any $|\mathcal{S}|$ free coefficients would remove from a residual spread evenly over the whitened coordinates ($|\mathcal{S}|/d = 160/1024$ and $320/1024$ for `resid_nc`'s 5×32 heads). A support drawn at random, at the same size per sample, removes more: 19% and 36% for the Ontologizer (ν random entries per head), and 20%, 13% and 4.6% for k160, `g160top1` and k32, against their own 8.5%,

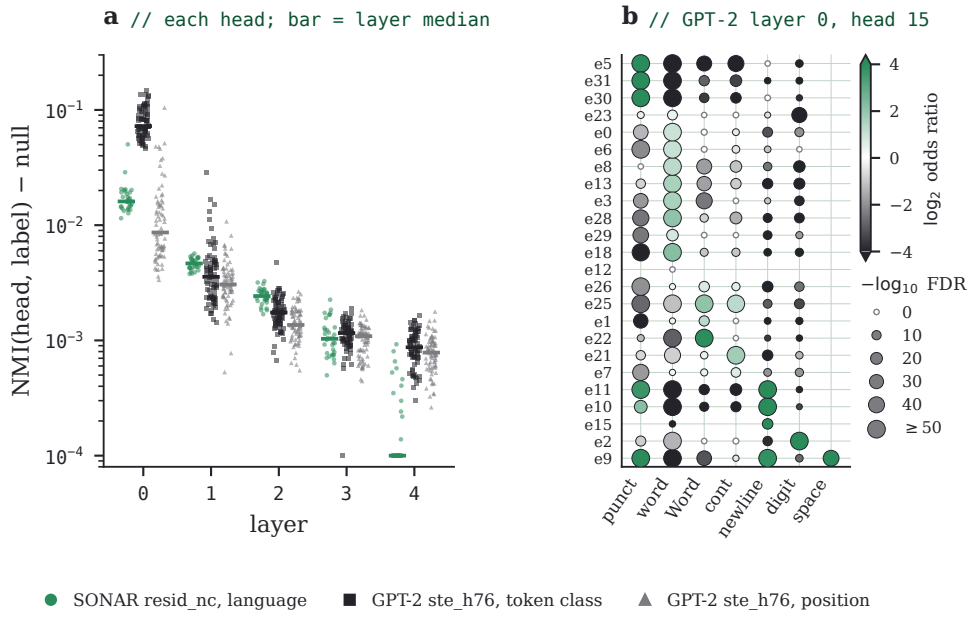


FIGURE 18 · **Label structure lives in layer 0.** (a) NMI between each head's winning entry and a label on the eval tail, less its mean over five label-shuffled draws, one point per head (values at or below that mean drawn at 10^{-4}) and a bar at the layer median: `resid_nc` against language, GPT-2 `ste_h76` against token class and position. (b) The GPT-2 layer-0 head with the most token-class NMI (head 15), each entry against each class: colour the Haldane-corrected $\log_2$ odds ratio, area $-\log_{10}$ FDR (capped at 50), filled where FDR < 0.05; entries grouped by the class they are most enriched for. One seed per model.

0.3% and 2.2% (`refit.py`). No code's own support removes more than chance, so nothing here shows the pin-and-renormalize convention mis-scaling what it keeps (supports of $\geq d$ coefficients are excluded as underdetermined). Separately, the gap between non-adaptive truncation (freeze soft P, truncate, decode linearly) and in-forward truncation (later layers reclassify against the truncated prefix residual) is large — 0.268 vs. 0.094 at ν=8: downstream layer compensation is worth ~3× under truncation. This was the first quantified reconstruction credit for the residual stack (Section 5.3 measures a larger one), and it is steering-relevant: the stack actively re-routes around code edits, which cuts both ways — robustness to small interventions, and, as Section 5.4 shows, widespread collateral.

C.8 Steering through the text round trip

Across the Ontologizer selection sweep, the straight-through arm and the k_{SAE}=32 SAE, at 64 features per run:

run	FVU_w	chrF	loss rec.	net @0.25	@0.5	@1.0	hit @1.0	rand. hit
top8	0.0370	0.416	0.941	0.203	0.404	**0.652**	0.734	0.359
top2	0.3516	0.246	0.521	0.072	0.277	0.509	0.578	0.109
top16	0.0052	0.557	0.987	0.027	0.110	0.363	0.381	0.156
top4_shm	0.0761	0.375	0.892	0.112	0.140	0.248	0.242	0.062
softmax_shm	0.0007	0.706	0.997	0.058	0.128	0.242	0.211	0.094
top2_shm	0.1911	0.301	0.742	0.018	0.071	0.196	0.150	0.016
top8_shm	0.0287	0.437	0.954	−0.015	0.074	0.176	0.330	0.094
top1	0.7190	0.170	0.126	0.047	0.104	0.064	0.102	0.109
ste_h76	0.2293	0.295	0.695	0.033	0.045	−0.020	0.055	0.125
SAE k32	0.4994	0.228	0.345	0.048	0.187	0.402	0.260	0.016

TABLE 16 · Text fidelity and text-cycle steering (`steerfid.py`, `--n-features 64`) across selection rules, and the k_{SAE}=32 SAE under the same protocol. Net effect is feature effect minus the random-direction control. A further run (`topk4_rev`, FVU_w 0.164) reads net 0.636 at 1.0.

Steerability does not track reconstruction. The best steerer is `top8` at FVU_w 0.037, the most faithful model (`softmax_shm`) steers at about a third of its rate, and every `_shm` variant steers about three times worse than its twin (`top2` 0.509 vs. `top2_shm` 0.196; `top8` 0.652 vs. 0.176). The straight-through arm is the weakest of all: its raw effect is flat in magnitude (0.018, 0.014, 0.027) and no larger than a random direction's, and its hit rate (0.055) does not clear its own random control (0.125). The SAE steers in text by the same measure (net 0.402 at 1.0, hit 0.260 against 0.016 for random). Two cautions on this instrument. Sixteen features is too few to quote, and even at 64 the straight-through arm's net effect changes sign with its random control: it was negative at every magnitude at 16 features and is +0.033, +0.045, −0.020 here, so its text-cycle effect is within the instrument's noise. And the hit

rate is not comparable across hard and soft codes: under a hard code an activation is exactly 0 or 1, so a "hit" demands the argmax land exactly on the target entry, where a soft code need only cross a fractional threshold.

The text cycle is too weak to support conclusions. Measured on an *unsteered* round trip (the last 512 cache rows), the cycle keeps part of each row's content but moves it off the corpus: $\cos \angle(\mathbf{x}, \text{cycle}(\mathbf{x}))$ is 0.382, against 0.091 to another row's cycle, while cycled embeddings sit at 0.160 to the corpus mean, where rows sit at 0.561. What it does not keep is the code. A head's argmax survives the round trip on 12.9% of rows for `ste_h76` and 10.8% for the softmax model, against 3.1% chance; for the hard code survival falls with depth, from 36.2% at layer 0 to 4.8% by layer 4. A perfect intervention could therefore be observed at most ~13% of the time — the same order as the hit rates the instrument reports. The protocol makes the cycle harder than it need be: rows are 512-token document openings, past the length at which SONAR's auto-encoding degrades, decoded into English whatever their language, through a decoder that small perturbations of the embedding can derail [7]. Reconstruction into embedding space is fine; generation is where the signal goes.

D Supplementary results: training, reproducibility and steering of the softmax Ontologizer

These results come from the project proposal's evaluation and concern the softmax Ontologizer. Two models recur. `resid_nc` (372,600 steps) is the reference model behind the softmax tables above. `resid_nc_hm` is the same configuration plus the batch mean-entropy bonus s_{KL_m}, trained to 4.38M steps; results here use its final checkpoint unless stated otherwise.

D.1 Head freezing and the mean-entropy bonus

A head is counted frozen when its top classification share is at least 0.95 and its churn at most 0.02. In `resid_nc` the top layer (layer 4 of 0–4) begins freezing near step 190k, well after the temperature anneal, reaches 27 of 32 heads by step 210k and 28 at its peak, and is still frozen at the last checkpoint (Figure 19). Section G.4 shows the same layer to be inert once the prefix error is small, which is why the freeze costs so little reconstruction. In `resid_nc_hm` the late freeze never occurs. What remains is an early annealing transient — 22 of 32 heads at layer 1 at step 10k, with 16, 11 and 4 of 32 at layers 2, 3 and 0 — that recovers completely by step 30k, leaving the final checkpoint unfrozen everywhere. **The mean-entropy bonus prevents the late freeze.**

Four retrains of the `resid_nc_hm` configuration (372k steps each) isolate the transient: a zero-change control and three single-knob variants. The control reproduces it head for head (22/32 at layer 1 at step 10k, zero everywhere by 30k)

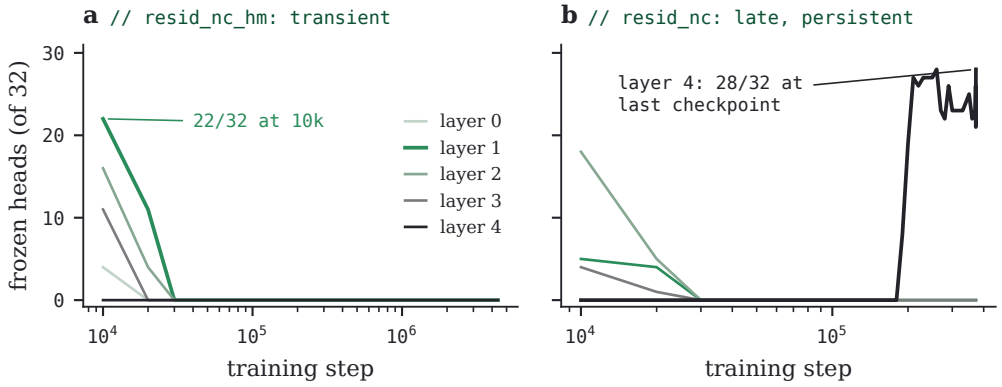


FIGURE 19 · Frozen heads per layer over training. (a) `resid_nc_hm`, with the mean-entropy bonus, has only an early annealing transient, peaking at 22/32 in layer 1 at step 10k and recovering to zero everywhere by step 30k. (b) `resid_nc` develops a post-anneal layer-4 freeze from ~190k that persists through its last checkpoint.

and shows no late freeze. Stretching the temperature anneal from 50k to 150k steps attenuates and spreads it (peak 14/32 at layer 1 at step 30k, clear by 50k). Raising winner dropout from 0.1 to 0.25 tracks the control to within two heads at the peak, and holding the classification-noise floor at its starting value changes nothing (Figure 20). The transient belongs to the temperature schedule alone, and every retrain ends unfrozen.

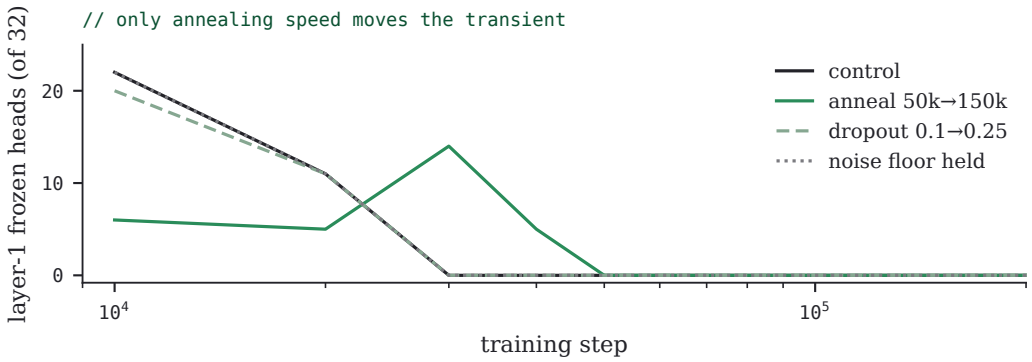


FIGURE 20 · The early transient under the four retrains, layer-1 frozen count. The noise-floor variant overlaps the control exactly and the dropout variant tracks it to within two heads at the peak; only stretching the temperature anneal moves the curve, trading peak height for duration.

An HSIC penalty in place of the auxiliary losses. The arms are named after the HSIC bottleneck [80], which trains each layer without backpropagation to depend little on its input and much on the class labels, both measured by normalized HSIC. An autoencoder's labels are its input, which collapses the two terms into

one, so neither is used here: reconstruction stays with the whitened MSE, and the only kernel term penalizes dependence among a layer's heads. Two arms on the `resid_nc_hm` configuration, trained to 372k steps, penalize it by the pairwise head CKA of Section E.9 under the biased estimator [81]: one replaces the four auxiliary terms with that penalty, the other adds it to them. Replacing them destroys head health: a late, persistent freeze takes 77 of 160 heads by the final checkpoint, one layer entirely (layer 3, 32/32 from step 80k), while reconstruction lands slightly behind the control (tail-mean training MSE 3.7×10^{-5} against 3.4×10^{-5}). Adding the penalty to the auxiliary terms is worse on both counts: MSE 4.4×10^{-5}, and 41 of 160 heads still frozen at the end, layer 3 fully frozen by 80k with only partial thaw (Figure 21). The failure is `resid_nc`'s late freeze rather than the transient, and it is the estimator's. The biased floor between independent heads falls with the number of entries a head uses and is zero for a frozen head (Section E.9), so the penalty can only be lowered by concentrating or freezing heads, which puts it in direct opposition to the mean-entropy bonus. Retrained on the current code, the replacement arm reproduces the freeze checkpoint for checkpoint (layer 3: 6, 29 and 32 of 32 frozen at 60k, 70k and 80k, against 7, 29 and 32) while its summed penalty falls by 80%; under the unbiased estimator the same arm freezes at most 5 heads, all in layer 4, by 93k steps (one seed each). With the corrected estimator the penalty has almost nothing to act on, and whether it could replace the auxiliary terms over a full run is untested.

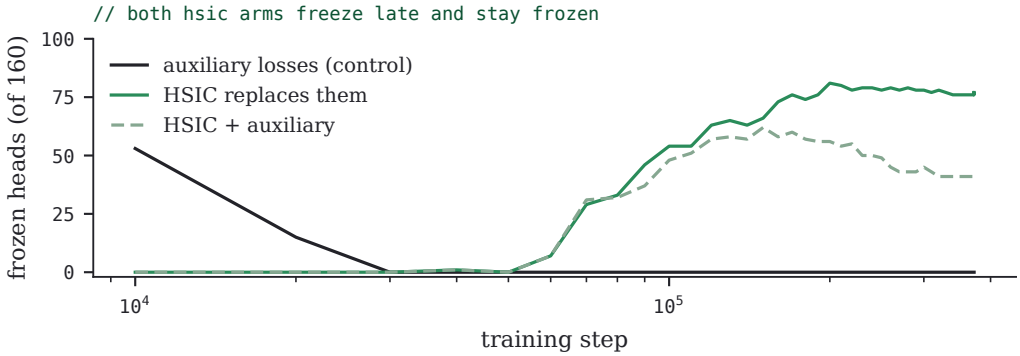


FIGURE 21 · Total frozen heads over training for the three loss configurations, all on the `resid_nc_hm` configuration at 372k steps. The auxiliary-loss control's freeze is the early transient and resolves; both HSIC arms, whose head-CKA penalty uses the biased estimator, develop the late, persistent freeze instead, and the auxiliary terms only partially thaw it.

D.2 Development over training

Tracking `resid_nc_hm`'s full 448-checkpoint history applies the z-score machinery of Section 4.3 to the refit end-to-end direction matrix rather than to raw dictionary

decode directions. The two are not directly comparable: raw decode directions on `resid_nc` repel (ζ_{cos}=−17.6), while refit directions on `resid_nc_hm`'s final checkpoint cohere (ζ_{cos}=+72); they differ in both instrument and substrate, and the refit frame folds in the shared decoder and cross-layer interactions. On that statistic the shape is rise, peak and partial relaxation (Figure 22): mean within-head ζ_{cos} climbs from 1.8 at step 10k to a peak of 119 near step 370k (median 123, every head above ζ_{cos}=2), then relaxes over the remaining 4M steps to 72 at the final checkpoint (median 105, 0.79 above ζ_{cos}=2). The code itself moves the other way after the annealing transient: per-sample entropy returns to 4.5 of a 5-bit maximum and usage KL falls toward uniform. Relative to initialization, the consolidation is in the geometry the classifications write, while the classifications stay soft.

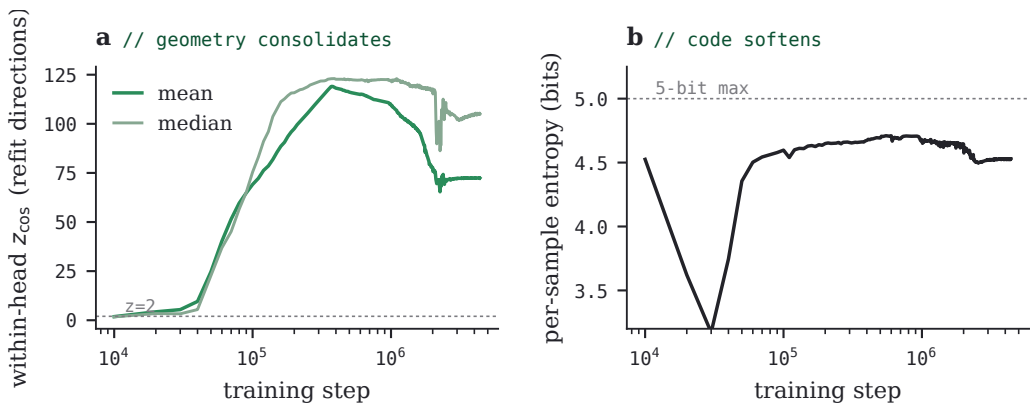


FIGURE 22 · The 448-checkpoint training history of `resid_nc_hm`. (a) Within-head coherence of refit end-to-end directions, z-scored against size-matched nulls; rise, peak near step 370k, partial relaxation. (b) Mean per-sample classification entropy returns toward the 5-bit maximum after the annealing transient.

Entry by entry, `resid_nc`'s dictionary settles after the anneal (Figure 23, `entrydrift.py`). In layers 0–3 a used entry's decoded atom has median cosine 0.40–0.51 with its final direction at step 10k, 0.73–0.82 at 60k, just after the anneal, and 0.99–1.00 from 320k. Layer 4 starts at 0.23, stalls near 0.53 from 60k to 160k and converges only after its freeze near 190k (Section D.1), keeping 142 of its 1,024 entries. Usage dips at the anneal's end instead: 40% of layer 0's entries fall out of use at 60k, and all but 7% are back in use by the end. One seed.

D.3 Scaling: a star sweep

Seven configurations around the shared h=32, k=32, l=5 shape, varying one axis at a time, each trained 4 epochs (62k steps against the retrain recipe's 372k), test whether head coherence rises with parameter count. No measured axis delivers the rise: coherence shows no positive relation to parameter count (Spearman +0.14 for mean ζ_{cos}, −0.54 for the fraction above ζ_{cos}=2), usage imbalance grows with size (+0.86), and the cleanest per-axis trend runs opposite to the prediction,

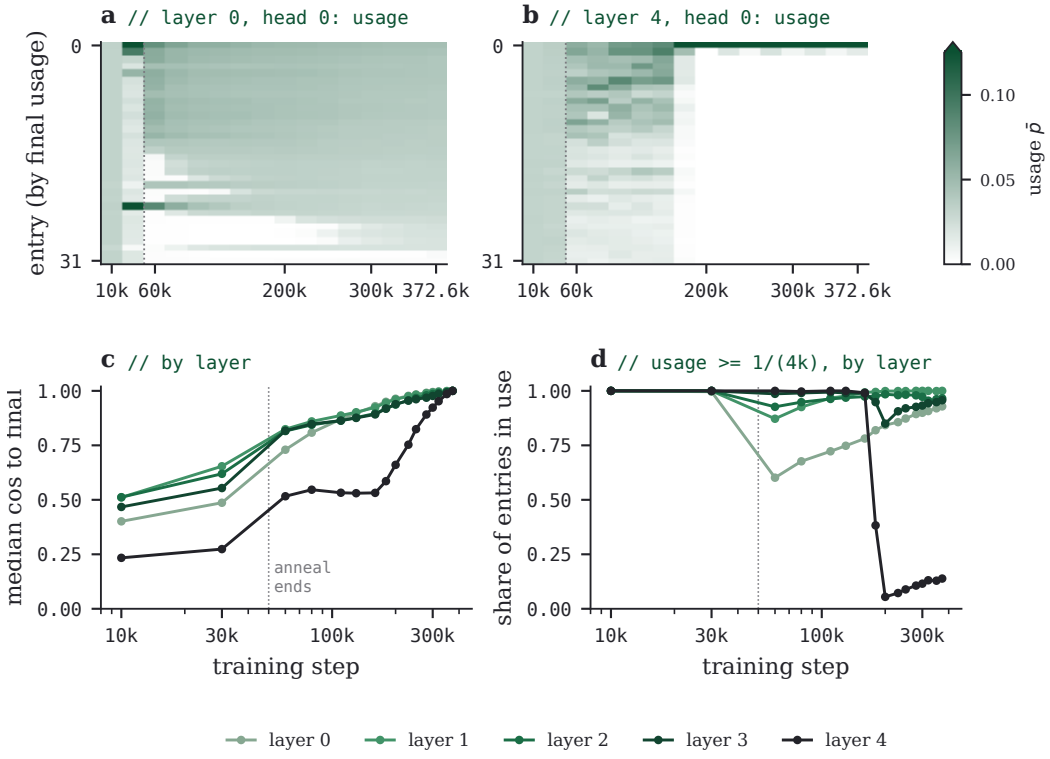


FIGURE 23 · **Entries settle after the anneal.** `resid_nc`'s 16 retained checkpoints on the eval tail. (a, b) Usage $\bar{p}_{ij}$ of head 0's 32 entries in layers 0 and 4, one column per checkpoint (not to scale in step), entries ordered by final usage; the dotted line marks the anneal's end. (c) Per layer, the median cosine between an entry's decoded atom and its final direction, over the entries in use at the last checkpoint. (d) Per layer, the share of entries in use ($\bar{p}_{ij} \geq 1/4k$). One seed.

coherence falling monotonically as entries per head grow (k=16/32/64: mean $\zeta_{\cos}$ 20.2/6.3/3.7). The widest model (h=64) has not converged at this budget (tail MSE 20× the others), so its collapse is not evidence (Figure 24). Seven points at short training are weak evidence either way; full-length retrains are the missing control.

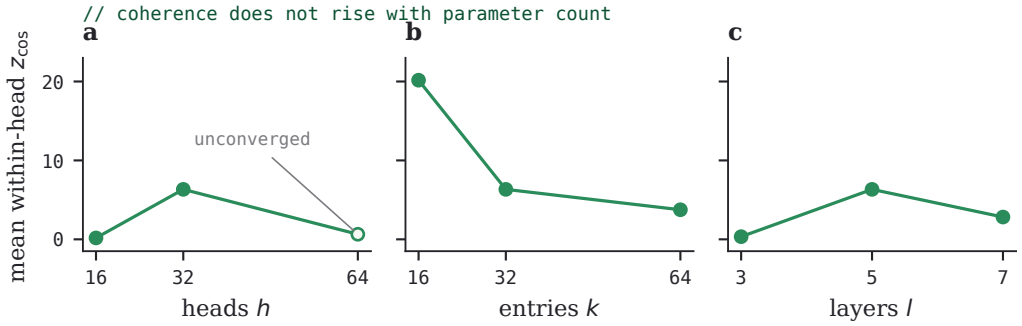


FIGURE 24 · Head coherence against each parameter axis of the star sweep: heads (a), entries (b) and layers (c), all runs at 4 epochs (62k steps). No axis delivers the predicted rise; the entries axis falls monotonically. The open marker in (a) is the unconverged h=64 run.

D.4 Seed stability of soft partitions

A second run of the `resid_nc_hm` configuration from a different seed, compared at matched training (370k steps on both sides), matches essentially nothing (Figure 25): no head pair and at most 0.3% of atoms clear a 0.3 cosine threshold on dictionary geometry, and activation-frame containment over a shared evaluation batch is at most 0.001 at any threshold, against 0.31/0.24/0.15/0.03 for the k32 SAE and its seed replica across the four thresholds (0.3–0.9). The metric has headroom: the same run against itself 2,600 steps apart scores 1.000 on geometry at every threshold and 0.65 at the loosest activation threshold (0.20 at the strictest). At matched training the softmax Ontologizer's partitions are less reproducible across seeds than SAE latents, the same verdict Section 5.5 reaches for the straight-through arm with partition and contribution measures. A replica carried to the full 4.4M steps is the remaining control.

D.5 Steering against supervised baselines

A matched-budget overlay on `resid_nc_hm` at step 4,380,000, the last checkpoint kept, compares forced classifications against difference-of-means and linear-probe directions on language steering, with a random-direction floor (Table 17). Each arm wins its own effect frame, as expected. On the shared difference-of-means frame, forcing a classification moves decodes along the target direction 0.64× as far as supervised steering does, and 4.2× as far as probe directions. No arm reliably reaches the target language: the median hit rate is 0.06 for the difference of means

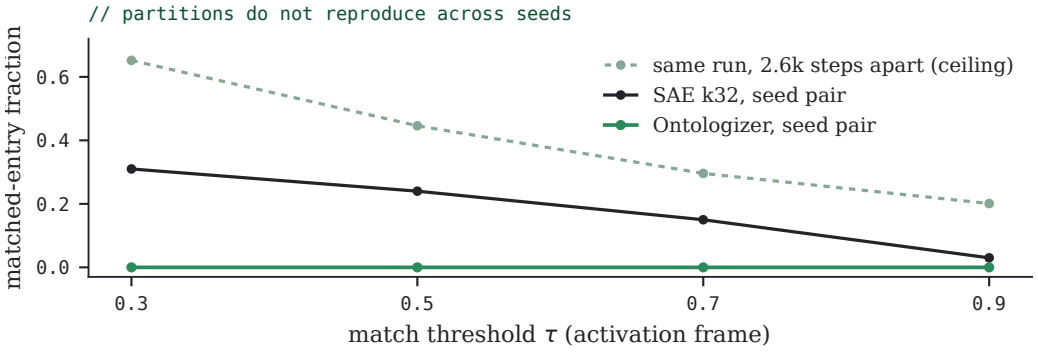


FIGURE 25 · Cross-seed matched-entry fraction on the activation frame by threshold. The same-run pair bounds what the metric can detect (0.65 at τ=0.3); the k32 SAE pair reproduces partially; the softmax Ontologizer's partitions reproduce not at all.

and 0 for the other arms. Every arm's collateral, supervised steering included, sits at the random-direction floor (0.70–0.78), the same floor the text round trip shows for SAE steering (Section C.8). On naturalness the arms are so far indistinguishable (Figure 26): chrF against unsteered decodes falls from ~0.37 at magnitude 0.25 to ~0.15 at 1.0 for the difference-of-means and forced-classification arms, with probe and random decaying in parallel from 0.38–0.40 to 0.19–0.20. A blinded human-rating sheet (120 items) is prepared but unscored.

arm	effect (native)	effect (dm frame)	hit rate	collateral
difference of means	0.474	0.474	0.062	0.752
classification forcing	0.399	0.302	0.000	0.776
linear probe	0.294	0.072	0.000	0.701
random direction	—	—	—	0.705

TABLE 17 · Language steering under matched intervention budgets on `resid_nc_hm` at step 4,380,000; medians over 24 language–magnitude conditions per arm. Effect is displacement along the arm's own direction (native) and along the shared difference-of-means direction (dm frame); hit rate is the fraction of steered decodes reaching the median projection of genuine target-language rows on the shared frame; collateral is content change on unrelated dimensions.

A classification as a training objective. A policy over steering directions on frozen `resid_nc_hm`, rewarded by one classification's quantile-normalized score at a matched intervention budget, converges cleanly in the differentiable in-embedding tier, saturating within roughly 40 steps (Figure 27). That tier trains by gradient ascent through the frozen model, as a ceiling for the decode-cycle tier, whose reward passes through text generation and so needs REINFORCE [82]; the curve does not show how fast REINFORCE would learn. Its collateral cannot move far

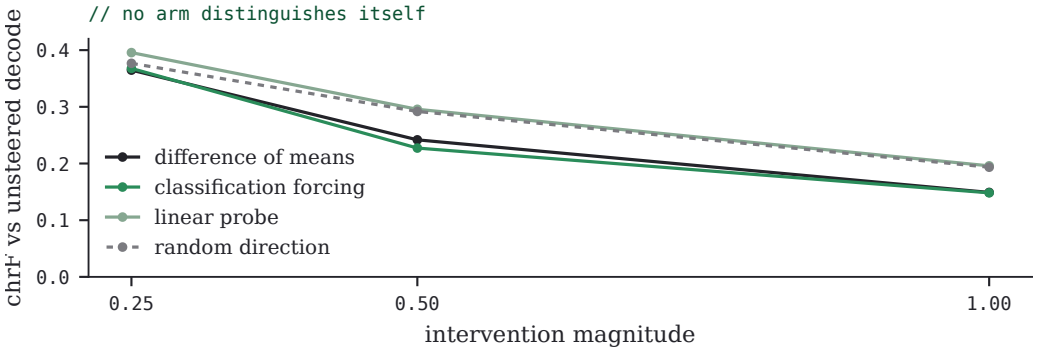


FIGURE 26 · chrF between steered and unsteered decodes by intervention magnitude. All four arms, the random direction included, damage the decode at nearly the same rate; no arm buys naturalness at matched magnitude.

either. The policy adds a direction to the unit-norm embedding at a fixed length of 0.5 and renormalizes, which caps the collateral at 0.134; it falls from 0.105 to 0.083 as the policy tilts its direction toward the row and so shortens its effective step. The policy's step is the residual form that latent-space steering of diffusion policies outperforms by acting on the generator's input instead [83], and a direction optimized against a classifier can realize its target through directions the data never vary along [68]. Neither the decode-cycle tier nor a convergence race against paired steering has been run.

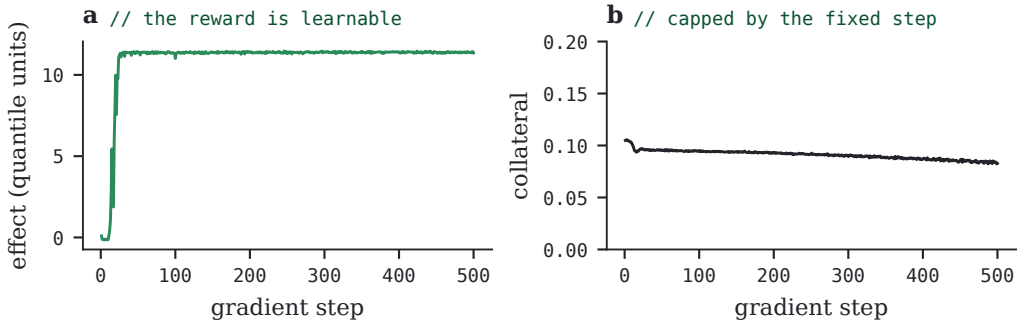


FIGURE 27 · A steering policy trained by gradient ascent on a single frozen classification's quantile-normalized score, in-embedding tier. The policy saturates the reward within roughly 40 steps (a); collateral, capped at 0.134 by the fixed step length, falls from 0.105 to 0.083 (b).

D.6 Head-level autointerp

The per-latent lenses of the description-specificity results mis-measure classifier codes, whose content lives in partitions, so a head-level pass over `resid_nc` (step

372,600) tests the claim directly; its frozen top layer (Section D.1) is a caveat for any layer-4 heads in the sample. Descriptions decoded from dictionary parameters fail at the partition level as they do per latent (median self accuracy 0.000, as is the null's). Activation-grounded head descriptions do no better than a null pairing at this sample size (20 heads per condition, median accuracy 0.083 against 0.125, where guessing one value per head scores about 0.11), and a head's detection accuracy is uncorrelated with the mean describability of its latents (Spearman +0.12 over 30 head–lens pairs; +0.25 under activations and +0.05 under parameters, 15 heads each, p=0.37 and 0.86). The two lenses disagree and neither shows above-chance head-level signal, so the mis-measurement argument predicts an advantage this sample does not deliver.

Describability against density. Across the six runs of the description-specificity table, the correlation between log nominal density and activation-description F1 is −0.894, so density alone carries about 80% of the variance. Fitting contrastive-description F1 linearly in log firing rate on the four `sae.py` runs with a nonzero firing rate (slope −0.069, correlation −0.766), and holding the Ontologizer out, predicts 0.421 against an actual 0.392, a residual of −1.8 standard errors: the Ontologizer sits on the trend, and `g160top1` lies 0.120 above it. The fit is thin. It rests on four runs and a weak correlation, it excludes `g160softmax`, whose thresholded firing rate is exactly zero, and the $2/k$ threshold is not monotone in true density.

Describability against language. Counting the distinct languages among each feature's 15 judged rows (`autointerp.py language`; Table 18) separates the sparse codes from the dense ones before any description is read. Within each sparse run F1 falls as the count grows (Spearman −0.47 to −0.67, activation descriptions), and within the dense runs it does not (−0.06 to +0.03). For `k32` and `g160top1` the modal language makes up 28–30% of the held-out positives the judge sees, against about 1% of the negatives and of the contrastive background. Refitted on the features spanning 13–15 languages, the contrastive trend in log firing rate puts `g160top1` 0.018 above it rather than 0.120, and the Ontologizer 0.049 above it rather than 0.029 below. Per feature, purity explains a fifth of the variance in F1 (0.20 for activation descriptions, 0.22 for contrastive ones) and log firing rate 1–4%. Holding language fixed needs matched rows on both sides, which `autointerp.py match` draws for a contrastive background (`describe --mode cacts_lang`) and for the negatives (`score --negatives lang`); neither has been scored.

	features by languages spanned					F1, all / 13–15 languages	
model	1–3	4–9	10–12	13–15	purity	acts	cacts
onto (resid_nc)	0	5	43	177	0.158	0.473 / 0.460	0.392 / 0.380
k32	63	55	68	64	0.466	0.645 / 0.564	0.603 / 0.482
k32_bl	9	9	10	47	0.277	0.578 / 0.502	0.470 / 0.338
g160top1	41	24	32	53	0.440	0.646 / 0.467	0.594 / 0.383
g160softmax	0	5	72	179	0.183	0.449 / 0.443	0.364 / 0.347
k5120 (dense ctrl)	0	2	18	236	0.131	0.470 / 0.470	0.332 / 0.322

TABLE 18 · Detection F1 of Table 4 by the number of distinct languages among each feature's 15 judged rows (the 10 a description was written from and the 5 held-out positives; 15 random rows span 13.8). Purity is the share of the description rows in their modal language. One seed per run.

E Supplementary results: the straight-through arm

E.1 Calibration

A naive port (live configuration, `select="ste"`) trains but the classifier learns nothing. The root cause is one property: **the argmax forward is scale-invariant, so here nothing pushed the logits up**; they sat at their initialization scale for the whole run (within-head standard deviation 0.0015, near the $2/(d{+}1) \approx 0.0020$ that the fan-in initialization gives a bilinear logit on a unit-norm shape with its constant coordinate — the variance argument behind scaled dot-product attention [49, fn. 4], at degree two). VQ-VAE meets the same gap, an encoder scale the code does not constrain, with a commitment loss [55]. Three settings follow from it. *Logit noise must be relative*: σ_K=0.02 is thirteen times the logit spread, so the argmax is chosen by noise, while σ_K=0 removes the only exploration and the codebook collapses; `--noise-k batchnorm` (Eq. (17)) scales noise by the logit RMS. *The surrogate temperature must track the logit scale*: at the live floor t=0.03, spread/t is 0.05 and the surrogate is flat, so what improves is the dictionary adapting to a fixed random partition. s_{KL_m} *must be rescaled*: 10^{-6} is calibrated against a whitened MSE of $\sim 3 \times 10^{-6}$, and a hard code trains at an MSE orders of magnitude larger, where the term is inert and heads collapse.

t	spread/t	max p	effective entries/head	raw FVU
0.01	0.15	0.043	23.4	178
0.0015	0.78	0.135	26.4	193
0.0005	3.38	0.681	14.0	59.2
0.00015	16.84	0.974	3.8	7.3

(4000 steps, lr 10^{-5}, `--noise-k batchnorm --sd-k 0.3`; one seed per arm, which ranks configurations but does not predict final quality.) Sharpening the surrogate is what makes the classifier train at all and improves reconstruction 25-fold, but it

trades directly against head usage, which s_{KL_m} exists to hold. At `sonar.py`'s learning rate 5×10^{-5} every sharp-surrogate arm collapsed abruptly after ~1700 steps; 10^{-5} is stable. The shipped arm is t=1.5 × 10^{-4}, lr 10^{-5}, `--noise-k batchnorm --sd-k 0.3`, s_{KL_m}=10^{-4}, trained 24 epochs (369,500 steps). The temperature is hand-matched to a logit scale measured once, and that scale moves: over training the shipped arm's layer-0 logits shrink 2.9-fold, so spread/t falls from 13 at initialization to 4.5, while its relative noise changes 20–29% of layer-0 assignments throughout (`logitscale.py`, on 2,048 tail rows). Scale invariance leaves the logits free to move either way, and the GPT-2 arms, at the naive port's settings, show the other way: `ste_h76`'s layer-0 logits, from a classifier without a gate, grew 452-fold, so its absolute noise, which changed more than half of the layer-0 assignments through 50,000 steps, changed about 1% from 150,000 on. The concatenated seed pair's sigmoid-gated classifiers kept their initial scale, and there the noise changed 71–90% of layer-0 assignments at every one of 16 checkpoints scored. The principled version normalizes logits per head or controls t against a spread/t setpoint.

E.2 Capacity beyond bits, and training dynamics

Distance from the Gaussian reference. Measured against the Gaussian reference at each code's own rate, the full stack sits further from it than one layer's prefix does. Each ratio bounds that code's distance from the optimum at its rate from below, since the optimum for real data may lie below the reference:

code	Gaussian reference	achieved	ratio
380 bits (one layer's prefix)	0.4208	0.537	1.28×
1900 bits, 5×76	0.0499	0.1535	3.08×
1900 bits, 5×380, k=2	0.0499	0.1844	3.70×

Bits are not the whole capacity. At identical bits, 380 binary heads reach 0.1844 against 76 thirty-two-way heads at 0.1535: bits go as $h \log_2 k$ but the codebook goes as $h \cdot k$ (3800 atoms against 12,160). The binary arm also trained at half the batch size (Table 10), so the gap mixes the larger codebook with a noisier gradient. A neural codec's residual quantizers point the same way: at a fixed 80 bits per frame, eight stages of 1024 codes beat eighty of two [6]. The binary arm gets within 17% on 2.9× fewer parameters, with features that are one bit each — the form that is actually nameable.

Realized bits, and what they bound. This paragraph follows information theory in writing random variables in uppercase. For a hard code and a random input $\mathbf{X}$, write $A_{\ell i}$ for the entry head (ℓ, i) chooses and $\mathbf{A}$ for all of them. *Realized bits* are the summed entropy of each head's usage over the data,

$$H_{\text{real}} := \sum_{\ell,i} H(A_{\ell i}) = \sum_{\ell,i} \left(\log_2 k - \mathrm{KL}(\bar{\mathbf{p}}_{\ell i} \,\|\, \text{uniform}) \right) \;\geq\; H(\mathbf{A}) = I(\mathbf{X}; \mathbf{A}), \quad (36)$$

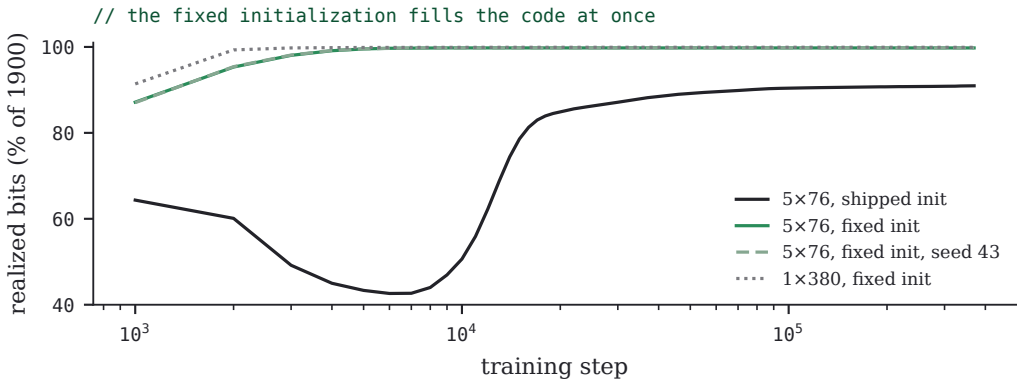


FIGURE 28 · Realized code bits over training, as a share of the nominal 1900, from the logged KL_m less its one-batch sampling floor. The shipped initialization's head usage collapses to under half before recovering to 90% (the curve's 91% over-corrects heads with dead entries); at the fixed initialization every head in the stack and the flat arm reaches near-uniform usage within 5k steps. An upper bound on the information carried, not a measure of it (Eq. (36)).

with $\bar{\mathbf{p}}_{\ell i}$ head (ℓ, i)'s usage distribution, its mean assignment; summed over layers this is $h(l \log_2 k - \mathrm{KL}_m)$ in the logged statistic's terms. Because the code is a deterministic function of the input, its joint entropy is exactly the information it carries about the input, and subadditivity puts the sum of marginal entropies above it, with equality only when the heads are independent. Realized bits therefore measure how evenly each head uses its entries, not how much the code says: two codes at the same H_{real} can carry different information if their heads are redundant to different degrees, and the joint entropy, which needs a density estimate over k^{lh} outcomes, is not measured here.

Training dynamics. On the shipped arm the training-log MSE (the deep-supervision prefix mean) is flat from step 237k while within-head row collinearity keeps falling (0.812 → 0.686, against ~0.64 for random non-negative rows). Its heads' usage collapses early and never evens out (Figure 28): realized bits fall to 43% of nominal near step 6k, recover to 85% by 20k as the s_{KL_m} rescale turns the usage collapse around, and end at 90%. At the fixed initialization every head's usage is near-uniform by step 5k (99% of nominal), in the stack and the flat arm alike, and stays there. Realized bits come from the logged KL_m, $h(l \log_2 k - \mathrm{KL}_m)$, less its sampling floor: KL_m is a plug-in estimate on one batch of 256 rows, which reads $(k-1)/(2b \ln 2) = 0.087$ bits per layer even for perfectly balanced usage, and on the converged fixed-init arms the logged value is almost entirely that floor. Corrected, the endpoints match the held-out measurement of Table 5 to within a bit; the correction assumes every entry is live, so it overstates the shipped arm by 14 bits.

E.3 Operational rate: every code quantized

Index bits and coefficient counts cannot rank a code that sends coefficients against one that does not. With everything each code sends quantized and entropy-coded (Eq. (26)), every code becomes a curve on one rate axis (Figure 29). At the stack's 1900 bits:

code	FVU_w unquantized	at 1900 bits entropy-coded	at 1900 bits fixed-length supports
straight-through, 5×76	0.1536	**0.155**	**0.158**
L1 SAE, $s_{L1}=3 \times 10^{-6}$	0.0052	**0.128**	0.179
softmax Ontologizer, soft code	0.0054	0.145	0.229
L1 SAE, $s_{L1}=10^{-5}$	0.048	0.173	0.179
`TopK` SAE, $k_{SAE}=5120$	0.0001	0.231	—
L1 SAE, $s_{L1}=3 \times 10^{-5}$	0.219	0.242	0.243
`TopK` SAE, $k_{SAE}=160$	0.258	0.259	0.259
`g160top1`	0.274	0.274	0.274
`g160softmax`	0.0044	0.332	0.326

Entropy-coded, a coarsely quantized L1 SAE beats the stack at the stack's own rate: 0.128 against 0.155, and the stack's pure-code 0.158 at 1655 bits. It sends about 440 of its roughly 1600 coefficients, at about 4.3 bits each with their support. L1 latents fire unevenly, so the support codes about 20% below its fixed-length cost, while the stack uses its entries almost uniformly (1895 of 1900 bits; Table 5) and has nothing to save. With supports at their fixed-length cost the stack leads every SAE (0.158 against 0.179). The `TopK` SAE flattens at its unquantized error from about 1900 bits on, consistent with its least-squares floor of 0.236, and is already at 0.275 at 1121 bits, so its coefficients carry little beyond which latents fire; the single-layer hard variant behaves the same way. The softmax model's own code, quantized around its mean assignment, reaches 0.145, and the same on rows no model trained on. The stack's gains cost 4.8 bits at a step of two standard deviations (taking it from 0.158 to 0.155) and 28 bits to recover the unquantized 0.1536. No code was trained for this codec: uniform usage is the most expensive kind to entropy-code, and no SAE was trained with quantized coefficients, so the curves rank these models, not their architectures.

E.4 Depth: eliminations and mechanism

Over training. Scored held out at every 10k checkpoint (Figure 30), the stack leads the flat arms from the first checkpoint on: the ratio to the matched flat arm is 1.59× at step 10k, peaks at 1.83× near 200k and ends at 1.79×. Neither shape has converged. Over the last quarter the stack improves 0.86% and the flat arms 2.3–2.75%, so the gap is narrowing, slowly; where it ends needs longer runs. Seed replicas overlap at every checkpoint.

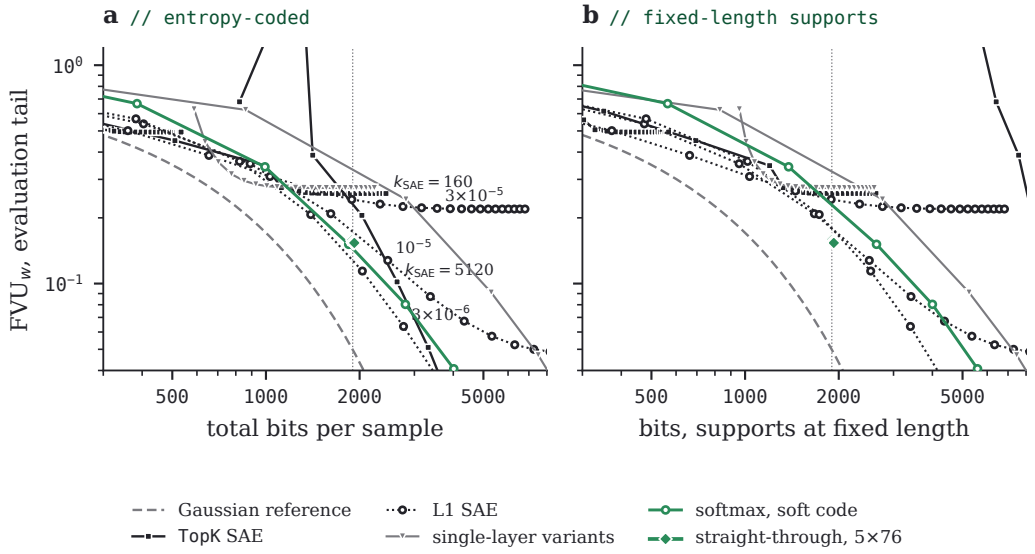


FIGURE 29 · Operational rate on the evaluation tail: everything each code sends quantized (Eq. (26)) at a swept step, (a) each slot's symbols entropy-coded and (b) each support charged its fixed-length cost, the values still entropy-coded. L1 SAEs are labeled by s_{L1}; the unlabeled `TopK` curve is k_{SAE}=32, and the single-layer variants are `g160top1` (level) and `g160softmax` (falling). The straight-through stack's curve spans 1895–1923 bits. The dotted line is 1900 bits and the dashed curve the Gaussian reference (Eq. (25)). The softmax model saw the tail in training; scored on rows no model saw, no curve's readout at 1900 bits moves by more than 0.0004 (Appendix B.1). One seed per curve.

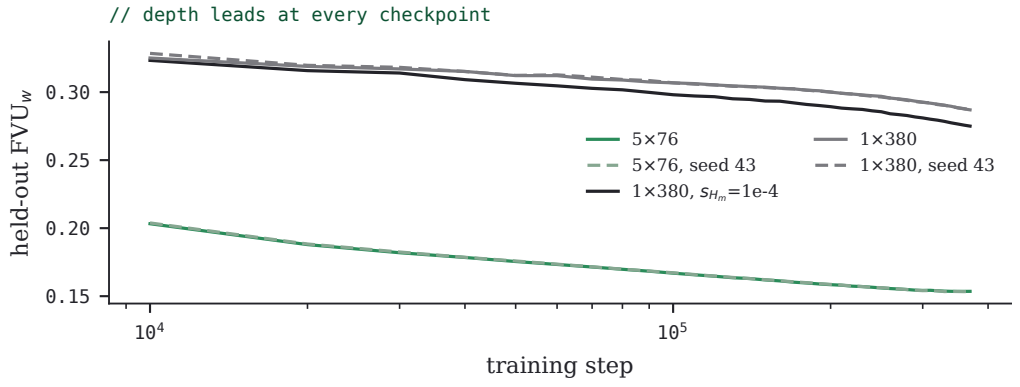


FIGURE 30 · Held-out FVU$_w$ over training for the 5×76 stack and the 1×380 flat arms at matched bits and parameters, with seed replicas. The flat arm with the stack's entropy pressure (s_{KL_m}=10^{-4}) is the one compared in Section 5.3.

The comparison went through two wrong versions first, and the eliminations are informative. On the shipped initialization the flat arm read 0.5822 at 484 realized bits (25.5%), a 2.5× gap. Correcting the per-head entropy pressure — KL_m averages over a layer's heads and the loss sums over layers, so per-head pressure scales as s_{KL_m}/h per layer and the flat arm had trained at a fifth of the stack's — took it to 1199 bits and 0.4307. A frozen ZCA whitener (`make_zca.py`) took the classifier's input from effective dimension 107 to 901 and the code to 1900 of 1900 bits, every head at 4.999 of 5 bits, more than the stack itself realized — and still reconstructed 1.7× worse (0.3842). Two survivors from that line: **usage is not what depth buys** (bits are not the objective either: dropping s_{KL_m} 100× costs the stack 176 bits and *improves* its error to 0.2095), and **a learned input transform collapses where a supplied one does not** — a jointly trained `Linear` encoder ends with every head on exactly two entries at FVU_w 0.6853, worse than no encoder.

The input each classifier sees. `inputgeom.py`, on the exact vector handed to each classifier (shipped arm):

layer	mean direction	mean pairwise cos	effective dim / 1024
0	0.5638	0.3146	107.5
1	0.0932	0.0087	611.3
2	0.1522	0.0233	596.3
3	0.1983	0.0396	548.8
4	0.2348	0.0553	507.1

Layer 0 (and every flat arm) classifies the raw cap; layers 1–4 classify residuals that span over half the space. On the shipped arm, layer 0 was the only layer whose heads left most entries unused (10.6 live entries, 216 bits, against ~375 for each later layer) — a within-model control on everything except the input.

What the cascade contributes: pairing. `blendablate.py` blends the subtracted term $\mathrm{dec}(\mathbf{r}_\ell)$ in layer ℓ's input toward a reference with the same marginals and none of the pairing — another row's reconstruction (shuffle), the mean, or Gaussian noise rescaled to the shuffle's per-sample magnitude — without retraining:

blend	shuffle	mean	noise	eff. dim L1 (shuffle)	(noise)
0.000	0.2292	0.2292	0.2292	499.1	499.1
0.125	0.2460	0.2374	0.2426	476.0	504.7
0.250	0.3084	0.2652	0.2838	394.2	518.3
0.500	0.6910	0.4100	0.4665	193.4	539.6
0.750	1.8154	0.7497	0.8502	103.0	527.6
1.000	4.6613	1.4302	1.5992	68.5	496.2

The pairing is load-bearing well below full severance — a quarter blend costs 35% of FVU_w — and another row's reconstruction does more damage than matched-magnitude noise at every level, so the damage is specific rather than generic perturbation sensitivity. The sweep cannot separate pairing from conditioning: effective dimension falls in lockstep with the error, because a residual is small and nearly isotropic *because* it is this sample's own error. That resolves the whitened arm rather than conflicting with it: whitening is conditioning without pairing, and it bought uniform usage of every entry and ~7% of FVU_w. (`forward="labels"`, which would test this by removing the residual entirely, has not been run at the fixed initialization. Forwarding the code *alongside* the residual, `forward="resid_labels"`, was a clear negative in a three-epoch pilot: held-out FVU_w 0.826 against 0.249, after leading early by 50× — the signature of a parameter advantage without an information advantage, since the residual is by construction what the code failed to explain.)

The code is not the whole channel. Each layer splits its input into a unit direction (all the classifier sees) and a norm that multiplies the layer's output, so one real number per layer per sample reaches the decoder outside the bottleneck. Pinning each gain to its held-out mean (`gainablate.py`) costs the stack +3.0% in total, +0.6 to +0.8% per layer (layer 0's gain is identically 1, since the cache is L2-normalized; SONAR's own embeddings are not [36]), against the 79% by which the matched flat arm trails it: the channel is real and is not the explanation. With the gains pinned the stack is a pure 1900-bit code (FVU_w 0.158), the form compared against SAEs at matched total rate in Section 5. It does matter as a confound for the whitened flat arm, whose whitener creates a varying layer-0 gain worth +4.5% — about 40% of whitening's apparent benefit.

The decoder is dense, and not because of the offset. Magnitude-pruning the trained decoder's smallest half takes FVU_w into the thousands, and its weight distribution looks like a Gaussian matrix's. The non-negative dictionary makes the accumulator's constant part dwarf its varying one ($\|\mathbb{E}\mathbf{r}\|/\mathbb{E}\|\mathbf{r} - \mathbb{E}\mathbf{r}\| = 11.6$), and the bias-free decoder does attenuate that constant to the embedding mean (norm 0.581 against 0.564). But a decoder bias (`--biased-dec`) is worth only −0.62% of training MSE at matched step (the run died at 23% of its schedule, so its held-out 0.2464 is not comparable to the converged arm's): pinning $W_{\mathrm{dec}}\,\mathbb{E}\mathbf{r}$ is 1024 linear constraints on a 1024×2048 matrix, 0.05% of its freedom. The decoder is dense because a full-rank map carrying a ~900-dimensional signal is dense.

How the layers' partitions relate. Each head's winning entry partitions the eval tail, and Figure 31 compares every pair of the 380 heads of `ste_h76` (shipped init; `headnmi.py`). Heads partition independently except within layer 0: the mean adjusted NMI within a layer is 0.0075 at layer 0 and 0.0006–0.0028 above it, and at

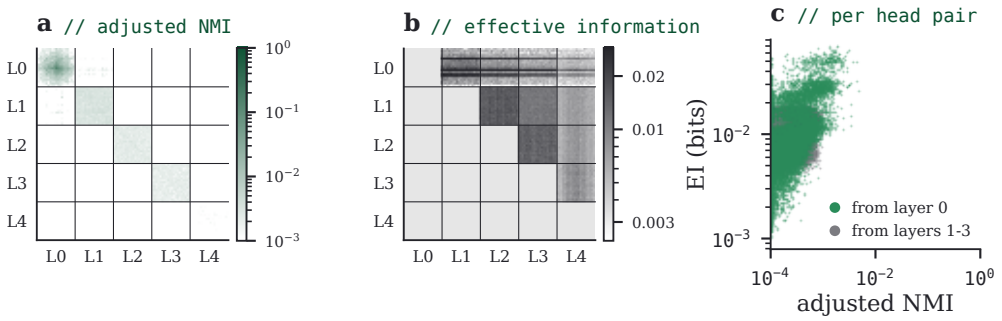


FIGURE 31 · **Partitions overlap, and track intervention, only in layer 0.** `ste_h76` (shipped init) on the eval tail, its 380 heads blocked by layer and ordered within a layer by hierarchical clustering. (a) Adjusted NMI between every pair of heads' winning-entry partitions (0 = chance), log scale, diagonal grey. (b) Effective information from the row head to the column head, with interventions uniform over the source's live entries; grey where the effect head is not downstream. (c) One against the other per head pair, by source layer. One seed.

most 0.0004 between layers (GPT-2's stack: 0.054 at layer 0, at most 0.004 elsewhere), so a later layer does not re-split what an earlier one split. Effective information, how far forcing one head moves another's choice, tracks that overlap only from layer 0: over head pairs its Spearman correlation with adjusted NMI is 0.76, 0.50, 0.38 and 0.23 from layer 0 to layers 1–4, and at most 0.08 from any later layer, whose heads move every downstream head by a similar 0.003–0.02 bits whatever their partitions share. One seed.

E.5 Steering by target layer, and head-health levers

Splitting the stack's steering by target layer (strength 1.0, 12 entries per layer) reverses the aggregate ranking of directions:

layer	decode	grad	margin	adjoint	random
0	0.216	**0.583**	0.547	0.237	0.034
1	0.464	0.089	0.367	0.000	0.023
2	0.742	0.130	0.216	0.000	0.055
3	**0.966**	0.130	0.091	0.000	0.034
4	0.362	0.047	0.036	0.005	0.047

Where the input reaches the classifier directly (layer 0), the classifier's own geometry (`grad`, `margin`) beats `decode` three- to tenfold. With depth it fails: a direction computed against layer ℓ's classifier input is only valid there, and the perturbation arrives after earlier layers have re-encoded it. The `decode` direction

moves the input toward what the model itself emits with the entry active, which is self-consistent down the stack. Layer 4 is hard for everything: its residual is nearly exhausted. The practical reading is to steer layer 0 by gradient and the middle layers by decode.

Head-health levers and steering. Three levers were tried on the shipped stack. *Mean-entropy pressure is not what costs steerability*: a twin trained at $s_{\mathrm{KL}_m}=10^{-6}$ has raw text-cycle effect 0.043 against 0.027, both an order of magnitude below `top2/top8` at the same weight, so hard selection, not the bonus, is the cost (removing the bonus does cost head usage: imbalance 5.00 bits against 2.71). *Head-independence pressure (HSIC/CKA) was chasing estimator bias* (Section E.9). *A row-collinearity setpoint is the one lever that bought steerability*: holding within-head row cosine at 0.5 (against an uncontrolled 0.774) improves embedding-space realization in every direction at lower collateral — `decode` +15%, `grad` +17%, `margin` +52%, oriented `adjoint` +122%, collateral 0.591 → 0.552 — with held-out FVU_w slightly better (0.2157 vs. 0.2293). The controller acts once (Figure 32): as the max row cosine over heads and layers climbs past 0.8, the multiplier rises to 4.9×10^{-3} near step 35k and pulls it down to the setpoint by 50k; from about 60k the multiplier is projected to zero and the statistic stays at or below 0.5 without pressure, ending at 0.473 against the uncontrolled 0.774. Its mechanism is unexplained: the statistic is computed in the e-dimensional dictionary space, where 74–78% of each atom's energy is in the decoder's null space (Section G.1), and decoded the atoms were already orthogonal. That the classifier-geometry directions gain most suggests an effect on the classifier rather than the dictionary, but nothing establishes it. The setpoint does not transport to the flat arm, whose steering is already saturated (0.992 vs. 0.966).

E.6 Supervised vectors at their own step size

Table 3 compares directions at an equal step. A real intervention has its own length: for a model, the output change of forcing the entry; for an SAE latent, its decoder direction at its mean activation; for the difference of means, the difference vector itself, which is the step activation addition takes. At those lengths (realized / collateral, median step as a fraction of $\|\mathbf{x}\|$, and each equal-length random step's collateral):

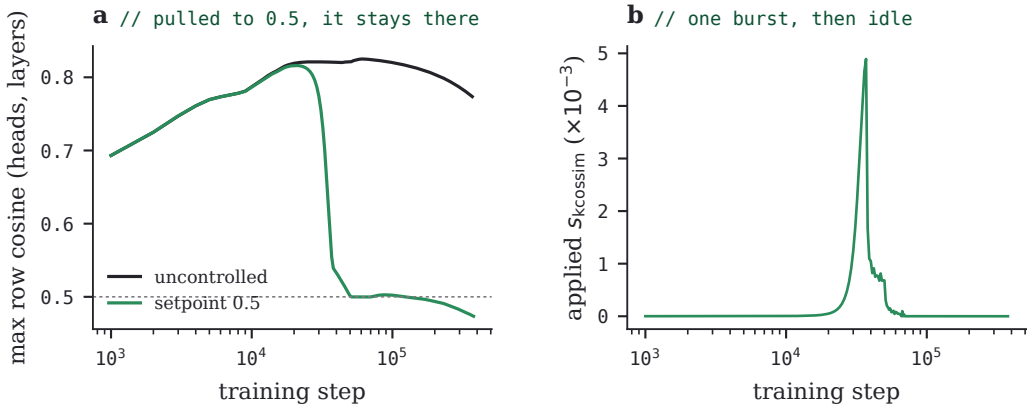


FIGURE 32 · The row-collinearity setpoint on the shipped stack, from the training log. (a) The max within-head row cosine over heads and layers, the statistic the controller watches, against the uncontrolled twin; the dotted line is the 0.5 setpoint. (b) The applied multiplier s_{kcossim}, averaged over 1000-step windows.

model	own (step)	diff. of means (step)	random: own / dm
softmax stack	0.19 / 0.25 (0.064)	0.18 / 0.25 (0.079)	0.25 / 0.30
hard-code stack	0.08 / 0.54 (0.129)	**0.27** / 0.47 (0.064)	0.53 / 0.48
hard-code flat	0.17 / 0.04 (0.028)	**0.54** / 0.12 (0.087)	0.05 / 0.17
`g160top1`	0.41 / **0.01** (0.186)	0.57 / 0.04 (0.053)	0.18 / 0.07
`g160softmax`	0.54 / 0.07 (0.050)	0.57 / 0.13 (0.107)	0.08 / 0.16
`TopK` SAE, k_{SAE}=32 (m=5120)	0.46 / 0.06 (0.072)	**0.77** / 0.15 (0.180)	0.07 / 0.17
`TopK` SAE, k_{SAE}=32 (m=11264)	0.50 / 0.08 (0.076)	**0.82** / 0.17 (0.172)	0.09 / 0.19
SAE, bilinear encoder	0.82 / 0.04 (0.078)	0.88 / 0.08 (0.121)	0.04 / 0.07

The difference of means realizes more in most models, and in the SAEs and the flat arm partly by taking 2–3 times longer steps. The hard-code stack is the clear case where direction, not length, decides: the supervised vector reaches its entries three times as often with half the step. The softmax stack's own intervention and the supervised vector tie. `g160top1`'s own intervention is the most precise step any method takes, disturbing 1.4% of other heads against 18% for random.

E.7 Data-shaped controls, and where a decode step realizes

Every steering curve is scored against an isotropic random step. SONAR's evaluation tail is anisotropic — 158 of its 1024 principal components hold half its variance, 474 hold 80% — so a unit step in raw coordinates is three times as long in the data's own (Mahalanobis) metric as a step shaped like the data, each relative to the distance between two tail rows, while the objective's diagonal whitening barely registers the difference (1.045 against 1.005). `steerembed.py` also scores three random controls of the same raw length: the isotropic draw shaped by the

covariance's square root, the same draw shaped by its inverse square root (toward what the data barely vary along), and the step toward another tail row. Collateral at strength 0.25, on the targets and rows of Table 3:

model	isotropic	shaped	to a row	inverse	decode	diff. of means
softmax stack	0.539	0.408	0.410	0.803	0.549	0.424
hard-code stack	0.653	0.634	0.640	0.754	0.658	0.629
hard-code flat	0.367	0.315	0.309	0.521	0.328	0.274

Against the data-shaped steps the softmax stack's decode direction disturbs 0.14 more heads, and its gradient, margin and adjoint directions 0.15–0.20 more, though all are within 0.07 of the isotropic step. In the hard-code stack every direction stays within 0.025 of the data-shaped steps. In the flat arm the decode direction and the difference of means stay within 0.041 of them, and the gradient-based directions sit 0.06–0.07 above, at the isotropic level. Steps toward what the data barely vary along disturb the most heads in every model.

Splitting the decode direction at the leading principal subspace holding 50% or 80% of the variance, and stepping each part at the full step's length, locates what realizes the target. In both stacks the part in the bottom fifth of the variance realizes 95% (hard) and 100% (softmax) of what the full step does, and the part in the top half 20% and 8%; the flat arm realizes through either part (0.958 from the top 80% alone, against 0.996). The decode direction leans toward the low-variance end: the data's variance along it is 1.6–1.8 times the mean eigenvalue, against 6.2–6.7 for a data-shaped step and 7.1–11.7 for the difference of means. That is the sensitivity to directions the data barely vary along which Ilyas et al. [84] tie to non-robust features. Equal raw length is the budget used throughout, so the split says where the classifiers are sensitive per unit of raw length; whether the SONAR decoder reads the low-variance part is untested. One seed per model.

E.8 What a removal does downstream

Removing a head shows what the cascade does with a change it did not choose (`textstrip.py --removals`). For 16 random heads per layer on 512 tail rows, each head's assignment is replaced by the uniform one, by the head's mean assignment, by its runner-up entry (what winner dropout trains on) or by another row's assignment, and the reconstruction is rebuilt either live, with later layers reclassifying the changed residual, or frozen, with only that head's contribution changed. Two readouts, whitened and each a share of the unablated reconstruction's error: the error the removal adds, and how far the output moves from the unablated reconstruction. In the hard-code stack the live forward cuts a layer-0 head's added error to 0.23 of the frozen value (0.48, 0.68 and 0.84 at layers 1–3), and cuts the runner-up replacement most (0.13), as backup heads take over a knocked-out head's role in IOI [85]. Yet the live output lands 19 times as far from the unablated

reconstruction as the frozen one (31, 19 and 7 times at layers 1–3): the later layers describe the input anew rather than restore the old description. Decoded to text, on 12 tail rows and the two heads per layer whose uniform removal moves the reconstruction furthest, a live uniform removal raises the decoder's per-token negative log-likelihood of the input's own decoding by nothing beyond the reconstruction's (−0.011 nats, against 0.036 frozen and 0.016 for a random step toward another row of the frozen change's length), though scored against the unablated reconstruction's text it looks like damage (0.139 against 0.040). The reference softmax model's later layers cancel a removal outright: a frozen layer-0 uniform removal adds 505% to the error and a live one 0.5%. These text runs compute the model in float32 on the CPU; JAX's default matmul precision on the GPU changes 226 of the hard stack's 4,560 codes on these 12 rows (159 of the 912 at layer 4), so cell-level comparisons across devices need matched precision. One seed per model.

E.9 Head independence: a measurement artifact

A head-independence penalty (`--s-hsic-heads`: the mean over a layer's head pairs of RBF-kernel CKA [86] between the heads' batch assignments, at the median-heuristic bandwidth) appeared to find substantial dependence. It was measuring bias: CKA as defined there normalizes the biased HSIC estimator [86, Eq. 3], which is positive for independent heads:

layer	0	1	2	3	4	mean
unbiased	0.0041	0.0005	0.0005	0.0004	0.0002	**0.0011**
biased	0.0315	0.1062	0.1018	0.0963	0.0976	0.0867

The biased estimator reads chance co-occurrence between independent categorical heads as dependence, with a floor that grows as the batch shrinks (0.109 at b=256, 0.029 at 1024, against an unbiased 0.0001 at every size). The floor also falls with the number of entries a head uses, and a frozen head, whose constant code depends on nothing, scores zero against every other: on synthetic independent heads at b=256 and k=32 it reads 0.108 at uniform usage, 0.016 at about four effective entries, and 0.026 with half the heads frozen (`cka_floor.py`). At the training batch the floor exceeded any real dependence, so the penalty had nothing to descend but the bias, and the bias descends only by concentrating or freezing heads. Reconstruction resisted that here; where the penalty replaced the softmax model's auxiliary losses, it froze a layer (Section D.1). `hsic.pairwise_head_cka` now normalizes Song et al.'s unbiased estimator [87, Eq. 5] instead ("debiased" CKA, as in the minibatch CKA of [88]); its expectation is exactly zero for independent heads at any batch size, and it reads 0.06 at 0.5 planted factor correlation and 0.41 at 0.8. Conditioning on language or script explains none of the residue. **This architecture produces near-independent heads with no pressure to do so**;

trained out under the corrected estimator the penalty changes nothing and costs localization (best-head language NMI 0.199 against 0.269). Two lessons generalize: before trusting a statistic as a training signal, measure it on a null it should score zero on; and loss-value share is not gradient share — at s_{HSIC}=10^{-4} the penalty was 12% of the loss value, while the measured MSE-to-CKA gradient ratio (1250 at the classifier, 221 at the dictionary) put equal pull at 8×10^{-4}.

E.10 Seed reproducibility, measure by measure

Partitions. The best-matched pair of heads reaches NMI 0.42, but the mean is 0.012 against a 0.005 null, and `splitting.py`'s entry containment reads 0.000. Neither is a metric artifact: comparing one run against its own checkpoint 19,500 steps earlier, the same partition measure scores 0.38 overall and 0.87 on layer 0. *Contributions.* Two heads could carve differently and add the same thing; they do not (centered cosine 0.0074 vs. 0.0015 null; uncentered it reads 0.205, which is the non-negative dictionary's shared offset). `headcontrib.py`, which fixes the measurement (decoded, whitened, centered contributions, Hungarian-matched), reads 0.019 against a 0.0017 null on the same pair at 4,096 rows; the earlier one-off figure's exact method was not recorded, and the conclusion is unchanged under either. *Subspaces.* Top-8 eigenspaces of per-layer contributions agree 25–33× above chance — but each model agrees with the *cache's own* principal directions by the same amount (e.g. layer 1: 123.7× between seeds, 101.3× and 101.2× to the data), so the agreement is the data's and nothing is left for the architecture. At layer 4 each seed tracks the data ~23× above chance while agreeing with the other only 9×: they pick *different* data-aligned directions. *Grams* (`atomgram.py`). Per-layer row collinearity agrees to four digits (0.2880 / 0.2891 at layer 0), but the Gram spectrum is nearly one-parameter — the top eigenvalue is exactly $1 + (k-1)\,\kappa_i$, carries 82% of the spectrum's norm, and is forced by non-negativity. Comparing two heads' Grams entrywise needs a correspondence between their entries. In canonical order (each head's entries sorted by their mean cosine to the rest of the head), Grams agree at chance against a null that pairs heads within the layer, across seeds (1.06–1.11×) and against the same run 19,500 steps earlier alike (1.05–1.14×). The second is a property of the order, not of the heads: with a nearly one-parameter Gram every entry's mean cosine is nearly the same, so the sort is set by small differences that change between checkpoints, and within one run it pairs only 33–66% of heads with themselves. Training never relabels entries (dead-entry revival is off in both runs), so within a run the stored order is the correspondence, and in it each head's Gram matches its earlier self at 8.5–10.7× chance (7.5–10.0× on decoded atoms), its decoded atoms at cosine 0.98–0.99 against 0.05–0.08 for the layer's other heads. Across seeds, with entries matched by linear assignment on their centered decoded atoms, which fixes the correspondence exactly, Grams stay at chance (1.05–1.19×), the matched atoms at cosine 0.07–0.15 against 0.06–0.10. Heads persist within a run and do not replicate across seeds.

E.11 What makes layer 0 reproduce: gain, atoms, and depth

Layer-0 heads are the largest and the most reproducible, so "layer 0 reproduces" might be "large heads reproduce". `headcontrib.py` measures a head's size as its centered contribution energy over the whitened target variance, and splits it exactly: a head's contribution on a row is its layer's input gain times the decoded atom it selected, so its size is gain^2 times atom spread^2 (the usage-weighted RMS distance of its decoded atoms from their mean); on both stacks the product reproduces the measured size to a log correlation of 1.000.

	GPT-2 concat			SONAR stack		
layer	agreement	gain	atom norm (10^{-4})	agreement	gain	atom norm
0	0.171	122.2	4.6	0.080	1.000	0.043
1	0.017	67.9	5.3	0.005	0.588	0.054
2	0.007	62.3	4.4	0.004	0.490	0.052
3	0.005	59.2	3.7	0.003	0.417	0.051
4	0.004	57.2	3.2	0.003	0.359	0.043

TABLE 19 · Head agreement (matched centered contribution cosine) and the two factors of head size, medians over each layer's heads; atoms decoded and whitened. SONAR: `ste_h76_init01` against `ste_h76_i01_s43`, 4,096 rows.

Layer 0's larger contributions are all gain. It receives the raw input — GPT-2 activations of norm ~122 against residuals of ~60; unit SONAR embeddings against residuals of 0.36–0.59 — while its atoms are no larger than layer 1's on either dataset, and on SONAR smaller than those of layers 1–3 and equal to layer 4's. Gain is one number per layer per row, shared by all of a layer's heads, so across layers "size" is the layer itself under another name. At equal atom spread, layer 0 still reproduces 14× better on GPT-2 and 23× better on SONAR (log agreement on log spread and a layer-0 indicator, coefficient of determination 0.979 and 0.941).

Against the flat arm at matched size. Within a stack, head size and layer do not overlap, so their effects cannot be separated there. The flat arm breaks the confound: all its heads see the raw input, with gain 1 and nothing downstream. On SONAR its own agreement tracks log head size at correlation 0.876 — from 0.008 for its smallest heads (0.057–0.060% of target variance) to 0.150 above 0.071%, and 0.78 and 0.56 for its two largest — so size matters a great deal on its own, and since its gain is constant it is an atom-magnitude curve. But the stack's heads fall far below that curve at every depth, layer 0 included:

stack layer	size, median	stack agreement	flat heads at that size (±10%)	gap
0	0.31%	0.080	0.67 (2 heads)	~8×
1	0.16%	0.005	0.12 (1 head)	~23×
2	0.11%	0.004	0.19 (3 heads)	~49×
3	0.073%	0.003	**0.059** (115 heads)	~**19**×
4	0.039%	0.003	below the flat range	—

The stack costs reproducibility at matched size, at every layer. Layer 0 sees exactly the flat arm's input and still reproduces several times worse than an equal-size flat head: in the stack, what a layer-0 head should carve depends on what the layers after it do with its error, so the input alone no longer pins it. Deeper layers lose more. This is the reproducibility face of the entanglement result (Section 5.4): the cascade that doubles a single-head intervention's collateral, to 66% of the other heads, also leaves each head underdetermined by the data. Read against the flat curve, the gain difference between layers 0 and 3 accounts for roughly a third of layer 0's lead in log terms; only layer 3's size match is solid — flat heads are almost all small, so layers 0–2 rest on one to three heads each — so the fraction is rough and the direction firm. The matched-size comparison is SONAR-only; GPT-2 has no flat arm.

Within layer 0 the atoms decide. SONAR's layer-0 gain is identically 1, so its within-layer size variation is pure atom magnitude, and there log agreement tracks atom norm at correlation 0.62 and spread at 0.64 (cosine against log size, 0.91); GPT-2's layer 0 shows 0.17 and 0.42. Past layer 1 the atoms predict nothing on either dataset. Spread and norm are one quantity here — their ratio is 0.99–1.08 in every layer, so a head's usage-weighted mean decoded atom is near zero: the non-negative dictionary's offset is cancelled by the decoder before it reaches the output.

E.12 How entries share a layer

An entry writes only where it wins its head, and then writes its layer's gain times its decoded atom, so its energy over the data is $\mathbb{E}\left[\mathbf{1}[\text{selected}]\,\text{gain}^2\right]\,\|\text{atom}\|^2$. Per layer, over 32,768 held-out rows (`entryshare.py`):

Layers 1–3 are near-equal; layer 0 is not, and its atoms are why. Deep entries have an effective number of 95–99.5% of the total and the largest is about twice the median; what inequality they have is mostly usage, itself mild (CV 0.14–0.18). Layer 0's usage is just as even — on SONAR every entry fires 2.1–4.5% of the time against $1/k = 3.1\%$ — but its atom norms vary five to ten times more (CV 0.34–0.52 against 0.02–0.10), mostly inside heads. The flat arm shows the same pattern more mildly, so it belongs to classifying the raw input. SONAR's last layer is unequal for the other reason: some atoms have shrunk almost to nothing (smallest/median 0.017), the inert final layer of Section G.4. Whole layers are unequal by gain: layer

run	layer	Gini	top 10% carry	max/ median	from atom norm	within heads
GPT-2 concat, k=256	0	**0.587**	42%	38.7	76%	60%
	1	0.155	16%	30.3	44%	100%
	2–4	0.11–0.12	14%	~2	~40%	100%
SONAR stack, k=32	0	**0.326**	32%	46.3	92%	83%
	1–3	0.06–0.10	12–14%	1.5–1.9	16–24%	~100%
	4	0.195	18%	3.2	77%	87%
SONAR flat, 1×380	—	0.190	23%	64.1	90%	76%

TABLE 20 · Inequality of entry energy within a layer. "From atom norm" is the covariance share of the variance of log entry energy (the remainder is usage; gain conditional on selection contributes ~0). "Within heads" is the share of that variance not explained by which head an entry is in. No entry is dead in any layer.

0 holds 67% of entry energy on GPT-2 and 49% on SONAR, layer 4 4–5%.

Where the large atoms sit differs completely between the models (`bigatoms.py`):

	GPT-2 concat, 20 heads	SONAR stack, 76 heads
within a head (atom norm / the head's own median, mean over heads)		
largest	3.7 (1.4–5.1 across heads)	2.0 (1.5–4.3 across heads)
2nd / 8th / 16th largest	3.2 / 2.0 / 1.7	1.7 / 1.2 / 1.0
largest entry's share of head energy	6.1% (uniform 0.4%)	9.9% (uniform 3.1%)
across the layer (atom norm / the layer median)		
atoms > 2×	1030 (20%), in all 20 heads	63 (2.6%), in 28 heads
atoms > 3×	58, in **all 20** heads (at most 15 each)	11, in **4** heads
atoms > 5×	1	5, **all in head 54**
largest head's energy share	15.5% (3× uniform)	9.2% (7× uniform)
usage		
Spearman(atom norm, usage) in heads	+0.81	+0.27 (−0.04 to +0.58)
usage of atoms > 3× / of the rest	0.0062 / 0.0037 ($1/k$ 0.0039)	0.033 / 0.031 ($1/k$ 0.031)
direction		
$\lvert\cos\angle\rvert$ between heads' largest atoms	0.022 (random pairs 0.048)	0.023 (random pairs 0.028)
energy in the data's top 10 PCs	4% (all atoms 13%)	9% (all atoms 3%)

On GPT-2 every head has a graded tail — its eighth-largest atom is still twice its median — and the large atoms are its frequent outcomes (1.6× uniform usage).

Agreement is fairly even across heads (0.135–0.235) and only loosely tied to atoms (Spearman 0.49 with energy share, 0.34 with the largest atom). On SONAR the large atoms are a few heads' property, and fire at the uniform rate: large, not frequent. Head 54 holds 8 of the 11 atoms above 3× and all 5 above 5×, carries 9% of the layer, and reproduces at **0.844** against 0.249 for the next head; across heads, agreement tracks the largest atom (Spearman 0.62) and how peaked the head's profile is (0.62), not its typical atom (−0.14). That resolves the within-layer-0 atom effect into a few heads with outsized atoms — but not the layer-0 lift: without head 54 the layer-0 mean falls only from 0.080 to 0.070, still twenty times the deep layers. Head 54 is the topic/genre head of Section 5.6, which recurs across seeds.

head	agreement	energy share	largest / layer median	atoms > 3×	largest / head median
SONAR stack, layer 0 (uniform share 1.3%)					
54	**0.844**	9.2%	6.81	8	3.41
46	0.249	2.5%	3.21	1	2.24
6	0.181	1.5%	2.76	0	2.83
18	0.150	1.9%	4.50	1	4.34
lowest four	0.008–0.021	1.0–1.1%	1.51–1.84	0	1.48–1.84
GPT-2 concat, layer 0 (uniform share 5%)					
6	0.235	3.1%	4.31	2	4.77
17	0.220	15.5%	4.99	5	1.90
19	0.215	3.4%	3.82	3	4.16
0	0.202	3.1%	3.95	1	4.19
lowest four	0.135–0.141	2.6–3.0%	3.13–4.17	1–2	3.45–4.56

TABLE 21 · Layer-0 heads by seed agreement (`bigatoms.py`; matched centered contribution cosine against the seed-43 twin). Spearman of agreement across heads, SONAR / GPT-2: with energy share +0.53 / +0.49; with the largest atom +0.62 / +0.34; with the head's peakedness (largest / head median) +0.62 / +0.09; with its median atom −0.14 / +0.16.

The two models differ in kind here. On SONAR one head stands out on every atom measure and alone reproduces strongly; the rest of layer 0 is a long, nearly flat tail, and agreement follows how peaked a head's atoms are. On GPT-2 no head stands out: every head has several large atoms, the spread of agreement is narrow, and the head with the largest share of the layer's energy (17) has a flat profile — uniformly large atoms rather than a few outsized ones — yet reproduces no better than its neighbors. Peakedness predicts nothing there (+0.09).

The large atoms are distinct directions: different heads' largest atoms are near orthogonal (median $|\cos \angle|$ 0.022–0.023, below random atom pairs). On SONAR they lean into the data's top ten principal directions (9% of their energy against 3% for a typical atom), consistent with cross-seed agreement being the data's dominant directions; on GPT-2 they lean the other way (4% against 13%). One seed pair per dataset throughout.

E.13 Language: the probe in detail

Held-out language identity over 86 languages (dense ridge ceiling 0.721, chance 0.012):

arm	shape	best-head NMI	less null	probe h'=1	h'=32
`ste_h76_init01`	5×76	0.180	0.169	0.103	0.420
seed 43	5×76	0.165	0.154	0.089	0.430
softmax reference	5×32	0.055	0.044	0.036	0.275
unfixed initialization					
`ste_h76`	5×76	0.269	0.261	0.165	0.385
+ collinearity setpoint	5×76	0.287	0.279	0.139	0.317
− mean-entropy bonus	5×76	0.085	0.078	0.038	0.321
+ head independence	5×76	0.199	0.189	0.097	0.389
flat	1×380	0.277	0.276	0.102	0.259
+ collinearity setpoint	1×380	0.247	0.246	0.080	0.273

TABLE 22 · `headlang.py`. Heads are ranked by NMI on the probe-training rows and scored on the held-out tail, and each probe's ridge penalty is chosen on a validation split of the training rows. The attainable NMI for a 5-bit head against the 6.43-bit label is 0.875. Head-health levers and the flat arm were measured only at the unfixed initialization.

As expected of a language-agnostic embedding, no head encodes language outright: the best reaches 0.18 of an attainable 0.875, and no head-health lever lifts any arm past 0.29 (a k=32 head could not hold 86 languages anyway, $\log_2 32 = 5 <$ 6.43 bits). What the probe does show is that partitions carry a known factor well above chance, and that **the hard code carries more of it**: 3.5–3.9 times the softmax reference on null-corrected best-head NMI, and 0.42–0.43 against 0.27 in a 32-head probe, consistently across seeds. The fixed initialization spreads language over more heads than the unfixed one did (null-corrected best head 0.15–0.17 against 0.26, 32-head probe 0.42–0.43 against 0.38). The stack beats the flat arm on probe accuracy at every head budget (0.385 vs. 0.259 at h'=32, both unfixed).

Completeness. The top 32 heads, all in layer 0, read 0.42–0.43, 73–74% of what all 380 read (0.57–0.58), but they are not complete in the sense of IOI's circuit criterion [85], under which removing them should leave the rest of the code unable to do the task: the other 348 heads still read 0.34–0.36, the other 44 layer-0 heads alone 0.26, and the 304 deeper heads together 0.13 (32 random deeper heads, 0.03). Language is spread over layer 0, not held by a few heads. (`summary.json`'s `info_share_best` favors the flat arm, but it divides by summed plug-in MI whose per-head bias differs eightfold between the shapes; do not read it across architectures.)

Script, cells, conjunctions. Script behaves the same way (best-head NMI 0.191 of a 0.671 ceiling; the same head, L0 h50, is best for both). A single cell is a strong

but permissive detector: the best cell per language reaches recall 0.74–0.98 at FPR 1.6–2.2%, but no language reaches precision 0.8. Conjunctions of top cells from distinct heads do not fix it: F1 peaks at h'=2 conjoined cells, 7% above a single cell, and the best precision is 0.598 at 4.8% recall. Recall combines nearly independently (0.234 against 0.262 predicted) but false positives do not (precision 0.43 at h'=2 against 0.83 if independent): many heads send the same confusable rows to the same wrong language — Finnish with Estonian, Japanese with Korean. The code encodes language *neighborhood* rather than identity.

Why neighborhoods, from the embedding geometry. The language-agnostic training leaves exactly the residue this pattern implies. Language explains 6.51% of embedding variance and script 2.26%; corrected for the number of classes (`headeta.py`) language is still a stronger partition than 90% of the heads (51× its random baseline, against 5× for the median head and 235× for the best), but its languages are weakly separated regions, not directions: within-language mean cosine is 0.3596 against 0.3160 between, and the shared mean direction accounts for 0.3178 of that, leaving a +0.044 excess, 0.30 sd of the pair distribution. A partition code cuts along local residual variance, and at 0.30 sd those cuts pass through the language clusters rather than around them. Layer-4 heads are near-random partitions (2× baseline), which is Section G.4's inert final layer seen from a different direction.

The topic head's entries. Decoding and autointerp were run on the shipped-initialization run's head 54, the head that holds the same index in the fixed-initialization run (Section 5.6). Decoded as deviations from the head's mean atom, its entries read as topic and genre. Contrastive autointerp (null 0.273) groups them as genre or topic (16 entries, F1 0.523), multilingual or code-switched passages (8, 0.476), text artifacts such as duplication and mojibake (3, 0.395), and script or region (5, 0.394); all four beat the null.

F Supplementary results: GPT-2 hard codes

F.1 What each design change buys

One-epoch runs (10M tokens) isolate them:

variant (800 bits unless noted)	FVU_w	loss recovered
STE alone (latent dictionary + decoder)	0.327	93.3%
+ gain	0.290	94.3%
+ gain, private heads	0.288	94.2%
+ gain, direct atoms	**0.217**	**96.95%**
seed 43	0.219	96.85%
k=64 (960 bits)	0.186	97.6%
k=128 (1120 bits)	0.165	97.9%

Gain-shape and direct atoms each help, and direct atoms most. Their advantage is partly speed: over 40M tokens private heads close the gap and finish slightly ahead (0.186 against 0.195). Without direct atoms and gain, more entries per head collapse rather than help (FVU_w 0.75 and 0.80 at k=64 and 128), and the other straight-through configurations froze their last layer at 4.8–5.0 bits, every head constant; the direct code's deepest layer never froze. The base result reproduces across seeds to 0.002 FVU_w. Annealing matters: straight-through at a constant t=1 reaches 0.50 and a scheduled soft-to-hard mix 0.39. Sparse encoders in front of the classifier never beat the dense one: at one epoch, 32- and 640-wide front-ends reach 0.25–0.29 against the dense classifier's 0.217, and a 4096-wide, 64-active encoder per head with dead-latent resampling reaches 0.225 against 0.195 for the dense front-end on the same 40M tokens, with 88% of its layer-1 latents dead at no cost to that layer.

The 40M-token runs, layer by layer. Every layer contributes: prefix FVU_w runs $0.472 \to 0.347 \to 0.278 \to 0.231 \to 0.195$ for the direct code and $0.451 \to 0.186$ for private heads. Entry usage stays near uniform in every layer (imbalance 0.01–0.16 bits of 5, and 0.30 at the private code's last layer), and the observed gain is close to the best per-sample scale for each layer's contribution (optimal rescale 0.89–1.09, coefficient of variation 0.05–0.08). The runs use two options this branch does not implement, `resid_first` and `gain_clip`, so they reproduce only on the `headline` branch; its direct atoms, private heads (this branch's `ConcatDictBlock`), fibers and head dropout have been ported.

G Supplementary results: dictionary geometry

G.1 Row collinearity is mostly gauge

Within-head row cosine κ_i (`cossim_k`, `rowcos`; $\bar{\kappa}$ its mean over heads) is computed on `dicts()` in the e-dimensional dictionary space. The decoder is $d \times e$ at full rank, so $e - d$ of those dimensions are its null space, and on the SONAR straight-through arm atoms put 74–78% of their energy there, where nothing they do reaches the output and no gradient reaches them. Decoded into output space the same statistic reads 0.023 at layer 0 and −0.015 to −0.020 below it, against 0.12–0.29 in e.

Dictionary statistics should be computed after decoding.

On GPT-2 layer 8 (e=1536, d=768) four tempting readings of the null-space mass each failed a check: head null components add coherently (3.3–11× over the cancelling expectation), so they are not interference debris; the fraction does not track feature noise; the per-atom visible fraction is flat at 22% across training and the decoder's row space rotates only 2.6–3.5°, so it is not a swap space; and a mixed-sign visible component needs no non-negative "lift". Two baselines also had to be corrected: for $|\mathcal{N}(0,1)|$ atoms the chance visible fraction is 18.26% (not 50%) and the chance alignment with $\mathbf{1}$ is 63.7% (not ∼0), so the trained dictionary is *less* null-heavy than chance; and a reported 49% of reconstruction gradient in $\ker(W_{\text{dec}})$ was a coordinate artifact, 3e-8 with respect to the atoms $|\mathsf{D}|$. How large "the null part" is depends entirely on the norm asked for: 2.43% of squared L2, 23.20% of L2, 56.55% of L1.

A random decoder's cone. Wendel's theorem [89] says how likely non-negativity is to bind at initialization. If the decoder's e columns are drawn from any sign-symmetric distribution in general position (an i.i.d. Gaussian initialization qualifies), they lie in a common open half-space through the origin — so the cone $\{W_{\text{dec}}\mathbf{r} : \mathbf{r} \geq 0\}$ misses directions of $\mathbb{R}^d$ — with probability $2^{1-e}\sum_{q<d}\binom{e-1}{q}$: 1 for a square decoder, exactly 1/2 at e=2d, switching over a window of width $O(\sqrt{d})$ (for d=768, 0.98 at e=1456 and 0.02 at e=1616). GPT-2's e=1536 and the SONAR softmax models' e=2048 both sit at the midpoint. This describes the random initialization only.

G.2 Heads share coordinates: no partition

Section 3.1.4's between-head orthogonality — non-negative heads that are orthogonal own disjoint coordinate blocks of the dictionary space, as a property of the weights — is the premise for reading a dictionary statically. Every model here trained before the support overlap ω existed, most with its predecessor penalized, so it had never been measured. Scored on final checkpoints (`headsupport.py`; one seed per row):

model	ω, dictionary (null)	spread	ω, decoded (null)
SONAR `resid_nc` (372.6k steps)	0.818 (0.815)	0.83	0.917 (0.910)
SONAR `resid_nc_hm` (8.27M steps)	0.113 (0.118)	0.13	0.892 (0.878)
SONAR `softmax_shm`	0.731 (0.738)	0.74	0.938 (0.930)
SONAR `top2_shm`	0.862 (0.860)	0.87	0.972 (0.958)
SONAR straight-through, shipped init	0.942 (0.942)	0.94	0.721 (0.719)
SONAR straight-through, 5×76	0.962 (0.961)	0.96	0.980 (0.965)
SONAR straight-through, 1×380	0.977 (0.977)	0.98	0.974 (0.963)
GPT-2 abs, e=1536	0.739 (0.643)	0.65	0.948 (0.925)
GPT-2 abs, e=768	0.596 (0.486)	0.52	0.875 (0.852)
GPT-2 signed, e=1536	0.937 (0.636)	0.64	0.964 (0.943)
GPT-2 signed, e=768	0.955 (0.950)	0.95	0.965 (0.943)
GPT-2 concat, d_{head}=32	0 by construction	—	0.846 (0.788)
`g160top1` (160 trained heads)	—	—	0.648 (0.644)
`g160softmax` (160 trained heads)	—	—	0.983 (0.975)
`k32` discovered groups (359)	—	—	0.899 (0.892)

No model partitions its dictionary space between heads. ω sits at or above its null everywhere, and each head's profile covers 52–98% of the coordinates where a partition into 32 heads needs about 3%. Where ω departs from the null it is *above* it: the GPT-2 arms place their heads on the same coordinates, the signed wide arm most. The one exception is `resid_nc_hm`, the reference configuration trained 22 times longer: its profiles are sparse and ω falls 4% below the null in four of five layers. That is long training sparsifying each head, not a partition — at spread 0.13 a head still uses four times its share — and the same configuration at 372.6k steps is dense and at its null. The predecessor penalty (s_{cossim_h}=10^{-6}, or 10^{-5} for `resid_nc`) did not produce the partition, and none of these models trained with ω itself as a loss term. `ConcatDictBlock` is the only route to disjoint support here, and it is disjoint by construction.

It is not the decoder's null space. Splitting each atom into its row-space (visible) and null-space parts and scoring each against the same null, the visible part is at chance on SONAR and above it on GPT-2 (the signed wide arm 0.955 against 0.706), and the e=768 arms, whose square decoder has no null space, sit at or above the null in the dictionary space itself. The row space is not coordinate-aligned, so the visible part's support is basis-dependent; the decoded ω is the nearest basis-free check, and it agrees. The decoded measure itself cannot separate architectures: every model's atoms are dense across SONAR's output coordinates, and all fifteen rows land within 7% of their nulls, the trained and discovered SAE groups included. The comparison ω was built for exists only in the dictionary space, which the SAE variants do not have.

Training with ω. Penalizing ω does produce the partition, but only where reconstruction cannot see it. On GPT-2 with a signed dictionary (e=1536; three

epochs, one seed per weight), it falls below its null from s_ω=10^{-4} (ratio 0.61, and 0.15 at 10^{-3}), at a cost of 10%, 21% and 43% of held-out FVU_w for s_ω=10^{-5}, 10^{-4} and 10^{-3}. Decoded, nothing changes: every arm sits 8–10% above its null, as the unpenalized one does. The penalized arms move 73–95% of atom energy into the decoder's null space, where each head's profile is a few large coordinates of its own; the visible part stays dense and overlaps more than its null as the weight rises. A square decoder (e=768) gives the same table to within a few percent, because it grows a near-null space instead (condition number 1.0×10^4 unpenalized, 5.1×10^5 at 10^{-3}), and each penalized head puts nearly all its energy in directions the decoder shrinks tenfold or more. Measured on the dictionary weights, ω can be gamed through any learned decoder.

G.3 Concatenated heads

`ConcatDictBlock` at d_{head}=32 gives each head a $(768, 32)$ decoder block: 32 non-negative generators in a 32-dimensional subspace, a simplicial cone, and the per-head null space is zero. The floor on a head's mean row cosine is $(k/d_{\text{head}}-1)/(k-1)$, which is 0 at d_{head}=k=32, so concat faces no raised floor; it faces a collapsed tradeoff, since κ_i=0 at d_{head}=k needs one coordinate per atom. It took the correlation, the lowest effective *and* numerical rank (rank99 17.1 of 32, against 30.2 summing), and the worst error (Table 7). Its extra cost is reachability, the one place the cone frame binds; it is not explained by $\bar{\kappa}$ alone (at step 90k it had slightly lower $\bar{\kappa}$ than `abs`/768, 0.571 against 0.578, and much worse held-out error, 0.806 against 0.463).

G.4 Selection rule, training duration, and the last layer

Softer codes do not raise the dictionary's rank. On the SONAR selection sweep, numerical rank is full (rank99 30.7–31.8 of 32) in all ten arms, and effective rank varies 7-fold but is not monotone in softness, with the two families (± mean-entropy bonus) ordering oppositely at convergence. Tracing each arm over training resolves it: **training duration dominates the selection rule by 3.8×** (mean $\bar{\kappa}$ falls 0.750 → 0.217 from step 10k to 369.5k, while the spread across rules at a fixed step averages 0.140), and the orderings cross near step 100k (Figure 33b, the `_shm` family). The soft arm's $\bar{\kappa}$ plateaus while its error keeps falling, and the hard arms do the reverse: from 150k to 369.5k `softmax` cuts its FVU_w by 85% while $\bar{\kappa}$ moves 3%, whereas `top8` moves FVU_w 3% while cutting $\bar{\kappa}$ by 45%, `top16`'s FVU_w does not improve while $\bar{\kappa}$ falls 29%, and `top2` cuts both, 12% and 34% (Figure 33). Under a hard code a head's output *is* one of its rows, so collinear rows make the output independent of which entry wins; collinearity costs a hard code information directly, and a soft code, with continuous coefficients, much less. At convergence the softer code has the *more* collinear dictionary. Matching arms on held-out FVU_w instead of step reverses that at moderate error: `softmax` has the lowest $\bar{\kappa}$ at FVU_w 0.60 and 0.40 (0.60 and 0.58, against 0.72–0.78 for the hard arms). Only at 0.25 does `top2` drop

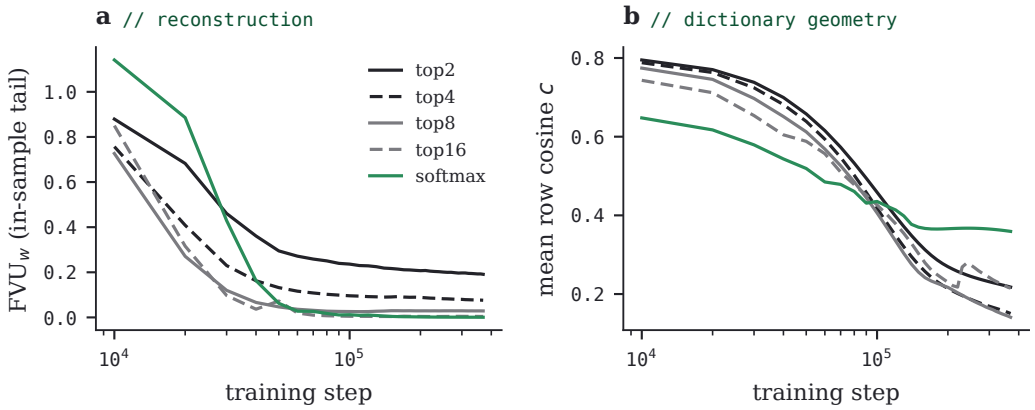


FIGURE 33 · The selection sweep's _shm arms over training: (a) FVU$_w$ on the (in-sample) tail, scored from each checkpoint, and (b) mean within-head row cosine $\bar{\kappa}$. The soft arm's $\bar{\kappa}$ flattens near 0.36 while its error keeps falling; the hard arms keep decorrelating.

below it (0.53 against 0.56), and it reaches that level at step 80k against 21–36k for the others, so its lower $\bar{\kappa}$ there is again duration. One seed per arm; these runs trained on the evaluation tail, so their FVU$_w$ is in-sample, though the softmax arm's moves by 0.13% on rows no model saw (Appendix B.1).

The last layer is inert once the error is small. Collinearity falls monotonically with depth in every hard arm, but soft arms spike at the last layer (`softmax_shm` layer 4: $\bar{\kappa}$=0.799, effective rank 1.54). `lastlayer.py` ablates the final layer's heads two ways: uniform (emit the mean atom) and zero (emit nothing):

arm	select	L4 $\bar{\kappa}$	L4 eff	prefix FVU$_w$ before L4	L4 adds	zero-ablate
top1	top1	+0.246	10.60	0.7511	+4.3%	+4.5% worse
top2_shm	top2	+0.188	14.89	0.2378	+19.7%	+24.5% worse
top8_shm	top8	—	—	0.0304	+5.5%	+5.8% worse
top16_shm	top16	+0.433	3.46	0.0043	+0.0%	+0.0%
softmax	softmax	+0.263	6.00	0.0023	−1.2%	−1.1% better
softmax_shm	softmax	+0.799	1.54	0.0007	−1.9%	−1.8% better

The last layer's contribution tracks the error left before it, not its collinearity: `softmax`'s layer 4 is not collapsed ($\bar{\kappa}$=0.263) and is just as useless. Below a prefix FVU$_w$ of about 0.005 the final layer is inert or mildly harmful — which accounts for `resid_nc`'s "mildly harmful" layer 4 (prefix 0.0051) without appeal to its frozen heads. The two collapses end differently: `top16_shm`'s layer 4 shrank its norms to 4% of the other layers' and emits nothing; `softmax_shm`'s is collinear at near-equal magnitude, emits a near-constant vector, and removing it helps — a small overfit, not a useful bias.

Partitions barely beat a plain smoother. The noise2self score that $L2_{pwak}$ would train can be read held out (Figure 34, `selection.py`): how well a head's co-classified neighbors predict a row's input, against the same heat kernel with every row in one cell. No arm of the sweep trained with the statistic. A head's partition improves on that smoother by at most 0.020, at layer 0 (`top4_shm`), by at most 0.005 at layers 1–3, and by 0.031 at layer 4 only in `softmax`, where the layer is inert anyway; a random partition with matched cell sizes is never better than the smoother, and a one-entry code (`top1`) is worse than it above layer 0. Since the score is not $\mathcal{J}$-invariant (Appendix B), the permuted partition is a floor only for partitions that ignore the input: a partition computed from the input, trained or not, can gain on it without clustering anything. One seed per arm; these runs trained on the evaluation tail.

Under a $\mathcal{J}$-invariant score, partitions show no more than locality. The split form (Appendix B) on `ste_h76_init01`, all 76 heads per layer, settles both confounds. It selects a bandwidth of 0.05 at layers 0 and 4 and 0.03 at layers 1–3, and under it the layer-0 smoother at 0.05 scores 0.112 below the smoother at Figure 34's 0.2, more than any head or gate gains at 0.2 (at most 0.060). A random-projection head is a locality-sensitive hash [90]: the chance it puts two rows in one cell falls as the angle between them grows, so gating the heat kernel by it sharpens the kernel the way a smaller bandwidth does. At a bandwidth of 0.2 one random head gains 0.017 over the smoother under the leave-one-out score and nothing under the split form (−0.001), so its gain was the leak, while the joint gate of eight keeps 0.042 and the collision kernel 0.047: they sharpen a kernel that is too flat. At the selected bandwidth single trained heads score worse than the smoother at every layer (by 0.040 at layer 0, 0.065–0.069 at layers 1–3 and 0.030 at layer 4), as single random heads do (0.030 and 0.063–0.068), and the layer's joint gate of trained heads gains 0.034 at layer 0 and 0.005–0.006 above it, where the random collision kernel gains 0.039 and 0.005–0.009. Trained heads read every coordinate, so these comparisons use the leave-one-out form; against random-projection heads, trained partitions show nothing beyond locality. On isotropic rows the split form holds its floor down to a bandwidth of 0.05, while the leave-one-out form falls below it by 0.010 at 0.2 and 0.090 at 0.02, which is its leak. One seed (`selection.py --jinv`).

G.5 Classifier geometry from the weights

Each classifier logit is one bilinear pair, $(\mathbf{w}\cdot\mathbf{u})(\mathbf{v}\cdot\mathbf{u})$, so $|\cos\angle(\mathbf{w},\mathbf{v})|$ says what kind of feature it is: near 1 the form is rank 1, proportional to $(\mathbf{w}\cdot\mathbf{u})^2$, even in $\mathbf{u}$ and so blind to the sign of the input along $\mathbf{w}$; near 0 it is a saddle, which sees whether its two projections agree in sign but is just as even, since $(\mathbf{w}\cdot(-\mathbf{u}))(\mathbf{v}\cdot(-\mathbf{u})) = (\mathbf{w}\cdot\mathbf{u})(\mathbf{v}\cdot\mathbf{u})$ on the semantic coordinates. Rank changes the shape of the region a feature responds to, not its sign-blindness, which only the constant coordinate breaks (protocol in Appendix B). Figure 35 reads it from the weights of the softmax stack (`resid_nc`) and the hard-code stack (`ste_h76_init01`).

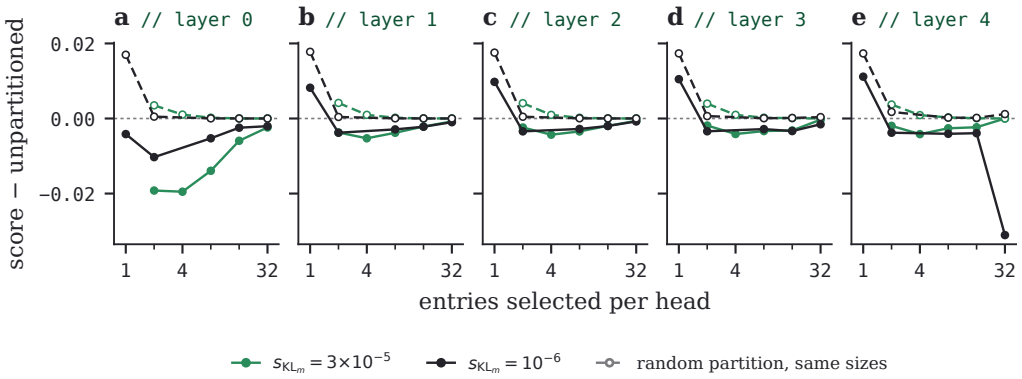


FIGURE 34 · **Partitions barely beat a plain smoother.** The selection sweep's held-out noise2self score per layer, against the number of entries each head selects (1 is argmax, 32 is softmax), less the same run's score with every row in one cell (below 0 the partition helps), for the two mean-entropy bonus settings. Filled: the heads' partitions, mean over heads; hollow, dashed: the same assignments with the rows permuted, a random partition with matched cell sizes. Heat kernel of bandwidth 0.2 on the layer's input, batches of 2,048 rows. One seed per arm, in-sample tail.

The softmax stack goes rank 1; the hard-code stack mostly does not. At the final checkpoint the softmax stack's median $|\cos \angle|$ is 0.94 in layer 1 and 0.52–0.80 in layers 2–3, where the hard-code stack's layers 1–3 stay at 0.10–0.16, against 0.021 for a random pair. Layer 0 sits at 0.54–0.64 in both, and the softmax stack's layer 4 falls to 0.18. The alignment is learned late: every layer starts at the random floor; the softmax stack's leave it between 30k and 50k steps, as its temperature anneal ends, and the hard-code stack's only from 80k (layer 0) and 220k (layers 3–4), with layers 1–2 above 0.05 only at the last checkpoint. The dominant eigenvalue is negative for 63–98% of the softmax stack's features and 97–99% of the hard-code stack's, so a rank-1 feature scores high where the input is orthogonal to $\mathbf{w}$, not along it: the quadratic invariances that bilinear layers learn in place of constants [52]. The constant coordinate can break the sign-blindness of a rank-1 feature for inputs weakly aligned with $\mathbf{w}$: in the softmax stack's layers 1–4 its weight is 0.18–0.24 of the rest of $\mathbf{w}$, in the hard-code stack's 0.017–0.025. One seed per run; weights only.

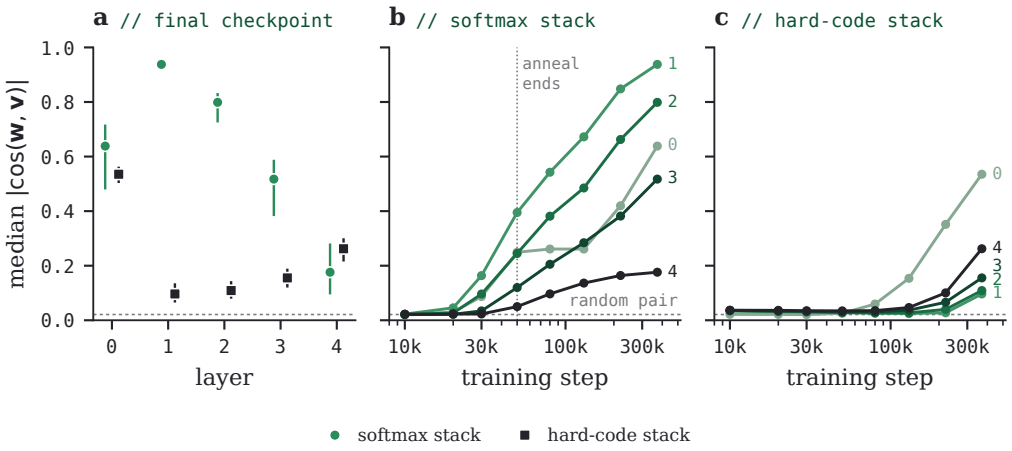


FIGURE 35 · **The softmax stack's classifier features go rank 1; the hard-code stack's mostly do not.** Median $|\cos \angle(\mathbf{w}, \mathbf{v})|$ of each layer's bilinear feature pairs (`bilinspec.py`), 1 for a rank-1 feature. (a) The final checkpoint by layer, with interquartile ranges: the softmax stack (`resid_nc`, step 372.6k) and the hard-code stack (`ste_h76_init01`, step 369.5k). (b, c) Over training, one line per layer, numbered at its end; the softmax stack's temperature anneal ends at 50k, and the hard-code stack's temperature is constant. Dashed: the median for a random pair in the 1024-dimensional semantic block, 0.021. One seed per run.